MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

SHIFT REGISTER, DIGITAL, KEY CODE CBF

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic shift register module.

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

* 2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 2.304 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be CBF.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/84-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

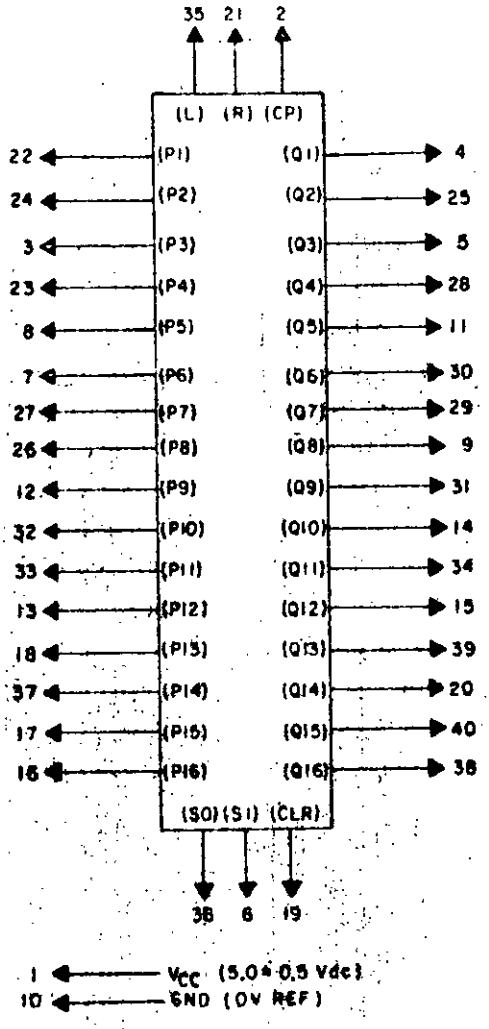
3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 V dc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 V dc ±1 percent.

3.10 Functional tests. All modules shall be subjected to functional tests which conform to the requirements stated in 3.10.1 through 3.10.3.

3.10.1 Test patterns. A set of unique input patterns shall be generated capable of detecting all nonredundant failures in the module logic network. The test patterns shall be generated by assuming "stuck at H state" and "stuck at L state" failures in each of the basic logic functions and applying input patterns to detect these failures. The unique test patterns necessary to satisfy these requirements shall be submitted with the qualification samples. If the tests of tables I and VIII satisfy or exceed these unique test patterns, no additional tests shall be added; otherwise, the needed tests shall be added to table VIII. "Stuck at H state" indicates a failure resulting in a steady high-level output. "Stuck at L state" indicates a failure resulting in a steady low-level output. Prior to any change in module design which would necessitate a different set of test patterns to detect a "stuck at H state" or a "stuck at L state", the vendor shall be required to submit to the procuring activity and SEMP Quality Assurance Activity the proposed new set of test patterns for approval.

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NOTES:
1. P1-P16 = Parrallel inputs.
2. Q1-Q16 = Outputs.
3. R = Shift right serial input.
4. L = Shift left serial input.
5. S0, S1 = Mode control inputs.
6. CP = Clock input.
7. CLR = Clear.

* FIGURE 1. CBF module function diagram, shift register.

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* TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A	Isolation	III	S	10	∞	1	∞	10	∞	1	∞	MΩ
B1-B16	V_{OUT} (low-loaded)	IV	S	0.01	1.00	0.01	1.00	0.01	1.00	0.01	1.00	V
C1-C18	Input breakdown current	V	S	0	1	0	1	0	1	0	1	mA
C19-C22	Input breakdown current	V	S	0	4	0	4	0	4	0	4	mA
D1-D16	Output leakage current	VI	X	0	1	0	1	0	1	0	1	mA
E1-E16	Output short circuit current	VII	X	-40	-100	-40	-100	-40	-100	-40	-100	mA
F1-F180	Functional	VIII	X	← Test vectors must produce output vectors as specified in test table VIII. →								
G1-G22	Input clamp diode voltage	IX	X	-0.4	-1.2	-0.4	-1.2	-0.4	-1.2	-0.4	-1.2	V
H1-H16	Output voltage, low level	X	X	0	500	0	500	0	500	0	500	mV
J1-J18	Output voltage, high level	XI	X	2.5	4.5	2.5	4.5	2.5	4.5	2.5	4.5	V
K1-K18	Input current, high level	XII	X	0	50	0	50	0	50	0	50	µA
K19-K22	Input current, high level	XII	X	0	200	0	200	0	200	0	200	µA
L1-L18	Input current, low level	XIII	X	-0.1	-2.2	-0.1	-2.2	-0.1	-2.2	-0.1	-2.2	mA
L19-L21	Input current, low level	XIII	X	-0.1	-8.8	-0.1	-8.8	-0.1	-8.8	-0.1	-8.8	mA
L22	Input current, low level	XIII	X	-3.0	-8.8	-3.0	-8.8	-3.0	-8.8	-3.0	-8.8	mA
M	Power supply current	XIV	X	135	488	135	488	135	488	135	488	mA
N1-N18	Transition time (t_{TLH}), C_L = 50 pF	XV	S	2	15	2	15	2	18	2	18	ns
N19-N36	Transition time (t_{TLH}), C_L = 150 pF	XV	Q	2	27	2	27	2	30	2	30	ns
P1-P18	Transition time (t_{THL}), C_L = 50 pF	XVI	S	2	8	2	8	2	10	2	10	ns
P19-P36	Transition time (t_{THL}), C_L = 150 pF	XVI	Q	2	11	2	11	2	13	2	13	ns

See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
Q1-Q18	Propagation delay time (t_{PLH}), C_L = 50 pF	XVII	S	5	14	5	14	5.0	15.5	5.0	15.5	ns
Q19-Q36	Propagation delay time (t_{PLH}), C_L = 150 pF	XVII	Q	5.0	19.5	5.0	19.5	5	21	5	21	ns
R1-R18	Propagation delay time (t_{PHL}), C_L = 50 pF	XVIII	S	5.0	18.5	5.0	18.5	5.0	20.5	5.0	20.5	ns
R19-R36	Propagation delay time (t_{PHL}), C_L = 150 pF	XVIII	Q	5	23	5	23	5	25	5	25	ns
S1-S18	Propagation delay time (t_{PHL}), C_L = 50 pF	XIX	S	5	21	5	21	5.0	22.5	5.0	22.5	ns
S19-S36	Propagation delay time (t_{PHL}), C_L = 150 pF	XIX	Q	5	25	5	25	5.0	26.5	5.0	26.5	ns
T1	Setup and hold time check (measuring Q1 output voltage, high level)	XX	S	2.5	4.5	2.5	4.5	2.5	4.5	2.5	4.5	V
T2	Setup and hold time check (measuring Q16 output voltage, low level)	XX	S	0	500	0	500	0	500	0	500	mV
T3	Setup and hold time check (measuring Q8 output voltage, high level)	XX	S	2.5	4.5	2.5	4.5	2.5	4.5	2.5	4.5	V
T4	Setup and hold time check (measuring Q9 output voltage, low level)	XX	S	0	500	0	500	0	500	0	500	mV
U1-U18	Maximum clock frequency check	XXI	S	36	44	36	44	36	44	36	44	ns
V1-V16	Output crosstalk, low level	XXII	Q	0.0	1.5	0.0	1.5	0.0	1.5	0.0	1.5	V
W1-W16	Output crosstalk duration, low level	XXII	Q	0	500	0	500	0	500	0	500	mV

See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
X1-X16	Output crosstalk, high level	XXIII	Q	0	700	0	700	0	700	0	700	mV
Y1-Y16	Output crosstalk duration, high level	XXIII	Q	0	500	0	500	0	500	0	500	mV
Z	Power supply transient amplitude	XXIV	Q	0	600	0	600	0	600	0	600	mV
AA	Power supply filter capacitance	XXV	S	3.7	7.2	3.7	7.2	3.7	7.2	3.7	7.2	µF
BB1-BB18	Input capacitance	XXVI	Q	5	15	5	15	5	15	5	15	pF
BB19,BB20, BB22	Input capacitance	XXVI	Q	5	60	5	60	5	60	5	60	pF
BB21	Input capacitance	XXVI	Q	5	72	5	72	5	72	5	72	pF
CC1-CC48	Noise sensitivity (PW)	XXVII	Q	450	550	450	550	450	550	450	550	ns
CC49-CC64	Noise sensitivity (PW)	XXVII	Q	0.0	1.5	0.0	1.5	0.0	1.5	0.0	1.5	V

1/ "X" parameters shall be 100 percent tested during module acceptance tests.

2/ "S" parameters shall be sample acceptance tested (1 percent, or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

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3.10.2 Input requirements. The following input voltage levels shall be used for test patterns:

Low level - Input voltage shall be 0.8 V ±1%.

High level - Input voltage shall be 2.0 V ±1%.

3.10.3 Output requirements. The output requirements for the test patterns shall be as follows:

Low level - Output voltage shall be 0 V minimum, 0.5 V maximum.

High level - Output voltage shall be 2.5 V minimum, V_{CC} maximum.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through CC shall be performed in any order with the following exceptions. If test conditions A, B, C, D, or E are part of a sequence of tests, the order of testing shall be as follows: A, B, C, D, E, where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through CC in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through AA shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 4.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 5. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition AA, shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests D through AA shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests D through CC shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

* 6.3 Cross reference of classification. Part number M28787/84-1 supersedes part number M28787/84. The two parts are interchangeable.

* 6.4 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table I.

Maximum power dissipation	2,684 mW maximum
Supply voltage	5 Vdc ±10 percent
Input voltage	
Low level	-0.5 V minimum, 0.8 V maximum
High level	2.0 V minimum, V_{CC} maximum
Input current P1-P16, R, L	
Low level (V_{IL} = 0.0 V)	-2.2 mA maximum
(V_{IL} = 0.5 V)	-2.0 mA maximum
High level (V_{IH} = 2.5 V)	50.0 µA maximum
(V_{IH} = 5.5 V)	1.0 mA maximum
Input current S0, S1, CP, CLR	
Low level (V_{IL} = 0.0 V)	-8.8 mA maximum
(V_{IL} = 0.5 V)	-8.0 mA maximum
High level (V_{IH} = 2.5 V)	200.0 µA maximum
(V_{IH} = 5.5 V)	4.0 mA maximum
Output voltage	
Low level (I_{OL} = 18 mA)	0.5 V maximum
High level (I_{OH} = -950 µA)	2.5 V minimum
Output drive current maximum capability	
Low level (V_{OL} = 0.5 V)	18 mA
High level (V_{OH} = 2.5 V)	-950 µA
Propagation delay time	
t_{PHL} - CP to Q	
C_L = 50 pF	5.0 ns minimum, 20.5 ns maximum
C_L = 150 pF	5.0 ns minimum, 25.0 ns maximum

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Propagation delay time (continued)

t_{PHL} - CLR to Q	
C_L = 50 pF	5.0 ns minimum, 22.5 ns maximum
C_L = 150 pF	5.0 ns minimum, 26.5 ns maximum
t_{PLH} - CP to Q	
C_L = 50 pF	5.0 ns minimum, 15.5 ns maximum
C_L = 150 pF	5.0 ns minimum, 21.0 ns maximum
Setup time	
t_{SETUP} (P1-P16, R, L)	8.0 ns minimum
t_{SETUP} (S0, S1)	11.0 ns minimum
Recovery time	
$t_{RECOVERY}$ (CLR)	10.0 ns minimum
Maximum clock frequency	
f_{MAX} (C_L = 50 pF)	50 MHz
Pulse width	
t_{PW} (CP)	7.0 ns minimum
t_{PW} (CL)	12.0 ns minimum
Hold time	
t_{HOLD} (P1-P16, R, L, S0, S1, CLR)	4.5 ns minimum

* 6.5 Module function. The module contains one 16-bit universal shift register. The register has four distinct modes of operation:

a. Parallel (broadside).

b. Shift right (in the direction Q1 → Q16).

c. Shift left (in the direction Q16 → Q1).

d. Inhibit clock (do nothing).

Synchronous parallel loading is accomplished by applying the 16 bits of data to P1-P16 inputs and taking both mode control inputs (S0 and S1) high. The data is loaded into a flip-flop and appears at the outputs after the positive transition of the clock input. During loading, serial data flow is inhibited.

Shift right is accomplished synchronously with the rising edge of the clock pulse when S0 is high and S1 is low. Serial data for this mode is entered at the shift-right serial input (R). When S0 is low and S1 is high, data shifts left synchronously and new data is entered at the shift-left serial input (L).

Clocking of the register is inhibited when both mode control inputs are low.

6.5.1 Input to output relationship. The 16-bit bidirectional universal shift register has the input to output relationship as shown in table II.

6.5.2 Parallel outputs. Paralleling of outputs is not permissible.

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6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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* TABLE II. Input to output relationship. 1/-7/

Function	Inputs							Outputs													
	Clear	Mode		Clock	Serial		Parallel														
	CLR	S1	S0	CP	L	R	P1-P16	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12	Q13	Q14	Q15	Q16
Clear	L	X	X	X	X	X	X-X	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L
	H	X	X	L	X	X	X-X	$Q1_o$	$Q2_o$	$Q3_o$	$Q4_o$	$Q5_o$	$Q6_o$	$Q7_o$	$Q8_o$	$Q9_o$	$Q10_o$	$Q11_o$	$Q12_o$	$Q13_o$	$Q14_o$	$Q15_o$	$Q16_o$
Parallel load	H	H	H	L-H	X	X	p1-p16	p1	p2	p3	p4	p5	p6	p7	p8	p9	p10	p11	p12	p13	p14	p15	p16
Right shift	H	L	H	L-H	X	H	X-X	H	$Q1_n$	$Q2_n$	$Q3_n$	$Q4_n$	$Q5_n$	$Q6_n$	$Q7_n$	$Q8_n$	$Q9_n$	$Q10_n$	$Q11_n$	$Q12_n$	$Q13_n$	$Q14_n$	$Q15_n$
	H	L	H	L-H	X	L	X-X	L	$Q1_n$	$Q2_n$	$Q3_n$	$Q4_n$	$Q5_n$	$Q6_n$	$Q7_n$	$Q8_n$	$Q9_n$	$Q10_n$	$Q11_n$	$Q12_n$	$Q13_n$	$Q14_n$	$Q15_n$
Left shift	H	H	L	L-H	H	X	X-X	$Q2_n$	$Q3_n$	$Q4_n$	$Q5_n$	$Q6_n$	$Q7_n$	$Q8_n$	$Q9_n$	$Q10_n$	$Q11_n$	$Q12_n$	$Q13_n$	$Q14_n$	$Q15_n$	$Q16_n$	H
	H	H	L	L-H	L	X	X-X	$Q2_n$	$Q3_n$	$Q4_n$	$Q5_n$	$Q6_n$	$Q7_n$	$Q8_n$	$Q9_n$	$Q10_n$	$Q11_n$	$Q12_n$	$Q13_n$	$Q14_n$	$Q15_n$	$Q16_n$	L
Inhibit	H	L	L	X	X	X	X-X	$Q1_o$	$Q2_o$	$Q3_o$	$Q4_o$	$Q5_o$	$Q6_o$	$Q7_o$	$Q8_o$	$Q9_o$	$Q10_o$	$Q11_o$	$Q12_o$	$Q13_o$	$Q14_o$	$Q15_o$	$Q16_o$

1/ H = High level (steady state).

2/ L = Low level (steady state).

3/ X = Irrelevant (any input, including transitions).

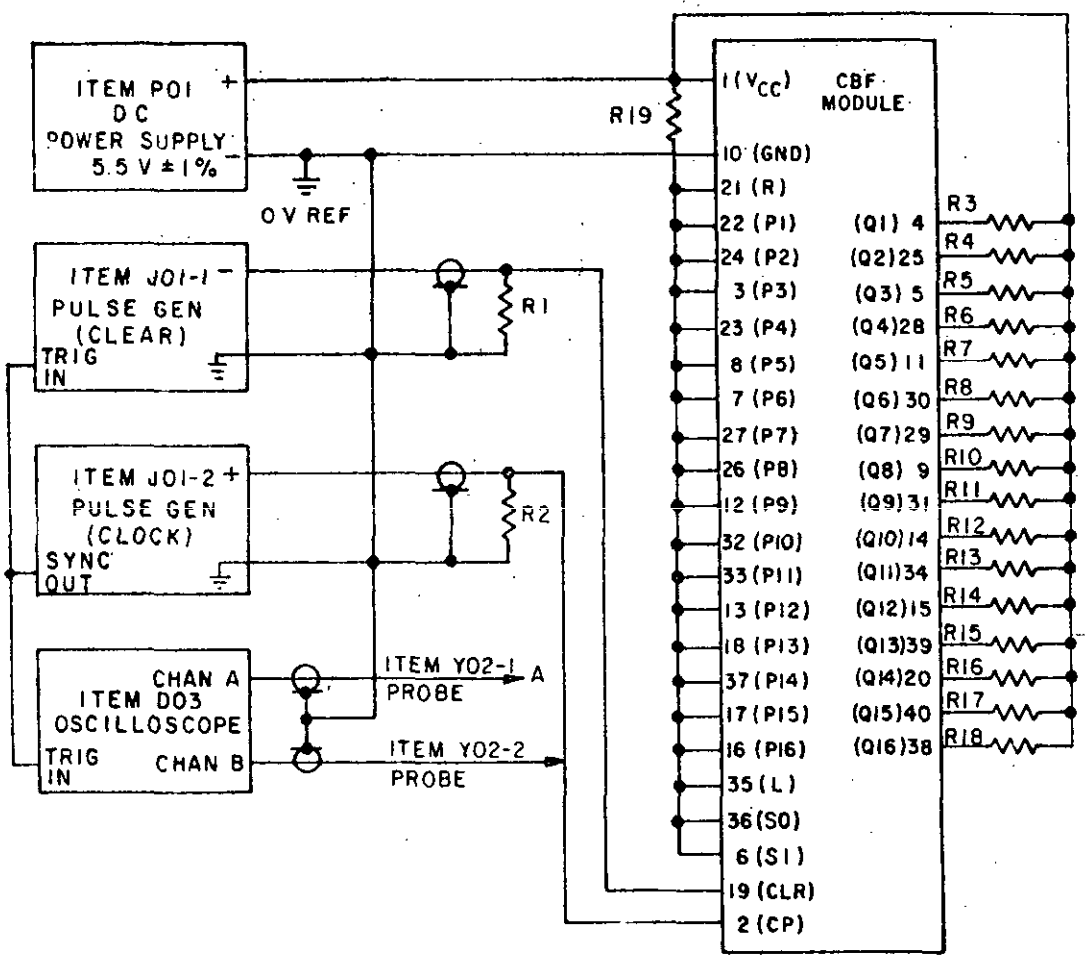
4/ L-H = Transition from low to high level.

5/ p1-p16 = The level of steady-state input at inputs P1-P16, respectively.

6/ $Q1_o$-$Q16_o$ = The level of Q1-Q16, respectively, before the indicated steady-state input conditions were established.

7/ $Q1_n$-$Q16_n$ = The level of Q1-Q16, respectively, before the most recent low to high transition of the clock.

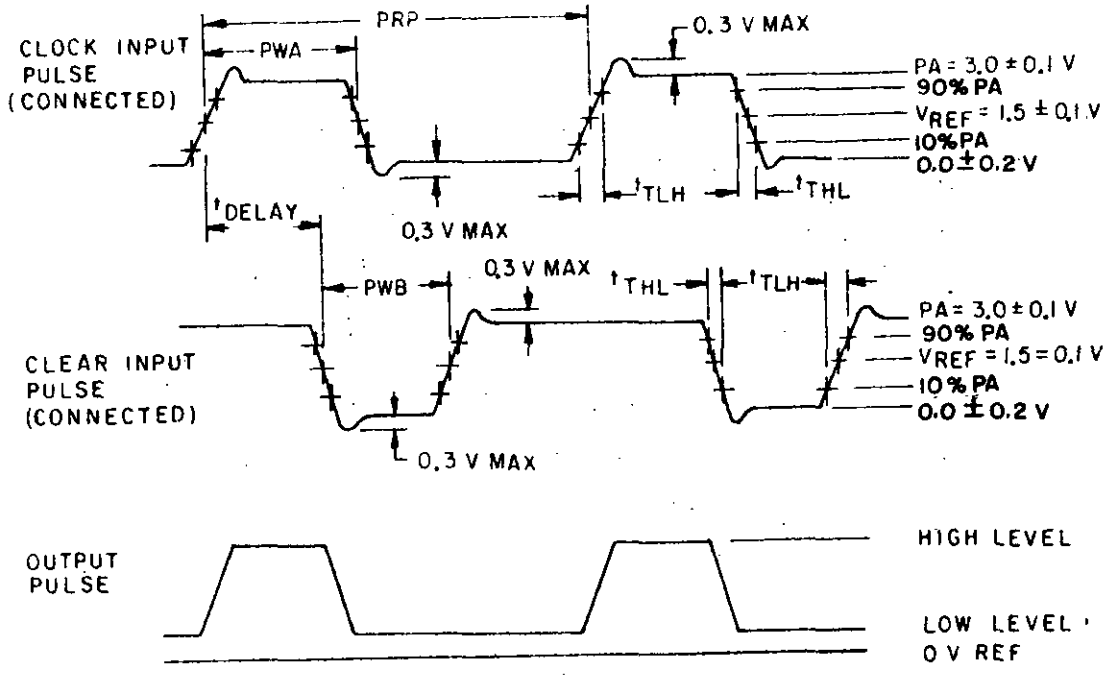
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* NOTES:
1. R1 = R2 = 51.1Ω ±1%, 1/4 W.
2. R3 - R18 = 274Ω ±1%, 1/4 W.
3. R19 = 1 kΩ ±5%, 1/2 W.
4. See figure 3 for waveforms.

FIGURE 2. Operating temperature and life test circuit.

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NOTES:
1. Clock input pulse parameters:

   $t_{TLH} = t_{THL} \leq 10$ ns.
   PWA = 500 ±100 ns.
   PRP = 1,000 ±200 ns.

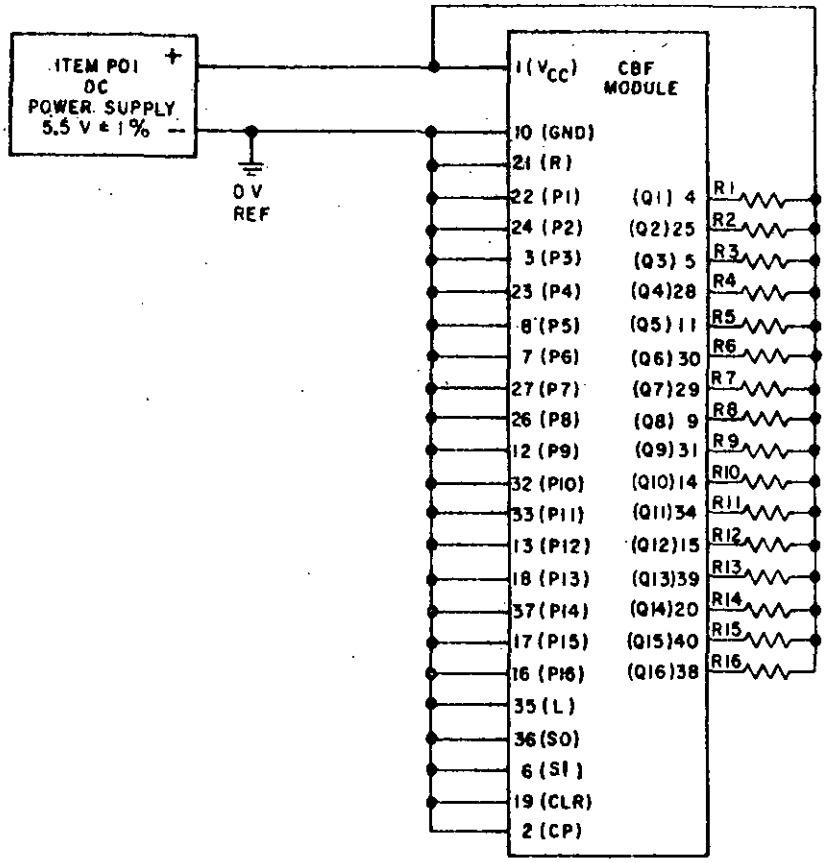
2. Clear input pulse parameters:

   $t_{TLH} = t_{THL} \leq 10$ ns, PWB = 400 ±100 ns.
   Negative output of item J01-1 is used for clear pulse. Adjust the DC offset and pulse amplitude to obtain the required waveform. Adjust the delay control of item J01-2 to obtain t_{DELAY} = 250 ±50 ns.

3. All voltages are referenced to ground.

* FIGURE 3. Operating temperature and life test circuit waveforms for figure 2.

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NOTE: R1 - R16 = 274Ω ±1%, 1/4 W.

FIGURE 4. Component temperature test circuit.

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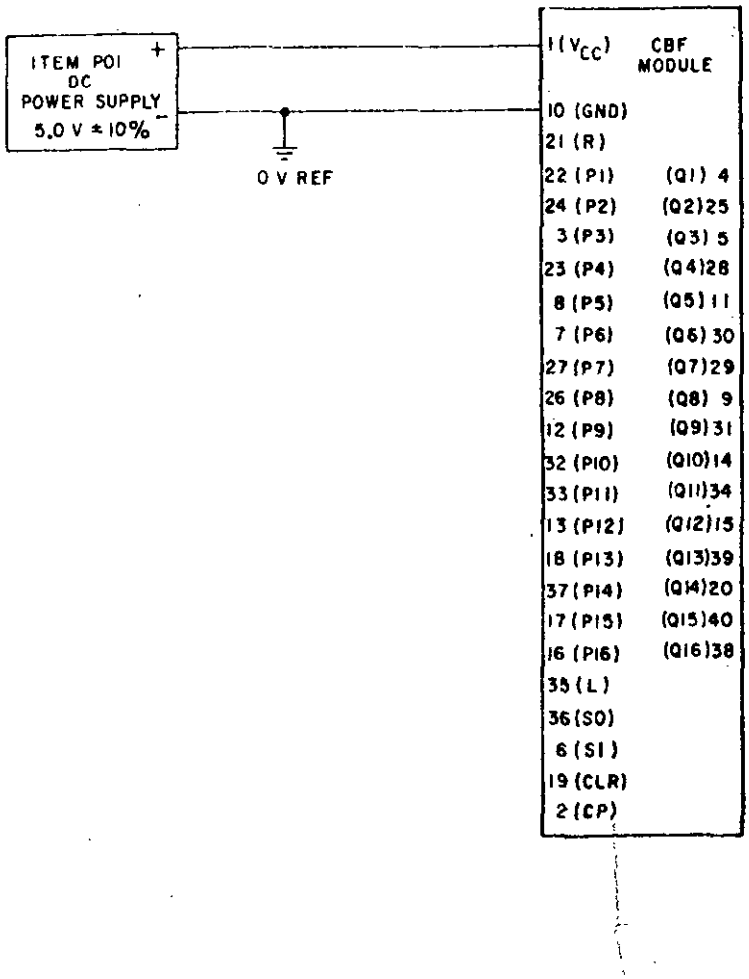


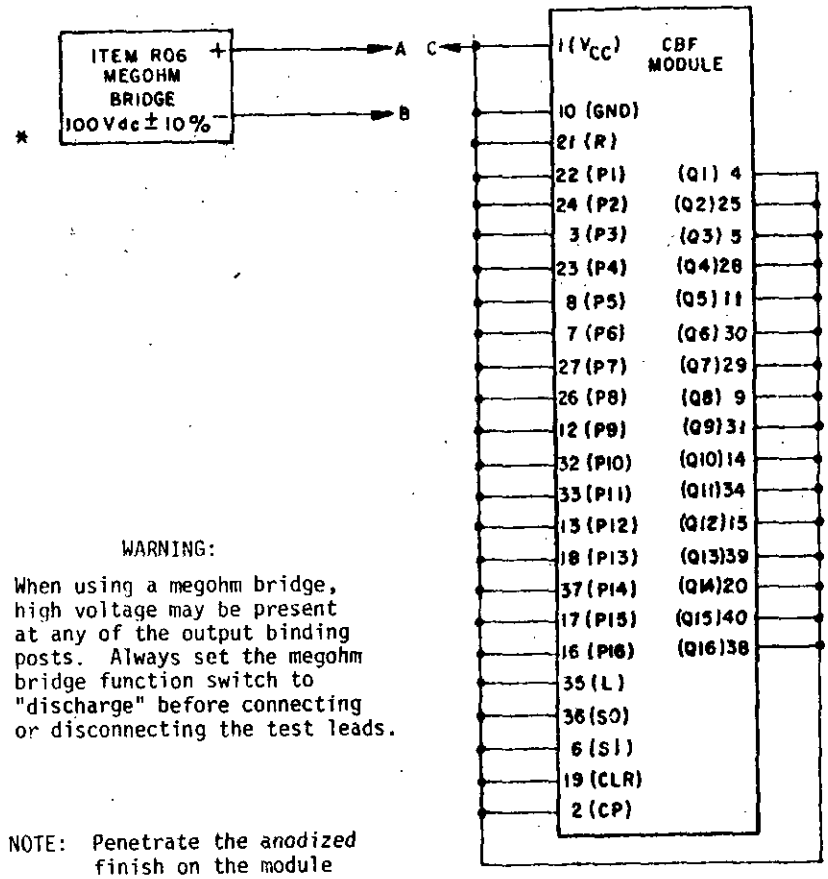
FIGURE 5. Vibration test circuit.

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TABLE III. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A	Isolation	C	Frame	10	∞	MΩ	6

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WARNING:

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

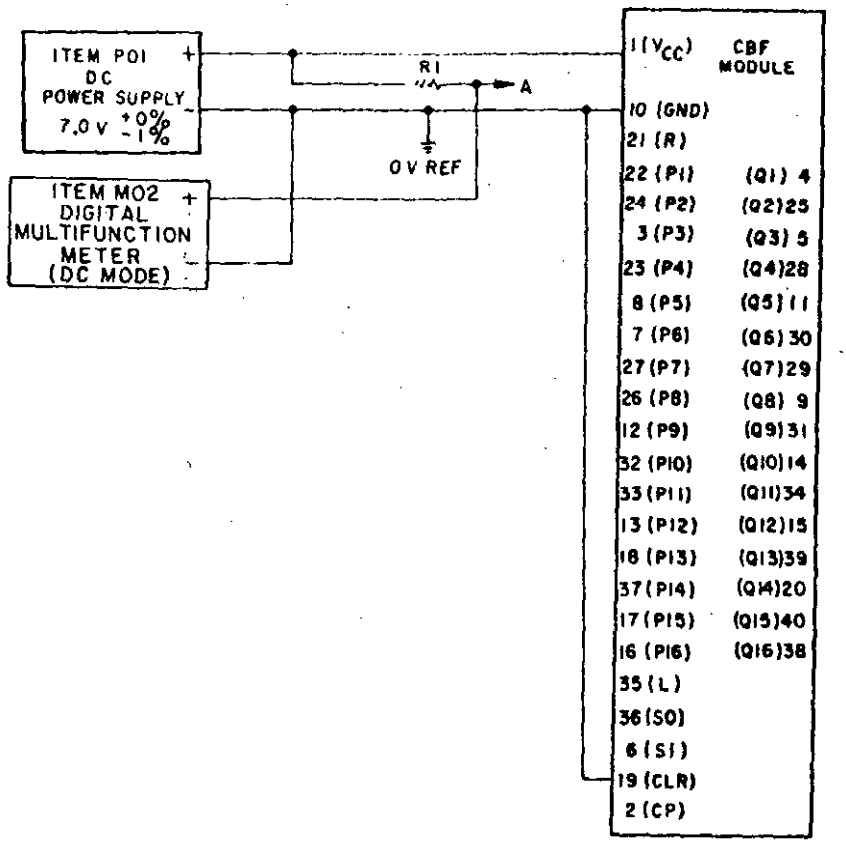
FIGURE 6. Isolation test circuit for table III.

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* TABLE IV. V_{OUT} (low-loaded) test.

Test condition	Parameter		Connect	Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	Min	Max		
B1	Q1	V_{OUT} (low loaded)	4	0.01	1.00	V	7
B2	Q2		25				
B3	Q3		5				
B4	Q4		28				
B5	Q5		11				
B6	Q6		30				
B7	Q7		29				
B8	Q8		9				
B9	Q9		31				
B10	Q10		14				
B11	Q11		34				
B12	Q12		15				
B13	Q13		39				
B14	Q14		20				
B15	Q15		40				
B16	Q16		38				

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NOTE: R1 = 274Ω ±1%, 1/4 W.

* FIGURE 7. V_{OUT} (low-loaded) test circuit for table IV.

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TABLE V. Input breakdown current test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C1	R Input breakdown current	2,3,6,7,8,12,13,16,17,18, 19,22,23,24,26,27,32,33, 35,36,37	21	0	1	mA	8
C2	P1	2,3,6,7,8,12,13,16,17,18, 19,21,23,24,26,27,32,33, 35,36,37	22				
C3	P2	2,3,6,7,8,12,13,16,17,18, 19,21,22,23,26,27,32,33, 35,36,37	24				
C4	P3	2,6,7,8,12,13,16,17,18, 19,21,22,23,24,26,27,32, 33,35,36,37	3				
C5	P4	2,3,6,7,8,12,13,16,17,18, 19,21,22,24,26,27,32,33, 35,36,37	23				
C6	P5	2,3,6,7,12,13,16,17,18, 19,21,22,23,24,26,27,32, 33,35,36,37	8				
C7	P6	2,3,6,8,12,13,16,17,18, 19,21,22,23,24,26,27,32, 33,35,36,37	7				
C8	P7	2,3,6,7,8,12,13,16,17,18, 19,21,22,23,24,26,32,33, 35,36,37	27				
C9	P8	2,3,6,7,8,12,13,16,17,18, 19,21,22,23,24,27,32,33, 35,36,37	26				

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TABLE V. Input breakdown current test - Continued.

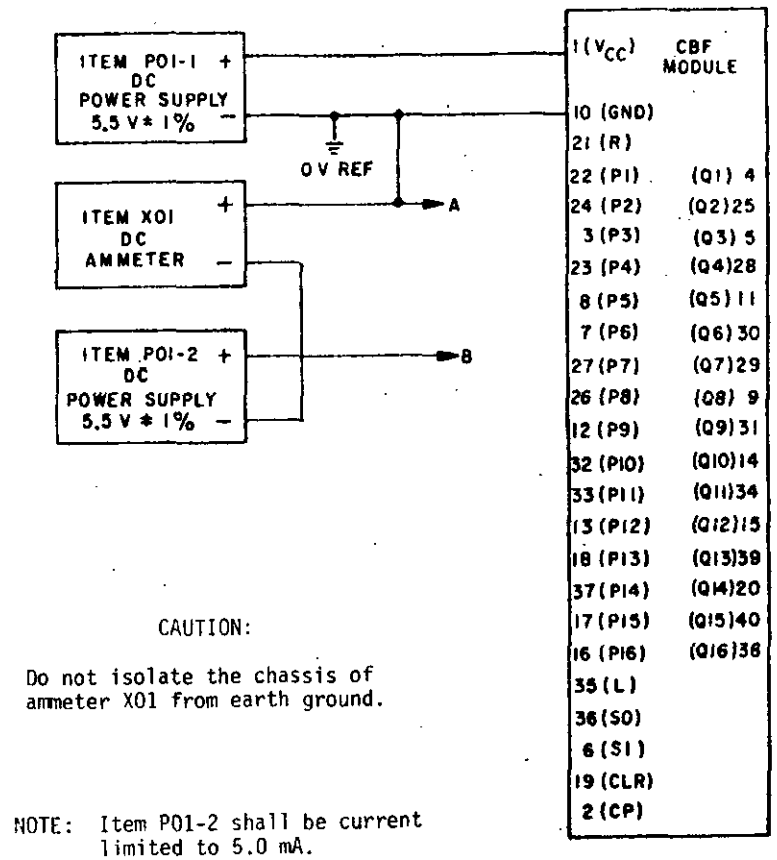
Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C10	P9 Input breakdown current	2,3,6,7,8,13,16,17,18,19,21,22,23,24,26,27,32,33,35,36,37	12	0	1	mA	8
C11	P10	2,3,6,7,8,12,13,16,17,18,19,21,22,23,24,26,27,33,35,36,37	32				
C12	P11	2,3,6,7,8,12,13,16,17,18,19,21,22,23,24,26,27,32,35,36,37	33				
C13	P12	2,3,6,7,8,12,16,17,18,19,21,22,23,24,26,27,32,33,35,36,37	13				
C14	P13	2,3,6,7,8,12,13,16,17,19,21,22,23,24,26,27,32,33,35,36,37	18				
C15	P14	2,3,6,7,8,12,13,16,17,18,19,21,22,23,24,26,27,32,33,35,36	37				
C16	P15	2,3,6,7,8,12,13,16,18,19,21,22,23,24,26,27,32,33,35,36,37	17				
C17	P16	2,3,6,7,8,12,13,17,18,19,21,22,23,24,26,27,32,33,35,36,37	16				
C18	L	2,3,6,7,8,12,13,16,17,18,19,21,22,23,24,26,27,32,33,36,37	35				

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TABLE V. Input breakdown current test - Continued.

Test condition	Parameter		Connect		Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
C19	S0	Input breakdown current	2,3,6,7,8,12,13,16,17,18, 19,21,22,23,24,26,27,32, 33,35,37	36	0	4	mA	8
C20	S1		2,3,7,8,12,13,16,17,18, 19,21,22,23,24,26,27,32, 33,35,36,37	6				
C21	CLR		2,3,6,7,8,12,13,16,17,18, 21,22,23,24,26,27,32,33, 35,36,37	19				
C22	CP		3,6,7,8,12,13,16,17,18, 19,21,22,23,24,26,27,32, 33,35,36,37	2				

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CAUTION:

Do not isolate the chassis of ammeter XO1 from earth ground.

NOTE: Item PO1-2 shall be current limited to 5.0 mA.

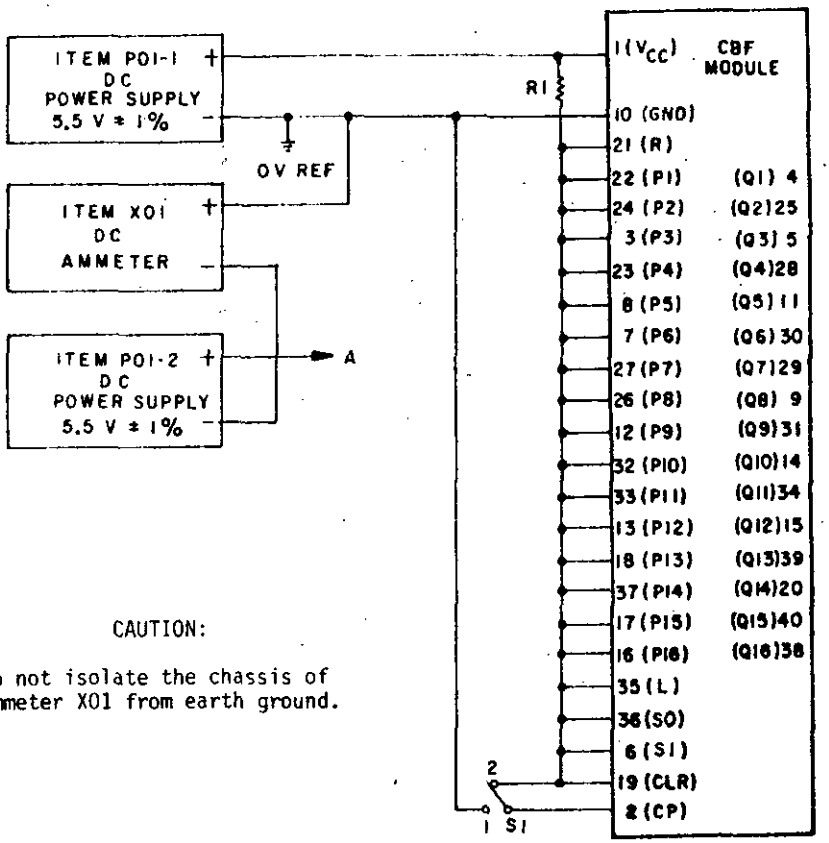
FIGURE 8. Input breakdown current test circuit for table V.

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TABLE VI. Output leakage current test.

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
D1	Q1 Output leakage current	4	0	1	mA	9
D2	Q2	25				
D3	Q3	5				
D4	Q4	28				
D5	Q5	11				
D6	Q6	30				
D7	Q7	29				
D8	Q8	9				
D9	Q9	31				
D10	Q10	14				
D11	Q11	34				
D12	Q12	15				
D13	Q13	39				
D14	Q14	20				
D15	Q15	40				
D16	Q16	38				

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CAUTION:

Do not isolate the chassis of ammeter XO1 from earth ground.

NOTES:

* 1. R1 = 1 kΩ ±5%, 1/2 W.
2. Switch S1 in position 1 for a minimum of 100 ns, then place switch S1 in position 2, connect point A and make measurements.
3. Item PO1-2 shall be current limited to 5.0 mA.

FIGURE 9. Output leakage current test circuit for table VI.

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TABLE VII. Output short circuit current test.

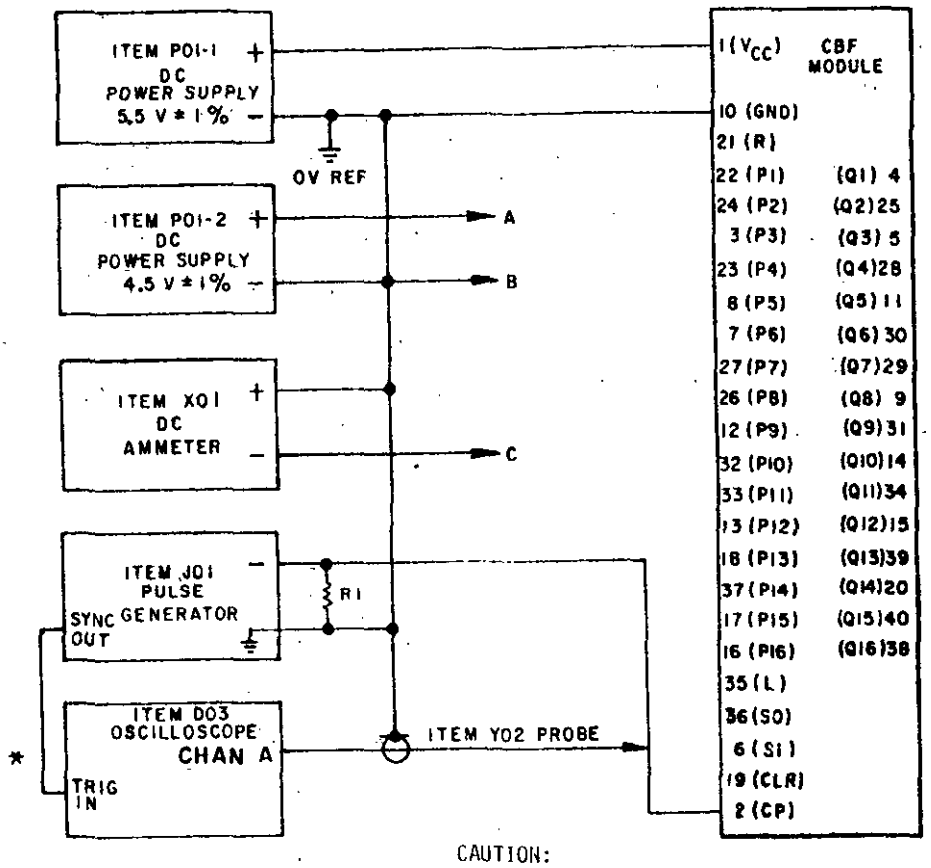
Test condition	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Initial limits 25°±5°C ambient: Min	Initial limits 25°±5°C ambient: Max	Unit	Test figure
E1	Q1 Output short circuit current	22	3,7,8,12,13,16,17,18,23, 24,26,27,32,33,37	4	-40	-100	mA	10,11
E2	Q2	24	3,7,8,12,13,16,17,18,22, 23,26,27,32,33,37	25				
E3	Q3	3	7,8,12,13,16,17,18,22,23, 24,26,27,32,33,37	5				
E4	Q4	23	3,7,8,12,13,16,17,18,22, 24,26,27,32,33,37	28				
E5	Q5	8	3,7,12,13,16,17,18,22,23, 24,26,27,32,33,37	11				
E6	Q6	7	3,8,12,13,16,17,18,22,23, 24,26,27,32,33,37	30				
E7	Q7	27	3,7,8,12,13,16,17,18,22, 23,24,26,32,33,37	29				
E8	Q8	26	3,7,8,12,13,16,17,18,22, 23,24,27,32,33,37	9				
E9	Q9	12	3,7,8,13,16,17,18,22,23, 24,26,27,32,33,37	31				
E10	Q10	32	3,7,8,12,13,16,17,18,22, 23,24,26,27,33,37	14				

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TABLE VII. Output short circuit current test - Continued.

Test condition	Parameter		Connect			Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
E11	Q11	Output short circuit current	33	3,7,8,12,13,16,17,18,22, 23,24,26,27,32,37	34	-40	-100	mA	10,11
E12	Q12	↓	13	3,7,8,12,16,17,18,22,23, 24,26,27,32,33,37	15	↓	↓	↓	↓
E13	Q13	↓	18	3,7,8,12,13,16,17,22,23, 24,26,27,32,33,37	39	↓	↓	↓	↓
E14	Q14	↓	37	3,7,8,12,13,16,17,18,22, 23,24,26,27,32,33	20	↓	↓	↓	↓
E15	Q15	↓	17	3,7,8,12,13,16,18,22,23, 24,26,27,32,33,37	40	↓	↓	↓	↓
E16	Q16	↓	16	3,7,8,12,13,17,18,22,23, 24,26,27,32,33,37	38	↓	↓	↓	↓

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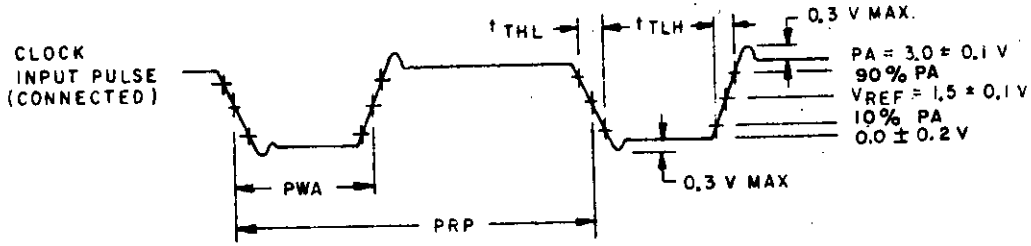
CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

NOTES:
1. R1 = 51.1Ω ±1%, 1/4 W.
2. After making all input circuit connections for a test condition, use the single pulse mode on item J01 (after initial setup) and apply a single pulse before making each measurement.
3. When performing output short circuit current test, only one output shall be shorted at a time, and the duration of the test shall not exceed 1 second.

FIGURE 10. Output short circuit current test circuit for table VII.

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NOTES:
1. Input pulse parameters:

   $t_{TLH} = t_{THL} = 50 \pm 25$ ns.
   PWA = 500 ±100 ns.
   PRP = 1,000 ±200 ns.

2. Use the negative output of item J01 and adjust the offset so that the output remains high after the pulse.

* FIGURE 11. Output short circuit current test circuit waveform for table VII.

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TABLE VIII. Functional test.

Test condition	Parameter	Connect				Output logic states																Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12	Q13	Q14	Q15	Q16	Min	Max		
F1	Functional	3,6,7,8,12,13,16, 17,18,21,22,23,24, 26,27,32,33,35,36, 37	2,19	2/ 3/ 4/	Open	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L		1/		12, 13
F2		3,6,7,8,12,13,16, 17,18,19,21,22,23, 24,26,27,32,33,35, 36,37	2		Open	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L				
F3		3,6,7,8,12,13,16, 17,18,19,21,23,24, 26,27,32,33,36,37	22,35		2	L	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H				
F4		3,6,7,8,12,13,16, 17,18,19,22,23,26, 27,32,33,36,37	21,24, 35			H	L	H	H	H	H	H	H	H	H	H	H	H	H	H	H				
F5		6,7,8,12,13,16,17, 18,19,22,23,24,26, 27,32,33,36,37	3,21, 35			H	H	L	H	H	H	H	H	H	H	H	H	H	H	H	H				
F6		3,6,7,8,12,13,16, 17,18,19,22,24,26, 27,32,33,36,37	21,23, 35			H	H	H	L	H	H	H	H	H	H	H	H	H	H	H	H				
F7		3,6,7,12,13,16,17, 18,19,22,23,24,26, 27,32,33,36,37	8,21, 35			H	H	H	H	L	H	H	H	H	H	H	H	H	H	H	H				
F8		3,6,8,12,13,16,17, 18,19,22,23,24,26, 27,32,33,36,37	7,21, 35			H	H	H	H	H	L	H	H	H	H	H	H	H	H	H	H				
F9		3,6,7,8,12,13,16, 17,18,19,22,23,24, 26,32,33,36,37	21,27, 35			H	H	H	H	H	H	L	H	H	H	H	H	H	H	H	H				
F10		3,6,7,8,12,13,16, 17,18,19,22,23,24, 27,32,33,36,37	21,26, 35			H	H	H	H	H	H	H	L	H	H	H	H	H	H	H	H				
F11		3,6,7,8,13,16,17, 18,19,22,23,24,26, 27,32,33,36,37	12,21, 35			H	H	H	H	H	H	H	H	L	H	H	H	H	H	H	H				
F12		3,6,7,8,12,13,16, 17,18,19,22,23,24, 26,27,33,36,37	21,32, 35			H	H	H	H	H	H	H	H	H	L	H	H	H	H	H	H				
F13		3,6,7,8,12,13,16, 17,18,19,22,23,24, 26,27,32,36,37	21,33, 35			H	H	H	H	H	H	H	H	H	H	L	H	H	H	H	H				
F14		3,6,7,8,12,16,17, 18,19,22,23,24,26, 27,32,33,36,37	13,21, 35			H	H	H	H	H	H	H	H	H	H	H	L	H	H	H	H				
F15		3,6,7,8,12,13,16, 17,19,22,23,24,26, 27,32,33,36,37	18,21, 35			H	H	H	H	H	H	H	H	H	H	H	H	L	H	H	H				
F16		3,6,7,8,12,13,16, 17,18,19,22,23,24, 26,27,32,33,36	21,35, 37			H	H	H	H	H	H	H	H	H	H	H	H	H	L	H	H				

See footnotes at end of table.

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TABLE VIII. Functional test - Continued.

Test condition	Parameter	Connect				Output logic states														Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12	Q13	Q14	Q15	Q16	Min	Max		
F17	Functional	3,6,7,8,12,13,16,18,19,22,23,24,26,27,32,33,36,37	17,21,35	2/ 3/ 4/	2	H	H	H	H	H	H	H	H	H	H	H	H	H	H	L	H	1/			12, 13
F18		3,6,7,8,12,13,17,18,19,22,23,24,26,27,32,33,36,37	16,21,35			H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	L				
F19		6,16,19,21,36	3,7,8,12,13,17,18,22,23,24,26,27,32,33,35,37			L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	H				
F20		6,17,19,21,35,36	3,7,8,12,13,16,18,22,23,24,26,27,32,33,37			L	L	L	L	L	L	L	L	L	L	L	L	L	L	H	L				
F21		6,19,21,35,36,37	3,7,8,12,13,16,17,18,22,23,24,26,27,32,33			L	L	L	L	L	L	L	L	L	L	L	L	L	H	L	L				
F22		6,18,19,21,35,36	3,7,8,12,13,16,17,22,23,24,26,27,32,33,37			L	L	L	L	L	L	L	L	L	L	L	L	H	L	L	L				
F23		6,13,19,21,35,36	3,7,8,12,16,17,18,22,23,24,26,27,32,33,37			L	L	L	L	L	L	L	L	L	L	L	H	L	L	L	L				
F24		6,19,21,33,35,36	3,7,8,12,13,16,17,18,22,23,24,26,27,32,37			L	L	L	L	L	L	L	L	L	L	H	L	L	L	L	L				
F25		6,19,21,32,35,36	3,7,8,12,13,16,17,18,22,23,24,26,27,33,37			L	L	L	L	L	L	L	L	L	H	L	L	L	L	L	L				
F26		6,12,19,21,35,36	3,7,8,13,16,17,18,22,23,24,26,27,32,33,37			L	L	L	L	L	L	L	L	H	L	L	L	L	L	L	L				
F27		6,19,21,26,35,36	3,7,8,12,13,16,17,18,22,23,24,27,32,33,37			L	L	L	L	L	L	L	H	L	L	L	L	L	L	L	L				
F28		6,19,21,27,35,36	3,7,8,12,13,16,17,18,22,23,24,26,32,33,37			L	L	L	L	L	L	H	L	L	L	L	L	L	L	L	L				
F29		6,7,19,21,35,36	3,8,12,13,16,17,18,22,23,24,26,27,32,33,37			L	L	L	L	L	H	L	L	L	L	L	L	L	L	L	L				

See footnotes at end of table.

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TABLE VIII. Functional test - Continued.

Test condition	Parameter	Connect				Output logic states																Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12	Q13	Q14	Q15	Q16	Min	Max		
F30	Functional	6,8,19,21, 35,36	3,7,12,13, 16,17,18,22, 23,24,26,27, 32,33,37	2/ 3/ 4/	2	L	L	L	L	H	L	L	L	L	L	L	L	L	L	L	L	1/			12, 13
F31		6,19,21,23, 35,36	3,7,8,12,13, 16,17,18,22, 24,26,27,32, 33,37			L	L	L	H	L	L	L	L	L	L	L	L	L	L	L	L				
F32		3,6,19,21, 35,36	7,8,12,13, 16,17,18,22, 23,24,26,27, 32,33,37			L	L	H	L	L	L	L	L	L	L	L	L	L	L	L	L				
F33		6,19,21,24, 35,36	3,7,8,12,13, 16,17,18,22, 23,26,27,32, 33,37			L	H	L	L	L	L	L	L	L	L	L	L	L	L	L	L				
F34		6,19,21,22, 35,36	3,7,8,12,13, 16,17,18,23, 24,26,27,32, 33,37			H	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L				
F35		6,19,21,35, 36	3,7,8,12,13, 16,17,18,22, 23,24,26,27, 32,33,37			L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L				
F36		3,7,8,12,13, 16,17,18,19, 21,22,23,24, 26,27,32,33, 35,37	6,36			L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L				
F37		3,6,7,8,12, 13,16,17,18, 19,22,23,24, 26,27,32,33, 36,37	21,35			H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H				
F38		19	3,6,7,8,12, 13,16,17,18, 21,22,23,24, 26,27,32,33, 35,36,37			H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H				
F39		19,22,36	3,6,7,8,12, 13,16,17,18, 21,23,24,26, 27,32,33,35, 37			L	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H				
F40		19,21,24,36	3,6,7,8,12, 13,16,17,18, 22,23,26,27, 32,33,35,37			H	L	H	H	H	H	H	H	H	H	H	H	H	H	H	H				

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TABLE VIII. Functional test - Continued.

Test condition	Parameter	Connect				Output logic states															Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12	Q13	Q14	Q15	Q16	Min	Max		
F41	Functional	3,19,21,36	6,7,8,12,13, 16,17,18,22, 23,24,26,27, 32,33,35,37	2/ 3/ 4/	2	H	H	L	H	H	H	H	H	H	H	H	H	H	H	H	H	1/			12, 13
F42		19,21,23,36	3,6,7,8,12, 13,16,17,18, 22,24,26,27, 32,33,35,37			H	H	H	L	H	H	H	H	H	H	H	H	H	H	H	H				
F43		8,19,21,36	3,6,7,12,13, 16,17,18,22, 23,24,26,27, 32,33,35,37			H	H	H	H	L	H	H	H	H	H	H	H	H	H	H	H				
F44		7,19,21,36	3,6,8,12,13, 16,17,18,22, 23,24,26,27, 32,33,35,37			H	H	H	H	H	L	H	H	H	H	H	H	H	H	H	H				
F45		19,21,27,36	3,6,7,8,12, 13,16,17,18, 22,23,24,26, 32,33,35,37			H	H	H	H	H	H	L	H	H	H	H	H	H	H	H	H				
F46		19,21,26,36	3,6,7,8,12, 13,16,17,18, 22,23,24,27, 32,33,35,37			H	H	H	H	H	H	H	L	H	H	H	H	H	H	H	H				
F47		12,19,21,36	3,6,7,8,13, 16,17,18,22, 23,24,26,27, 32,33,35,37			H	H	H	H	H	H	H	H	L	H	H	H	H	H	H	H				
F48		19,21,32,36	3,6,7,8,12, 13,16,17,18, 22,23,24,26, 27,33,35,37			H	H	H	H	H	H	H	H	H	L	H	H	H	H	H	H				
F49		19,21,33,36	3,6,7,8,12, 13,16,17,18, 22,23,24,26, 27,32,35,37			H	H	H	H	H	H	H	H	H	H	L	H	H	H	H	H				
F50		13,19,21,36	3,6,7,8,12, 16,17,18,22, 23,24,26,27, 32,33,35,37			H	H	H	H	H	H	H	H	H	H	H	L	H	H	H	H				
F51		18,19,21,36	3,6,7,8,12, 13,16,17,22, 23,24,26,27, 32,33,35,37			H	H	H	H	H	H	H	H	H	H	H	H	L	H	H	H				
F52		19,21,36,37	3,6,7,8,12, 13,16,17,18, 22,23,24,26, 27,32,33,35			H	H	H	H	H	H	H	H	H	H	H	H	H	L	H	H				
F53		17,19,21,36	3,6,7,8,12, 13,16,18,22, 23,24,26,27, 32,33,35,37			H	H	H	H	H	H	H	H	H	H	H	H	H	H	L	H				

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TABLE VIII. Functional test - Continued.

Test condition	Parameter	Connect				Output logic states															Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12	Q13	Q14	Q15	Q16	Min	Max		
F54	Functional	16,19,21,36	3,6,7,8,12, 13,17,18, 22,23,24, 26,27,32, 33,35,37	2/ 3/ 4/	2	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	L		1/		12, 13
F55		19,21,36	3,6,7,8,12, 13,16,17, 18,22,23, 24,26,27, 32,33,35, 37			H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H				
F56		6,19,21,35,36	3,7,8,12, 13,16,17, 18,22,23, 24,26,27, 32,33,37			L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L				
F57		3,6,7,8,12, 13,17,18,19, 21,22,23,24, 26,27,32,33, 35,37	16,36			L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	H				
F58		3,6,7,8,12, 13,16,18,19, 21,22,23,24, 26,27,32,33, 37	17,35,36			L	L	L	L	L	L	L	L	L	L	L	L	L	L	H	L				
F59		3,6,7,8,12, 13,16,17,18, 19,21,22,23, 24,26,27,32, 33	35,36,37			L	L	L	L	L	L	L	L	L	L	L	L	L	H	L	L				
F60		3,6,7,8,12, 13,16,17,19, 21,22,23,24, 26,27,32,33, 37	13,35,36			L	L	L	L	L	L	L	L	L	L	L	L	H	L	L	L				
F61		3,6,7,8,12, 16,17,18,19, 21,22,23,24, 26,27,32,33, 37	13,35,36			L	L	L	L	L	L	L	L	L	L	L	H	L	L	L	L				
F62		3,6,7,8,12, 13,16,17,18, 19,21,22,23, 24,26,27,32, 37	33,35,36			L	L	L	L	L	L	L	L	L	L	H	L	L	L	L	L				

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TABLE VIII. Functional test - Continued.

Test condition	Parameter	Connect				Output logic states														Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12	Q13	Q14	Q15	Q16	Min	Max		
F63	Functional	3,6,7,8,12, 13,16,17,18, 19,21,22,23, 24,26,27,33, 37	32,35,36	2/ 3/ 4/	2	L	L	L	L	L	L	L	L	L	H	L	L	L	L	L	L	1/			12, 13
F64		3,6,7,8,13, 16,17,18,19, 21,22,23,24, 26,27,32,33, 37	12,35,36			L	L	L	L	L	L	L	L	H	L	L	L	L	L	L	L				
F65		3,6,7,8,12, 13,16,17,18, 19,21,22,23, 24,27,32,33, 37	26,35,36			L	L	L	L	L	L	L	H	L	L	L	L	L	L	L	L				
F66		3,6,7,8,12, 13,16,17,18, 19,21,22,23, 24,26,32,33, 37	27,35,36			L	L	L	L	L	L	H	L	L	L	L	L	L	L	L	L				
F67		3,6,8,12,13, 16,17,18,19, 21,22,23,24, 26,27,32,33, 37	7,35,36			L	L	L	L	L	H	L	L	L	L	L	L	L	L	L	L				
F68		3,6,7,12,13, 16,17,18,19, 21,22,23,24, 26,27,32,33, 37	8,35,36			L	L	L	L	H	L	L	L	L	L	L	L	L	L	L	L				
F69		3,6,7,8,12, 13,16,17,18, 19,21,22,24, 26,27,32,33, 37	23,35,36			L	L	L	H	L	L	L	L	L	L	L	L	L	L	L	L				
F70		6,7,8,12,13, 16,17,18,19, 21,22,23,24, 26,27,32,33, 37	3,35,36			L	L	H	L	L	L	L	L	L	L	L	L	L	L	L	L				
F71		3,6,7,8,12, 13,16,17,18, 19,21,22,23, 26,27,32,33, 37	24,35,36			L	H	L	L	L	L	L	L	L	L	L	L	L	L	L	L				
F72		3,6,7,8,12, 13,16,17,18, 19,23,24,26, 27,32,33,37	21,22,35,36			H	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L				
F73		3,6,7,8,12, 13,16,17,18, 19,21,22,23, 24,26,27,32, 33,37	35,36			L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L				

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TABLE VIII. Functional test - Continued.

Test condition	Parameter	Connect				Output logic states																Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12	Q13	Q14	Q15	Q16	Min	Max		
F74	Functional	3,6,8,12,17, 18,19,22,27, 33,35,36	7,13,16,21, 23,24,26, 32,37	2/ 3/ 4/	2	H	L	H	L	H	L	H	L	H	L	H	L	H	L	H	L		1/		12, 13
F75		3,8,12,17,18, 19,22,27,33, 36	6,7,13,16, 21,23,24, 26,32,35,37			L	H	L	H	L	H	L	H	L	H	L	H	L	H	L	H				
F76		3,8,12,17,18, 19,22,27,33, 35	6,7,13,16, 21,23,24, 26,32,36,37			L	H	L	H	L	H	L	H	L	H	L	H	L	H	L	H				
F77		6,7,13,16,19, 23,24,26,32, 37	3,8,12,17, 18,21,22, 27,33,35,36			H	L	H	L	H	L	H	L	H	L	H	L	H	L	H	L				
F78		7,13,16,19, 21,23,24,26, 32,37	3,6,8,12, 17,18,22, 27,33,35,36			H	L	H	L	H	L	H	L	H	L	H	L	H	L	H	L				
F79		6,7,13,16,19, 21,23,24,26, 32,36,37	3,8,12,17, 18,22,27, 33,35			L	H	L	H	L	H	L	H	L	H	L	H	L	H	L	H				
F80		3,6,8,12,17, 18,22,27,33, 35,36	7,13,16,19, 21,23,24, 26,32,37			L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L				
F81		0r	2,3,6,7,8, 12,13,16, 17,18,19, 21,22,23, 24,26,27, 32,33,35, 36,37		Open	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L				
F82		19	3,6,7,8,12, 13,16,17, 18,21,22, 23,24,26, 27,32,33, 35,36,37			L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L				
F83		Open	2,3,6,7,8, 12,13,16, 17,18,19, 21,22,23, 24,26,27, 32,33,35, 36,37			L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L				
F84		6,19,22,36	3,7,8,12, 13,16,17, 18,21,23, 24,26,27, 32,33,35, 37		2	H	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L				
F85		6,19	3,7,8,12, 13,16,17, 18,21,22, 23,24,26, 27,32,33, 35,36,37		2	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L				

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TABLE VIII. Functional test - Continued.

Test condition	Parameter	Connect				Output logic states															Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12	Q13	Q14	Q15	Q16	Min	Max		
FB6	Functional	Open	2,3,6,7,8,12,13,16,17,18,19,21,22,23,24,26,27,32,33,35,36,37	2/ 3/ 4/	Open	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	1/			12, 13
F87		6,19,35	3,7,8,12,13,16,17,18,21,22,23,24,26,27,32,33,36,37		2	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	H				
F88		19	3,6,7,8,12,13,16,17,18,21,22,23,24,26,27,32,33,35,36,37		2	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	H				
F89		3,6,7,8,12,13,16,17,18,19,22,23,24,26,27,32,33,36,37	21,35		2	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H				
F90		3,6,7,8,12,13,16,17,18,21,22,23,24,26,27,32,33,35,36,37	19		2	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L				
F91		3,6,7,8,12,13,16,17,18,21,22,23,24,26,27,32,33,35,36,37	2,19		Open	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L				
F92		3,6,7,8,12,13,16,17,18,19,21,22,23,24,26,27,32,33,35,36,37	2		Open	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L				
F93		3,6,7,8,12,13,16,17,18,19,21,23,24,26,27,32,33,36,37	22,35		2	L	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H				
F94		3,6,7,8,12,13,16,17,18,19,22,23,26,27,32,33,36,37	21,24,35		2	H	L	H	H	H	H	H	H	H	H	H	H	H	H	H	H				
F95		6,7,8,12,13,16,17,18,19,22,23,24,26,27,32,33,36,37	3,21,35		2	H	H	L	H	H	H	H	H	H	H	H	H	H	H	H	H				
F96		3,6,7,8,12,13,16,17,18,19,22,24,26,27,32,33,36,37	21,23,35		2	H	H	H	L	H	H	H	H	H	H	H	H	H	H	H	H				

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TABLE VIII. Functional test - Continued.

Test condition	Parameter	Connect				Output logic states																Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12	Q13	Q14	Q15	Q16	Min	Max		
F97	Functional	3,6,7,12,13, 16,17,18,19, 22,23,24,26, 27,32,33,36,37	8,21,35	2/ 3/ 4/ 7	2	H	H	H	H	L	H	H	H	H	H	H	H	H	H	H	H	1/			12,13
F98		3,6,8,12,13, 16,17,18,19, 22,23,24,26, 27,32,33,36,37	7,21,35			H	H	H	H	H	L	H	H	H	H	H	H	H	H	H	H				
F99		3,6,7,8,12,13, 16,17,18,19, 22,23,24,26, 32,33,36,37	21,27,35			H	H	H	H	H	H	L	H	H	H	H	H	H	H	H	H				
F100		3,6,7,8,12,13, 16,17,18,19, 22,23,24,27, 32,33,36,37	21,26,35			H	H	H	H	H	H	H	L	H	H	H	H	H	H	H	H				
F101		3,6,7,8,13, 16,17,18,19, 22,23,24,26, 27,32,33,36,37	12,21,35			H	H	H	H	H	H	H	H	L	H	H	H	H	H	H	H				
F102		3,6,7,8,12,13, 16,17,18,19, 22,23,24,26, 27,32,36,37	21,32,35			H	H	H	H	H	H	H	H	H	L	H	H	H	H	H	H				
F103		3,6,7,8,12,13, 16,17,18,19, 22,23,24,26, 27,32,36,37	21,33,35			H	H	H	H	H	H	H	H	H	H	L	H	H	H	H	H				
F104		3,6,7,8,12, 16,17,18,19, 22,23,24,26, 27,32,33,36,37	13,21,35			H	H	H	H	H	H	H	H	H	H	H	L	H	H	H	H				
F105		3,6,7,8,12, 13,16,17,19, 22,23,24,26, 27,32,33,36,37	18,21,35			H	H	H	H	H	H	H	H	H	H	H	H	L	H	H	H				
F106		3,6,7,8,12, 13,16,17,18, 19,22,23,24, 26,27,32,33,36	21,35,37			H	H	H	H	H	H	H	H	H	H	H	H	H	L	H	H				
F107		3,6,7,8,12, 13,16,18,19, 22,23,24,26, 27,32,33,36,37	17,21,35			H	H	H	H	H	H	H	H	H	H	H	H	H	H	L	H				
F108		3,6,7,8,12, 13,17,18,19, 22,23,24,26, 27,32,33,36,37	16,21,35			H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	L				
F109		6,16,19,21,36	3,7,8,12, 13,17,18, 22,23,24, 26,27,32, 33,35,37			L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	H				

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TABLE VIII. Functional test - Continued.

Test condition	Parameter	Connect				Output logic states														Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12	Q13	Q14	Q15	Q16	Min	Max		
F110	Functional	6,17,19,21, 35,36	3,7,8,12, 13,16,18, 22,23,24, 26,27,32, 33,37	2/ 3/ 4/	2	L	L	L	L	L	L	L	L	L	L	L	L	L	L	H	L	1/			12,13
F111		6,19,21,35, 36,37	3,7,8,12, 13,16,17, 18,22,23, 24,26,27, 32,33			L	L	L	L	L	L	L	L	L	L	L	L	L	H	L	L				
F112		6,18,19,21, 35,36	3,7,8,12, 13,16,17, 22,23,24, 26,27,32, 33,37			L	L	L	L	L	L	L	L	L	L	L	L	H	L	L	L				
F113		6,13,19,21, 35,36	3,7,8,12, 16,17,18, 22,23,24, 26,27,32, 33,37			L	L	L	L	L	L	L	L	L	L	L	H	L	L	L	L				
F114		6,19,21,33, 35,36	3,7,8,12, 13,16,17, 18,22,23, 24,26,27, 32,37			L	L	L	L	L	L	L	L	L	L	H	L	L	L	L	L				
F115		6,19,21,32, 35,36	3,7,8,12, 13,16,17, 18,22,23, 24,26,27, 33,37			L	L	L	L	L	L	L	L	L	H	L	L	L	L	L	L				
F116		6,12,19,21, 35,36	3,7,8, 13,16,17, 18,22,23, 24,26,27, 32,33,37			L	L	L	L	L	L	L	L	H	L	L	L	L	L	L	L				
F117		6,19,21,26, 35,36	3,7,8,12, 13,16,17, 18,22,23, 24,27,32, 33,37			L	L	L	L	L	L	L	H	L	L	L	L	L	L	L	L				
F118		6,19,21,27, 35,36	3,7,8,12, 13,16,17, 18,22,23, 24,26,32, 33,37			L	L	L	L	L	L	H	L	L	L	L	L	L	L	L	L				
F119		6,7,19,21, 35,36	3,8,12, 13,16,17, 18,22,23, 24,26,27, 32,33,37			L	L	L	L	L	H	L	L	L	L	L	L	L	L	L	L				
F120		6,8,19,21, 35,36	3,7,12, 13,16,17, 18,22,23, 24,26,27, 32,33,37			L	L	L	L	H	L	L	L	L	L	L	L	L	L	L	L				
F121		6,19,21,23, 35,36	3,7,8,12, 13,16,17, 18,22, 24,26,27, 32,33,37			L	L	L	H	L	L	L	L	L	L	L	L	L	L	L	L				

See footnotes at end of table.

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TABLE VIII. Functional test - Continued.

Test condition	Parameter	Connect				Output logic states															Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12	Q13	Q14	Q15	Q16	Min	Max		
F122	Functional	3,6,19,21, 35,36	7,8,12,13, 16,17,18, 22,23,24, 26,27,32, 33,37	2/ 3/ 4/	2	L	L	H	L	L	L	L	L	L	L	L	L	L	L	L	L	1/			12,13
F123		6,19,21,24, 35,36	3,7,8,12, 13,16,17, 18,22,23, 26,27,32, 33,37			L	H	L	L	L	L	L	L	L	L	L	L	L	L	L	L				
F124		6,19,21,22, 35,36	3,7,8,12, 13,16,17, 18,23,24, 26,27,32, 33,37			H	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L				
F125		6,19,21,35, 36	3,7,8,12, 13,16,17, 18,22,23, 24,26,27, 32,33,37			L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L				
F126		3,7,8,12, 13,16,17,18, 19,21,22,23, 24,26,27, 32,33,35,37	6,36			L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L				
F127		3,6,7,8,12, 13,16,17,18, 19,22,23, 24,26,27, 32,33,35,37	21,35			H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H				
F128		19	3,6,7,8, 12,13,16, 17,18,21, 22,23,24, 26,27,32, 33,35,36, 37			H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H				
F129		19,22,36	3,6,7,8, 12,13,16, 17,18,21, 23,24,26, 27,32,33, 35,37			L	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H				
F130		19,21,24,36	3,6,7,8, 12,13,16, 17,18,22, 23,26,27, 32,33,35, 37			H	L	H	H	H	H	H	H	H	H	H	H	H	H	H	H				
F131		3,19,21,36	6,7,8,12, 13,16,17, 18,22,23, 24,26,27, 32,33,35, 37			H	H	L	H	H	H	H	H	H	H	H	H	H	H	H	H				
F132		19,21,23,36	3,6,7,8, 12,13,16, 17,18,22, 24,26,27, 32,33,35, 37			H	H	H	L	H	H	H	H	H	H	H	H	H	H	H	H				

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TABLE VIII. Functional test - Continued.

Test condition	Parameter	Connect				Output logic states														Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12	Q13	Q14	Q15	Q16	Min	Max		
F133	Functional	8,19,21,36	3,6,7, 12,13,16, 17,18,22, 23,24,26, 27,32,33, 35,37	2/ 3/ 4/	2	H	H	H	H	L	H	H	H	H	H	H	H	H	H	H	H	1/			12,13
F134		7,19,21,36	3,6,8, 12,13,16, 17,18,22, 23,24,26, 27,32,33, 35,37			H	H	H	H	H	L	H	H	H	H	H	H	H	H	H	H				
F135		19,21,27,36	3,6,7,8, 12,13,16, 17,18,22, 23,24,26, 32,33,35, 37			H	H	H	H	H	H	L	H	H	H	H	H	H	H	H	H				
F136		19,21,26,36	3,6,7,8, 12,13,16, 17,18,22, 23,24,27, 32,33,35, 37			H	H	H	H	H	H	H	L	H	H	H	H	H	H	H	H				
F137		12,19,21,36	3,6,7,8, 13,16,17, 18,22,23, 24,26,27, 32,33,35, 37			H	H	H	H	H	H	H	H	L	H	H	H	H	H	H	H				
F138		19,21,32,36	3,6,7,8, 12,13,16, 17,18,22, 23,24,26, 27,33,35, 37			H	H	H	H	H	H	H	H	H	L	H	H	H	H	H	H				
F139		19,21,33,36	3,6,7,8, 12,13,16, 17,18,22, 23,24,26, 27,32,35, 37			H	H	H	H	H	H	H	H	H	H	L	H	H	H	H	H				
F140		13,19,21,36	3,6,7,8, 12,16, 17,18,22, 23,24,26, 27,32,33, 35,37			H	H	H	H	H	H	H	H	H	H	H	L	H	H	H	H				
F141		18,19,21,36	3,6,7,8, 12,13,16, 17,22, 23,24,26, 27,32,33, 35,37			H	H	H	H	H	H	H	H	H	H	H	H	L	H	H	H				
F142		19,21,36,37	3,6,7,8, 12,13,16, 17,18,22, 23,24,26, 27,32,33, 35			H	H	H	H	H	H	H	H	H	H	H	H	H	L	H	H				

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TABLE VIII. Functional test - Continued.

Test condition	Parameter	Connect				Output logic states															Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12	Q13	Q14	Q15	Q16	Min	Max		
F143	Functional	17,19,21,36	3,6,7,8, 12,13,16, 18,22,23, 24,26,27, 32,33,35, 37	2/ 3/ 4/	2	H	H	H	H	H	H	H	H	H	H	H	H	H	H	L	H	1/			12,13
F144		16,19,21,36	3,6,7,8, 12,13,17, 18,22,23, 24,26,27, 32,33,35, 37			H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	L				
F145		19,21,36	3,6,7,8, 12,13,16, 17,18,22, 23,24,26, 27,32,33, 35,37			H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H				
F146		6,19,21,35,36	3,7,8, 12,13,16, 17,18,22, 23,24,26, 27,32,33, 37			L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L				
F147		3,6,7,8,12, 13,17,18,19, 21,22,23,24, 26,27,32,33, 35,37	16,36			L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	H				
F148		3,6,7,8,12, 13,16,18,19, 21,22,23,24, 26,27,32,33, 37	17,35,36			L	L	L	L	L	L	L	L	L	L	L	L	L	L	H	L				
F149		3,6,7,8,12, 13,16,17,18,19, 21,22,23,24, 26,27,32,33	35,36,37			L	L	L	L	L	L	L	L	L	L	L	L	L	H	L	L				
F150		3,6,7,8,12, 13,16,17,19 21,22,23,24, 26,27,32,33,37	16,35,36			L	L	L	L	L	L	L	L	L	L	L	L	H	L	L	L				
F151		3,6,7,8,12, 16,17,18,19, 21,22,23,24, 26,27,32,33,37	13,35,36			L	L	L	L	L	L	L	L	L	L	L	H	L	L	L	L				
F152		3,6,7,8,12, 13,16,17,18,19, 21,22,23,24, 26,27,32,37	33,35,36			L	L	L	L	L	L	L	L	L	L	H	L	L	L	L	L				
F153		3,6,7,8,12, 13,16,17,18,19, 21,22,23,24, 26,27,33,37	32,35,36			L	L	L	L	L	L	L	L	L 4/	H	L	L	L	L	L	L				
F154		3,6,7,8,13, 16,17,18,19, 21,22,23,24, 26,27,32,33,37	12,35,36			L	L	L	L	L	L	L	L	H	L	L	L	L	L	L	L				

See footnotes at end of table.

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TABLE VIII. Functional test - Continued.

Test condition	Parameter	Connect				Output logic states														Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12	Q13	Q14	Q15	Q16	Min	Max		
F155	Functional	3,6,7,8,12,13, 16,17,18,19, 21,22,23,24, 27,32,33,37	26,35,36	2/ 3/ 4/	2	L	L	L	L	L	L	L	H	L	L	L	L	L	L	L	L	1/			12,13
F156		3,6,7,8,12,13, 16,17,18,19, 21,22,23,24, 26,32,33,37	27,35,36			L	L	L	L	L	L	H	L	L	L	L	L	L	L	L	L				
F157		3,6,8,12,13, 16,17,18,19, 21,22,23,24, 26,27,32,33,37	7,35,36			L	L	L	L	L	H	L	L	L	L	L	L	L	L	L	L				
F158		3,6,7,12,13, 16,17,18,19, 21,22,23,24, 26,27,32,33,37	8,35,36			L	L	L	L	H	L	L	L	L	L	L	L	L	L	L	L				
F159		3,6,7,8,12,13, 16,17,18,19, 21,22,24, 26,27,32,33,37	23,35,36			L	L	L	H	L	L	L	L	L	L	L	L	L	L	L	L				
F160		6,7,8,12,13, 16,17,18,19, 21,22,23,24, 26,27,32,33,37	3,35,36			L	L	H	L	L	L	L	L	L	L	L	L	L	L	L	L				
F161		3,6,7,8,12,13, 16,17,18,19, 21,22,23, 26,27,32,33,37	24,35,36			L	H	L	L	L	L	L	L	L	L	L	L	L	L	L	L				
F162		3,6,7,8,12,13, 16,17,18,19, 23,24,26,27, 32,33,37	21,22,35, 36			H	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L				
F163		3,6,7,8,12,13, 16,17,18,19, 21,22,23,24, 26,27,32,33,37	35,36			L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L				
F164		3,6,8,12,17, 18,19,22,27, 33,35,36	7,13,16,21, 23,24,26, 32,37			H	L	H	L	H	L	H	L	H	L	H	L	H	L	H	L				
F165		3,8,12,17, 18,19,22,27, 33,36	6,7,13,16, 21,23,24, 26,32,35, 37			L	H	L	H	L	H	L	H	L	H	L	H	L	H	L	H				
F166		3,8,12,17, 18,19,22,27, 32,33,35	6,7,13,16, 21,23,24, 26,36,37			L	H	L	H	L	H	L	H	L	H	L	H	L	H	L	H				
F167		6,7,13,16, 19,23,24,26, 32,37	3,8,12,17, 18,21,22, 27,33,35,36			H	L	H	L	H	L	H	L	H	L	H	L	H	L	H	L				
F168		7,13,16,19, 21,23,24,26, 32,37	3,6,8,12, 17,18,22, 27,33,35,36			H	L	H	L	H	L	H	L	H	L	H	L	H	L	H	L				
F169		6,7,13,16,19, 21,23,24,26, 32,36,37	3,8,12,17, 18,22,27, 33,35			L	H	L	H	L	H	L	H	L	H	L	H	L	H	L	H				

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TABLE VIII. Functional test - Continued.

Test condition	Parameter	Connect				Output logic states															Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12	Q13	Q14	Q15	Q16	Min	Max		
F170	Functional	3,6,8,12,17, 18,22,27,33, 35,36	7,13,16, 19,21,23, 24,26,32, 37	2/ 3/ 4/	2	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	1/		12,13
F171		Open	2,3,6,7, 8,12,13, 16,17,18, 19,21,22, 23,24,26, 27,32,33, 35,36,37		Open	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L			
F172		19	3,6,7, 8,12,13, 16,17,18, 21,22, 23,24,26, 27,32,33, 35,36,37		2	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L			
F173		Open	2,3,6,7, 8,12,13, 16,17,18, 19,21,22, 23,24,26, 27,32,33, 35,36,37		Open	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L			
F174		6,19,22,36	3,7,8,12, 13,16,17, 18,21,23, 24,26,27, 32,33,35, 37		2	H	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L			
F175		6,19	3,7,8,12, 13,16,17, 18,21,22, 23,24,26, 27,32,33, 35,36,37		2	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L			
F176		Open	2,3,6,7,8, 12,13,16, 17,18,19, 21,22,23, 24,26,27, 32,33,35, 36,37		Open	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L			
F177		6,19,35	3,7,8,12, 13,16,17, 18,21,22, 23,24,26, 27,32,33, 36,37		2	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	H			

See footnotes at end of table.

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TABLE VIII. Functional test - Continued.

Test condition	Parameter	Connect				Output logic states																Initial limits 25°-+5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12	Q13	Q14	Q15	Q16	Min	Max		
F178	Functional	19	3,6,7,8, 12,13,16, 17,18,21, 22,23,24, 26,27,32, 33,35,36, 37	2/ 3/ 4/	2	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	H	1/			12,13
F179	Functional	3,6,7,8,12, 13,16,17,18, 19,22,23,24, 26,27,32,33, 36,37	21,35	2/ 3/ 4/	2	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	1/			12,13
F180	Functional	3,6,7,8,12, 13,16,17,18, 21,22,23,24, 26,27,32,33, 35,36,37	19	2/ 3/ 4/	2	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	1/			12,13

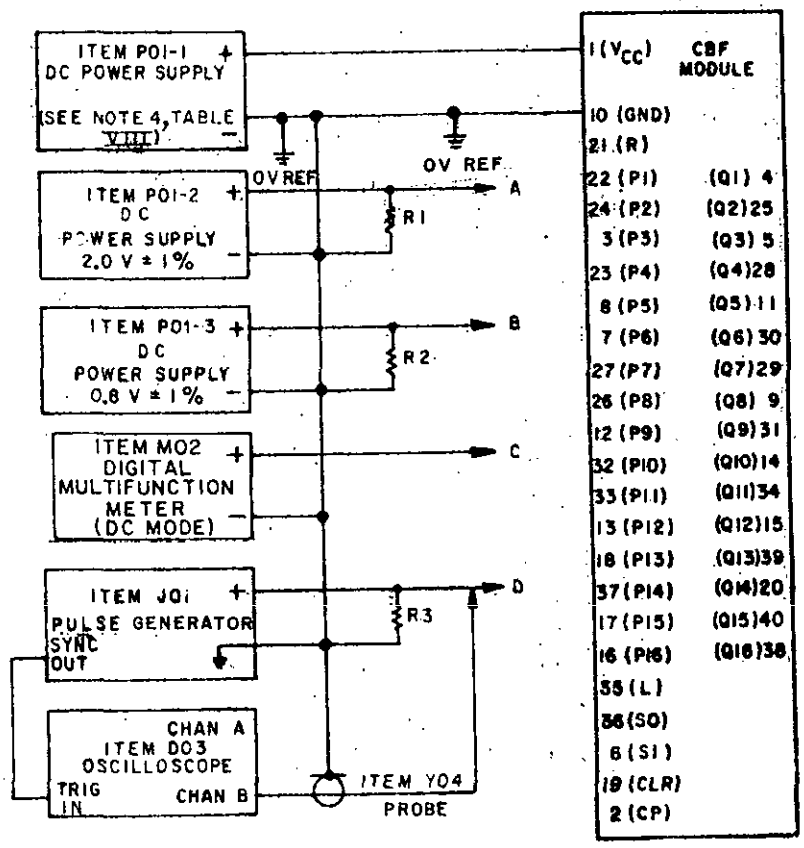
1/ Output logic states:
L = Low level = 0.0 to 0.5 V.
H = High level = 2.5 to 4.5 V.

2/ For test conditions F1-F180 verification of the output logic states in footnote 1 is obtained by connecting item M02, point C to each output pin in succession after input conditions have been accomplished. Tests F1-F180 are to be performed in sequence.

3/ Q1-Q16 output logic states are measured at output pins 4, 25, 5, 28, 11, 30, 29, 9, 31, 14, 34, 15, 39, 20, 40, and 38 respectively.

4/ Test conditions:
a. For test conditions F1 through F90 set item P01-1 to 4.5 V dc ±1%.
b. For test conditions F91 through F180 set item P01-1 to 5.5 V dc ±1%.

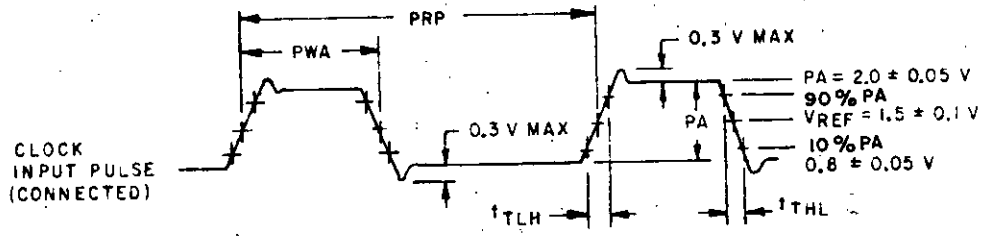
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1/2 W.
2. R3 = 51.1Ω ±1%, 1/4 W.
3. After making all input circuit connections for a test condition, use the single pulse mode on item J01 (after initial setup) and apply one pulse before making each measurement. Tests F1-F180 are to be performed in sequence.

* FIGURE 12. Functional test circuit for table VIII.

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NOTES:
1. Clock input pulse parameters:

   $t_{TLH} = t_{THL} = 50 \pm 25$ ns.
   PWA = 500 ±100 ns.
   PRP = 1,000 ±200 ns.

2. Clock threshold level setup: Make circuit connections per table VIII. Place item J01 in single pulse mode. Use item D03 and adjust DC offset of item J01 for low level of 0.8 ±0.05 V (clock low-level threshold). Adjust item J01 to obtain the required PWA and PRP. Use item D03 and adjust item J01 amplitude for a high level of 2.0 ±0.05 V (clock high-level threshold).

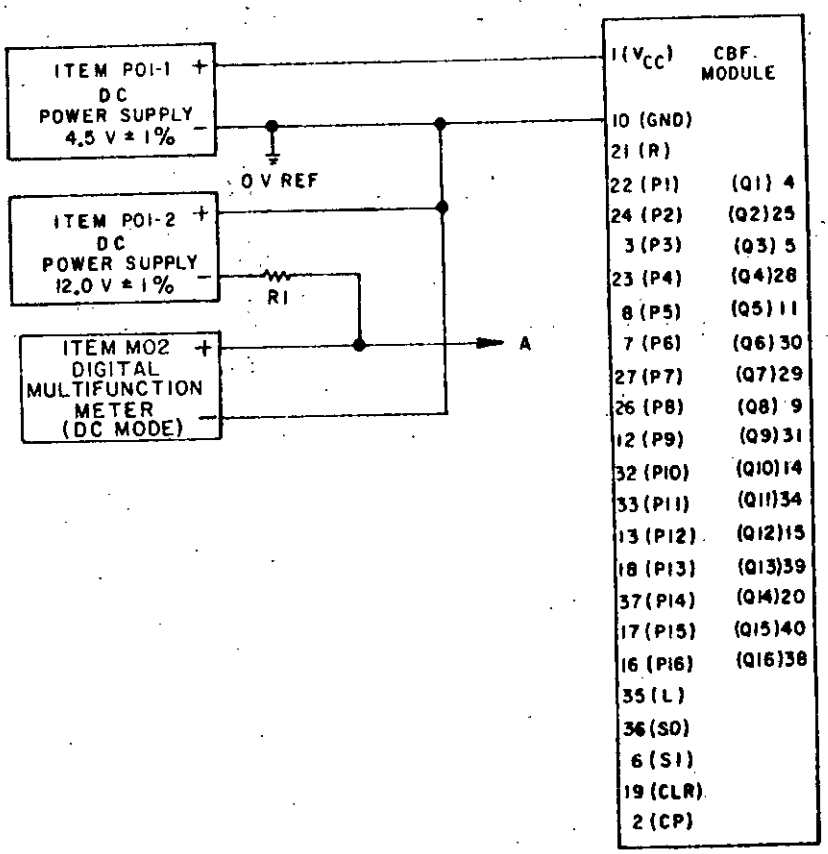
* FIGURE 13. Functional test circuit waveform for table VIII.

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* TABLE IX. Input clamp diode voltage test.

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
G1	R Input clamp diode voltage	21	-0.4	-1.2	V	14
G2	P1	22				
G3	P2	24				
G4	P3	3				
G5	P4	23				
G6	P5	8				
G7	P6	7				
G8	P7	27				
G9	P8	26				
G10	P9	12				
G11	P10	32				
G12	P11	33				
G13	P12	13				
G14	P13	18				
G15	P14	37				
G16	P15	17				
G17	P16	16				
G18	L	35				
G19	S0	36				
G20	S1	6				
G21	CLR	19				
G22	CP	2				

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NOTE: R1 = 590Ω ±1%, 1/4 W.

* FIGURE 14. Input clamp diode voltage test circuit for table IX.

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* TABLE X. Output voltage (low-level) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
H1	Q1 Output voltage, low level	6,19,36	3,7,8,12, 13,16,17, 18,22,23, 24,26,27, 32,33,37	4	0	500	mV	15,16
H2	Q2			25				
H3	Q3			5				
H4	Q4			28				
H5	Q5			11				
H6	Q6			30				
H7	Q7			29				
H8	Q8			9				
H9	Q9			31				
H10	Q10			14				
H11	Q11			34				
H12	Q12			15				
H13	Q13			39				
H14	Q14			20				
H15	Q15			40				
H16	Q16			38				

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* TABLE XI. Output voltage (high-level) test.

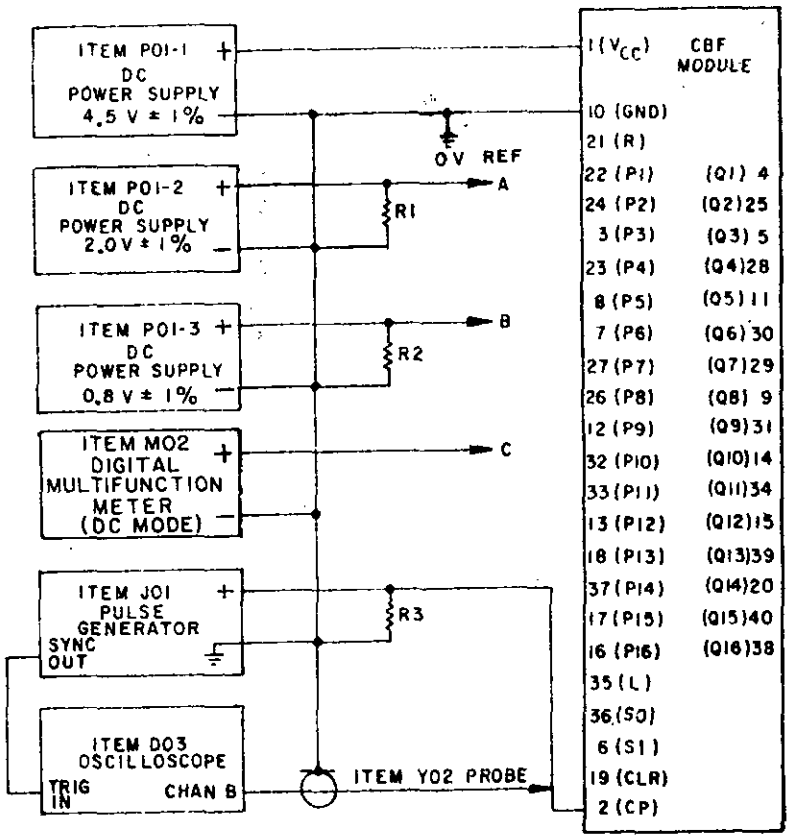
Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
J1	Q1 Output voltage, high level (R input threshold)	19,21,36	3,6,7,8,12,13,16,17,18, 22,23,24,26,27,32,33, 35,37	4	2.5	4.5	V	15,16
J2	Q1 Output voltage, high level	6,19,22, 36	3,7,8,12,13,16,17,18, 21,23,24,26,27,32,33, 35,37	4				
J3	Q2	6,19,24, 36	3,7,8,12,13,16,17,18, 21,22,23,26,27,32,33, 35,37	25				
J4	Q3	3,6,19,36	7,8,12,13,16,17,18,21, 22,23,24,26,27,32,33, 35,37	5				
J5	Q4	6,19,23, 36	3,7,8,12,13,16,17,18, 21,22,24,26,27,32,33, 35,37	28				
J6	Q5	6,8,19,36	3,7,12,13,16,17,18,21, 22,23,24,26,27,32,33, 35,37	11				
J7	Q6	6,7,19,36	3,8,12,13,16,17,18,21, 22,23,24,26,27,32,33, 35,37	30				
J8	Q7	6,19,27, 36	3,7,8,12,13,16,17,18, 21,22,23,24,26,32,33, 35,37	29				
J9	Q8	6,19,26, 36	3,7,8,12,13,16,17,18, 21,22,23,24,27,32,33, 35,37	9				

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* TABLE XI. Output voltage (high-level) test - Continued.

Test condition	Parameter		Connect			Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
J10	Q9	Output voltage, high level	6,12,19, 36	3,7,8,13,16,17,18,21, 22,23,24,26,27,32,33, 35,37	31	2.5	4.5	V	15,16
J11	Q10		6,19,32, 36	3,7,8,12,13,16,17,18, 21,22,23,24,26,27,33, 35,37	14				
J12	Q11		6,19,33, 36	3,7,8,12,13,16,17,18, 21,22,23,24,26,27,32, 35,37	34				
J13	Q12		6,18,19, 36	3,7,8,12,16,17,18,21, 22,23,24,26,27,32,33, 35,37	15				
J14	Q13		6,18,19, 36	3,7,8,12,13,16,17,21, 22,23,24,26,27,32,33, 35,37	39				
J15	Q14		6,19,36, 37	3,7,8,12,13,16,17,18, 21,22,23,24,26,27,32, 33,35	20				
J16	Q15		6,17,19, 36	3,7,8,12,13,16,18,21, 22,23,24,26,27,32,33, 35,37	40				
J17	Q16		6,16,19, 36	3,7,8,12,13,17,18,21, 22,23,24,26,27,32,33, 35,37	38				
J18	Q1	Output voltage, high level (L input threshold)	6,19,35	3,7,8,12,13,16,17,18, 21,22,23,24,26,27,32, 33,36,37	38				

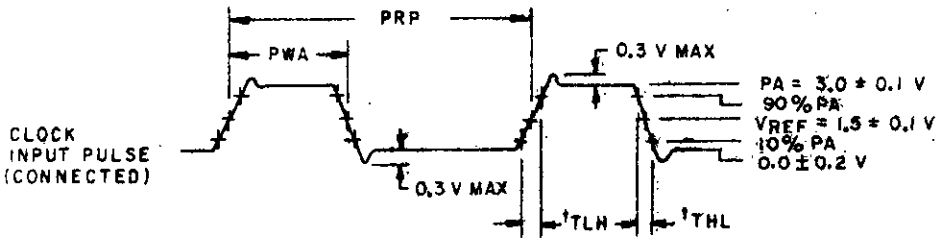
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1/2 W.
2. R3 = 51.1Ω ±1%, 1/4 W.
3. After making all input circuit connections for a test condition, use the single pulse mode on item J01 (after initial setup) and apply a single pulse before making each measurement.
4. Load all outputs as specified on figure 17.

* FIGURE 15. Output voltage (low-level and high-level) test circuit for tables X and XI.

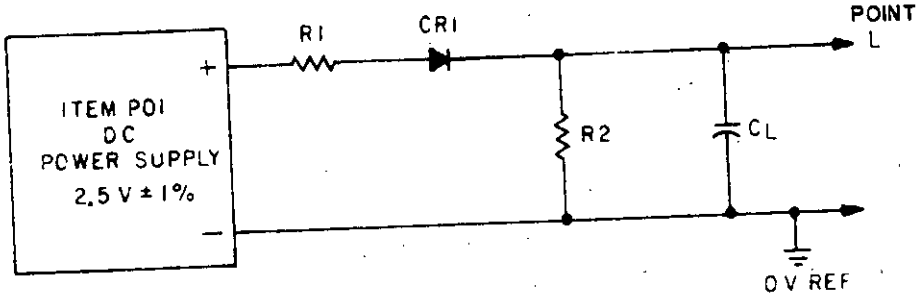
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NOTE: Input pulse parameters:
$t_{TLH} = t_{THL} = 50 \pm 25$ ns.
PWA = 500 ±100 ns.
PRP = 1,000 ±200 ns.

* FIGURE 16. Output voltage (low-level and high-level) test circuit waveform for tables X and XI.

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NOTES:
1. CR1 = 1N3600, 1N4150, or equivalent.
2. R1 = Select value for a current flow of 18.0 mA ±5% with point L held at 0.5 V.
3. R2 = 2.61 kΩ ±1%, 1/8 W.
4. Test type S:
   C_L = 50 ±5 pF (including probe and fixture capacitance)
   Test type Q:
   C_L = 150 ±5 pF (including probe and fixture capacitance).
5. Capacitor C_L is not required for dc electrical test conditions H1-H18 and J1-J18.
6. Lead length from load circuit to module output pin under test is to be less than 1 inch (25.4 mm).

* FIGURE 17. CBF logic load circuit.

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TABLE XII. Input current (high level) test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
K1	R Input current, high level	21	2,3,6,7,8,12,13,16,17, 18,19,22,23,24,26,27,32, 33,35,36,37	0	50	µA	18
K2	P1	22	2,3,6,7,8,12,13,16,17, 18,19,21,23,24,26,27,32, 33,35,36,37				
K3	P2	24	2,3,6,7,8,12,13,16,17, 18,19,21,22,23,26,27,32, 33,35,36,37				
K4	P3	3	2,6,7,8,12,13,16,17,18, 19,21,22,23,24,26,27,32, 33,35,36,37				
K5	P4	23	2,3,6,7,8,12,13,16,17, 18,19,21,22,24,26,27,32, 33,35,36,37				
K6	P5	8	2,3,6,7,12,13,16,17,18, 19,21,22,23,24,26,27,32, 33,35,36,37				
K7	P6	7	2,3,6,8,12,13,16,17,18, 19,21,22,23,24,26,27,32, 33,35,36,37				
K8	P7	27	2,3,6,7,8,12,13,16,17, 18,19,21,22,23,24,26,32, 33,35,36,37				
K9	P8	26	2,3,6,7,8,12,13,16,17, 18,19,21,22,23,24,27,32, 33,35,36,37				

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TABLE XII. Input current (high level) test - Continued.

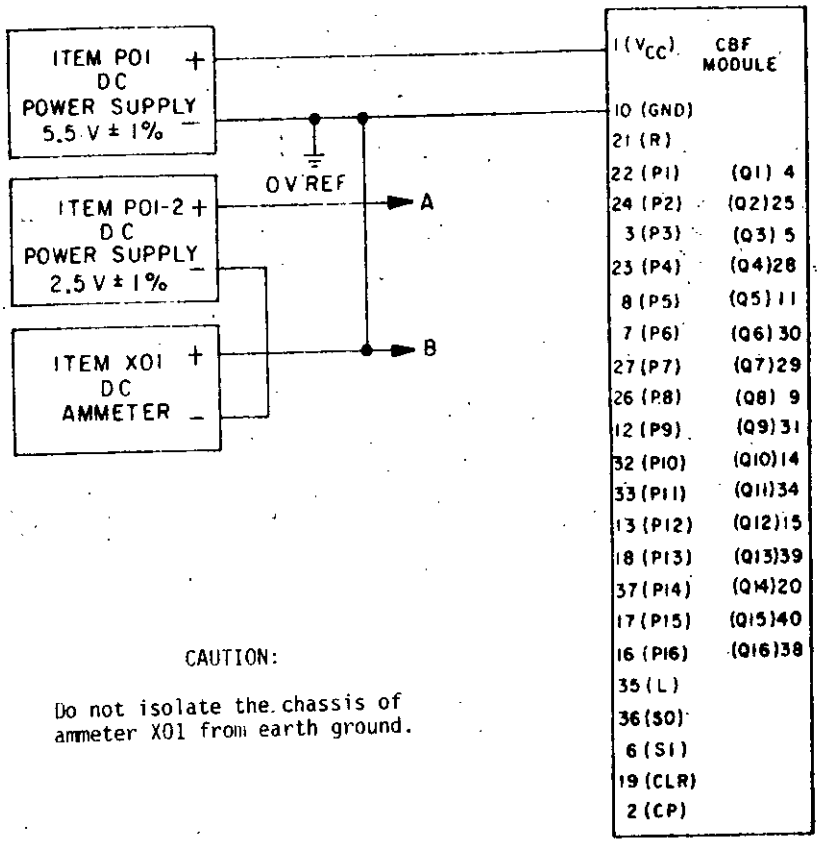
Test condition	Parameter		Connect		Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
K10	P9	Input current, high level	12	2,3,6,7,8,13,16,17,18, 19,21,22,23,24,26,27,32, 33,35,36,37	0	50	µA	18
K11	P10		32	2,3,6,7,8,12,13,16,17, 18,19,21,22,23,24,26,27, 33,35,36,37				
K12	P11		33	2,3,6,7,8,12,13,16,17, 18,19,21,22,23,24,26,27, 32,35,36,37				
K13	P12		13	2,3,6,7,8,12,16,17,18, 19,21,22,23,24,26,27,32, 33,35,36,37				
K14	P13		18	2,3,6,7,8,12,13,16,17, 19,21,22,23,24,26,27,32, 33,35,36,37				
K15	P14		37	2,3,6,7,8,12,13,16,17, 18,19,21,22,23,24,26,27, 32,33,35,36				
K16	P15		17	2,3,6,7,8,12,13,16,18, 19,21,22,23,24,26,27,32, 33,35,36,37				
K17	P16		16	2,3,6,7,8,12,13,17,18, 19,21,22,23,24,26,27,32, 33,35,36,37				

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TABLE XII. Input current (high level) test - Continued.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
K18	IIL Input current, high level	35	2,3,6,7,8,12,13,16,17, 18,19,21,22,23,24,26,27, 32,33,36,37	0	50	μA	18
K19	SO	36	2,3,6,7,8,12,13,16,17, 18,19,21,22,23,24,26,27, 32,33,35,37		200		
K20	S1	6	2,3,7,8,12,13,16,17,18, 19,21,22,23,24,26,27,32, 33,35,36,37				
K21	CLR	19	2,3,6,7,8,12,13,16,17, 18,21,22,23,24,26,27,32, 33,35,36,37				
K22	CP	2	3,6,7,8,12,13,16,17,18, 19,21,22,23,24,26,27,32, 33,35,36,37				

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

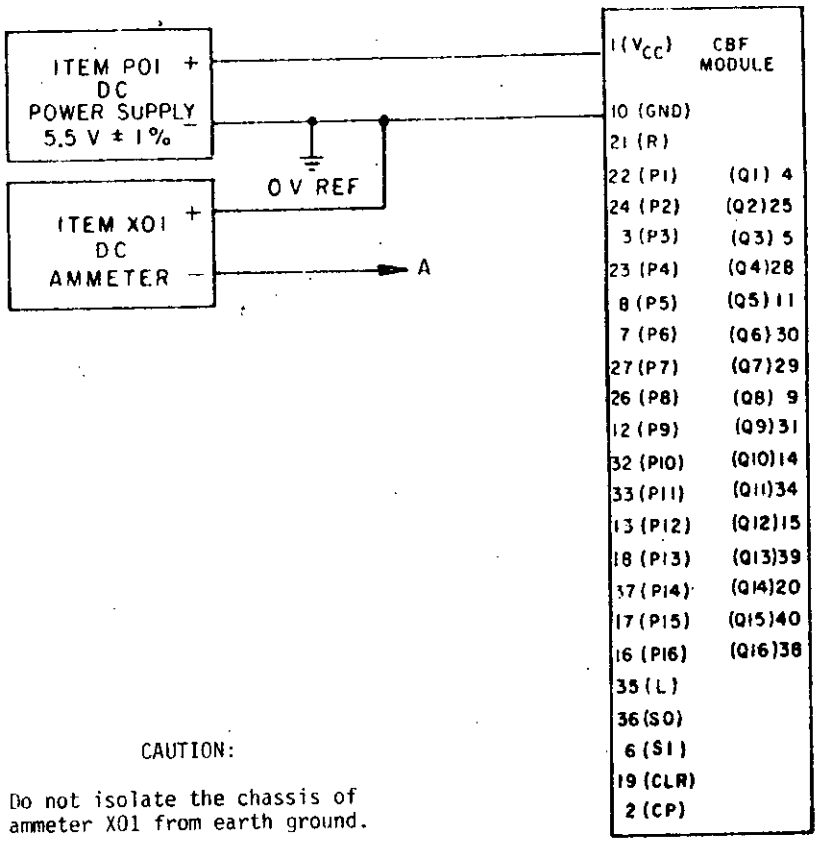
FIGURE 18. Input current (high-level) test circuit for table XII.

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TABLE XIII. Input current (low-level) test.

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
L1	R Input current, low level	21	-0.1	-2.2	mA	19
L2	P1	22				
L3	P2	24				
L4	P3	3				
L5	P4	23				
L6	P5	8				
L7	P6	7				
L8	P7	27				
L9	P8	26				
L10	P9	12				
L11	P10	32				
L12	P11	33				
L13	P12	13				
L14	P13	18				
L15	P14	37				
L16	P15	17				
L17	P16	16				
L18	L	35				
L19	SO	36		-8.8		
L20	S1	6				
L21	CLR	19				
L22	CP	2	-3.0			

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

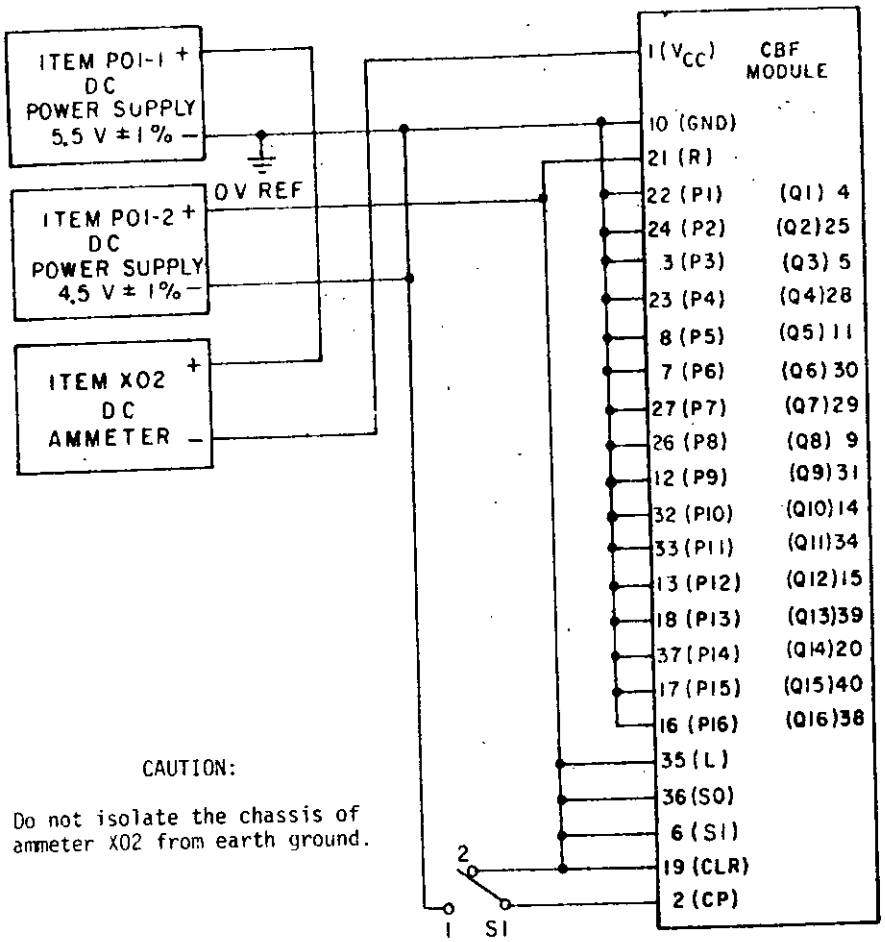
FIGURE 19. Input current (low-level) test circuit for table XIII.

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TABLE XIV. Power supply current test.

Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
M	Power supply current	135	488	mA	20

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NOTE: After making all circuit connections, place switch S1 in position 1 for a minimum of 100 ns, then place switch S1 in position 2 and make measurement.

FIGURE 20. Power supply current test circuit for table XIV.

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* TABLE XV. Transition time (t_{TLH} and minimum clear pulse width check) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
N1	Q1 Transition time (t_{TLH}) parallel mode, C_L = 50 pF	6,36	Open	4	2	15	ns	21,22
N2	Q2			25				
N3	Q3			5				
N4	Q4			28				
N5	Q5			11				
N6	Q6			30				
N7	Q7			29				
N8	Q8			9				
N9	Q9			31				
N10	Q10			14				
N11	Q11			34				
N12	Q12			15				
N13	Q13			39				
N14	Q14			20				
N15	Q15			40				
N16	Q16			38				
N17	Q1 Transition time (t_{TLH}) right shift mode, C_L = 50 pF	36	6	4				
N18	Q16 Transition time (t_{TLH}) left shift mode, C_L = 50 pF	6	36	38				

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* TABLE XV. Transition time (t_{TLH} and minimum clear pulse width check) test - Continued.

Test condition		Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
N19	Q1	Transition time (t_{TLH}) parallel mode, C_L = 150 pF	6,36	Open	4	2	27	ns	21,22
N20	Q2				25				
N21	Q3				5				
N22	Q4				28				
N23	Q5				11				
N24	Q6				30				
N25	Q7				29				
N26	Q8				9				
N27	Q9				31				
N28	Q10				14				
N29	Q11				34				
N30	Q12				15				
N31	Q13				39				
N32	Q14				20				
N33	Q15				40				
N34	Q16				38				
N35	Q1	Transition time (t_{TLH}) right shift mode, C_L = 150 pF	36	6	4				
N36	Q16	Transition time (t_{TLH}) left shift mode, C_L = 150 pF	6	36	38				

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* TABLE XVI. Transition time (t_{THL} and minimum clear pulse width check) test.

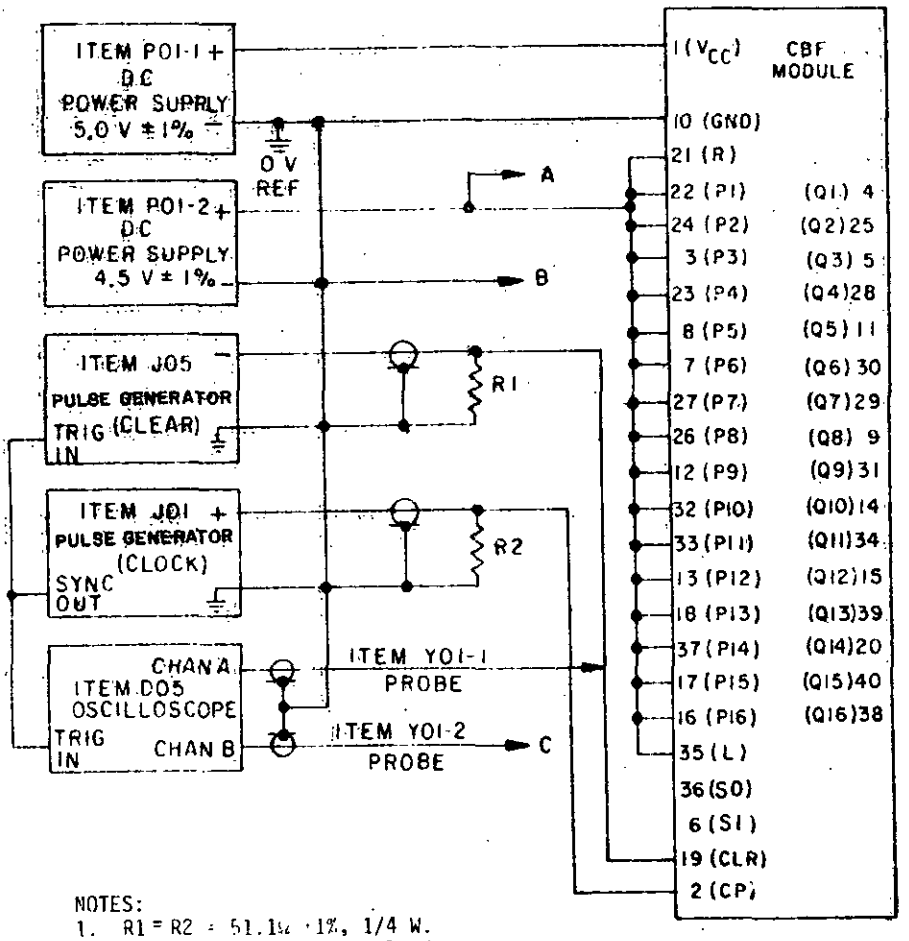
Test condition	Parameter		Connect			Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
P1	Q1	Transition time (t_{THL}) parallel mode, C_L = 50 pF	6,36	Open	4	2	8	ns	21,22
P2	Q2				25				
P3	Q3				5				
P4	Q4				28				
P5	Q5				11				
P6	Q6				30				
P7	Q7				29				
P8	Q8				9				
P9	Q9				31				
P10	Q10				14				
P11	Q11				34				
P12	Q12				15				
P13	Q13				39				
P14	Q14				20				
P15	Q15				40				
P16	Q16				38				
P17	Q1	Transition time (t_{THL}) right shift mode, C_L = 50 pF	36	6	4				
P18	Q16	Transition time (t_{THL}) left shift mode, C_L = 50 pF	6	36	38				

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* TABLE XVI. Transition time (t_{THL} and minimum clear pulse width check) test - Continued.

Test condition		Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
P19	Q1	Transition time (t_{THL}) parallel mode, C_L = 150 pF	6,36	Open	4	2	11	ns	21,22
P20	Q2				25				
P21	Q3				5				
P22	Q4				28				
P23	Q5				11				
P24	Q6				30				
P25	Q7				29				
P26	Q8				9				
P27	Q9				31				
P28	Q10				14				
P29	Q11				34				
P30	Q12				15				
P31	Q13				39				
P32	Q14				20				
P33	Q15				40				
P34	Q16				38				
P35	Q1	Transition time (t_{THL}) right shift mode, C_L = 150 pF	36	6	4				
P36	Q16	Transition time (t_{THL}) left shift mode, C_L = 150 pF	6	36	38				

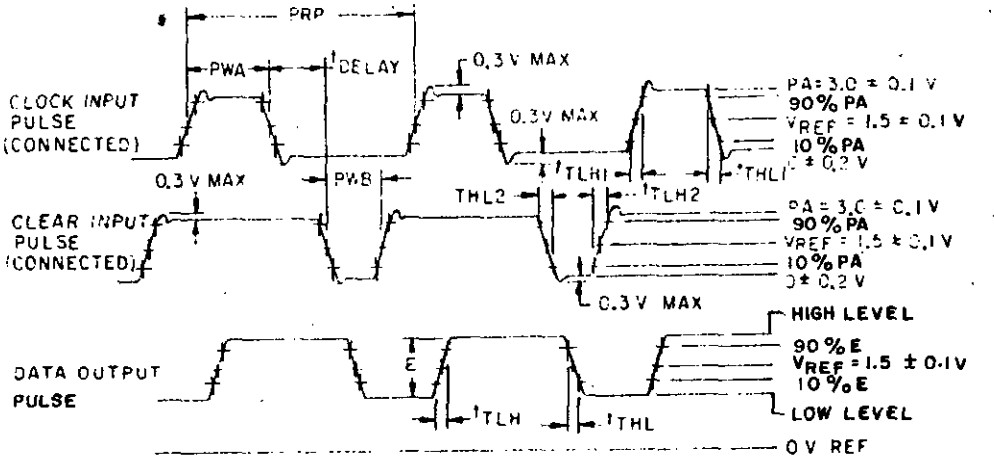
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NOTES:
1. R1 = R2 = 51.1Ω ±1%, 1/4 W.
2. Load all outputs with load as specified on figure 17.

* FIGURE 21. Transition time (t_{TLH} and t_{THL} and clear pulse width check) test circuit for tables XV and XVI.

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NOTES:
1. Clock input pulse parameters: PWA = 100 ±20 ns; PRP = 500 ±100 ns; $t_{TLH1} = t_{THL1} = 4 \pm 2$ ns.
2. Clear input pulse parameters: Clear pulse is taken from the negative output of item J05. Adjust the dc offset level of item J05 to obtain the required pulse voltage levels. Adjust the pulse width controls of item J05 to obtain PWB = 12 +1.0,-0.0 ns (clear minimum pulse width). Adjust the delay control of item J01 to obtain $t_{DELAY} = 100 \pm 20$ ns; $t_{TLH2} = t_{THL2} = 4 \pm 2$ ns.
3. Verify all outputs have a high-level $\geq$ 2.5 V and low-level $\leq$ 0.5 V before making measurements.
4. Physical length of electrical conductors between resistors R1 and R2 and input pins; between items Y01-1 and Y01-2 and input pins; and between load circuits and output pins, shall be less than 1 inch (25.4 mm).
5. All voltages are referenced to ground.

* FIGURE 22. Transition time (t_{TLH} and t_{THL} and clear pulse width check) test circuit waveforms for tables XV and XVI.

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TABLE XVII. Propagation delay time (t_{PLH}) test.

Test condition	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Connect: D to pin	Initial limits 25°±5°C ambient: Min	Initial limits 25°±5°C ambient: Max	Unit	Test figure
Q1	Q1 Propagation delay time (t_{PLH}), clock, C_L = 50 pF	6,36	3,7,8,12,13,16,17,18,21,23, 24,26,27,32,33,35,37	22	4	5	14	ns	23,24
Q2	Q2		3,7,8,12,13,16,17,18,21,22, 23,26,27,32,33,35,37	24	25				
Q3	Q3		7,8,12,13,16,17,18,21,22,23, 24,26,27,32,33,35,37	3	5				
Q4	Q4		3,7,8,12,13,16,17,18,21,22, 24,26,27,32,33,35,37	23	28				
Q5	Q5		3,7,12,13,16,17,18,21,22,23, 24,26,27,32,33,35,37	8	11				
Q6	Q6		3,8,12,13,16,17,18,21,22,23, 24,26,27,32,33,35,37	7	30				
Q7	Q7		3,7,8,12,13,16,17,18,21,22, 23,24,26,32,33,35,37	27	29				
Q8	Q8		3,7,8,12,13,16,17,18,21,22, 23,24,27,32,33,35,37	26	9				
Q9	Q9		3,7,8,13,16,17,18,21,22,23, 24,26,27,32,33,35,37	12	31				
Q10	Q10		3,7,8,12,13,16,17,18,21,22, 23,24,26,27,33,35,37	32	14				
Q11	Q11		3,7,8,12,13,16,17,18,21,22, 23,24,26,27,32,35,37	33	34				
Q12	Q12		3,7,8,12,16,17,18,21,22,23. 24,26,27,32,33,35,37	13	15				

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* TABLE XVII. Propagation delay time (t_{PLH}) test - Continued.

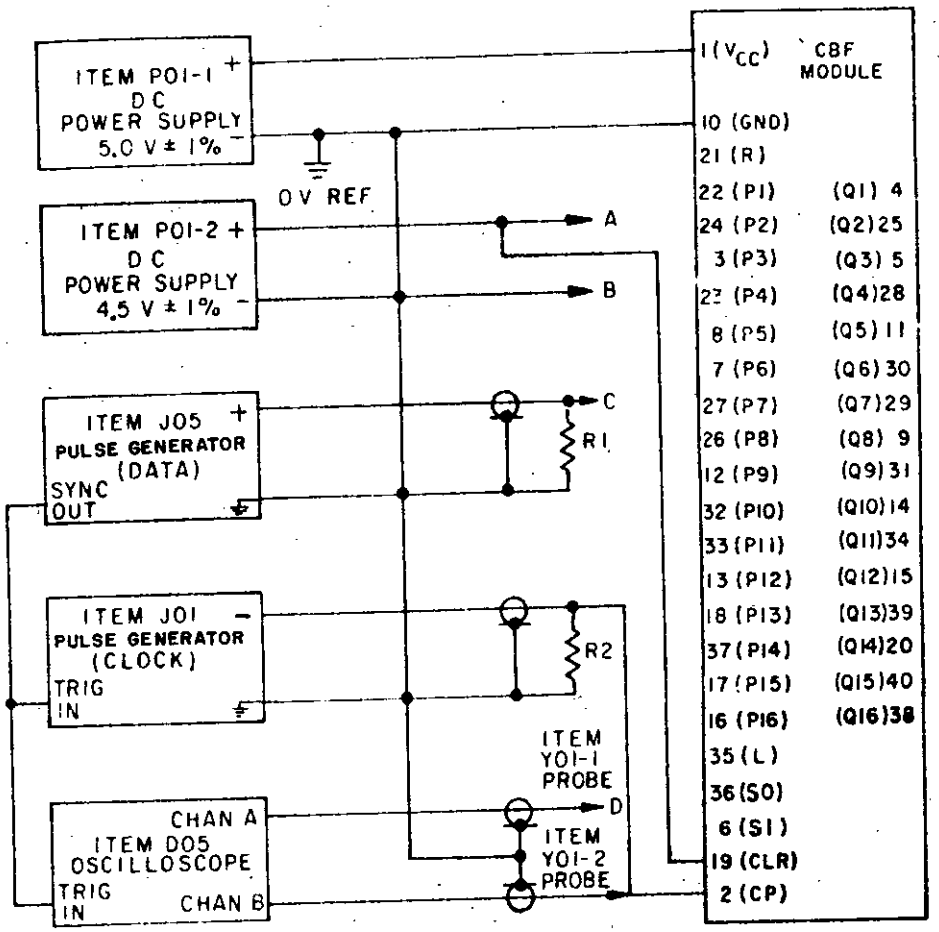
Test condition	Parameter		Connect: A to pin	Connect: B to pin	Connect: C to pin	Connect: D to pin	Initial limits 25°±5°C ambient: Min	Initial limits 25°±5°C ambient: Max	Unit	Test figure
Q13	Q13	Propagation delay time (t_{PLH}), clock, C_L = 50 pF	6,36	3,7,8,12,13,16,17,21,22,23, 24,26,27,32,33,35,37	18	39	5	14	ns	23,24
Q14	Q14			3,7,8,12,13,16,17,18,21,22, 23,24,26,27,32,33,35	37	20				
Q15	Q15			3,7,8,12,13,16,18,21,22,23, 24,26,27,32,33,35,37	17	40				
Q16	Q16			3,7,8,12,13,17,18,21,22,23, 24,26,27,32,33,35,37	15	38				
Q17	Q1	Propagation delay time (t_{PLH}), clock, right shift mode C_L = 50 pF	36	3,6,7,8,12,13,16,17,18,22,23, 24,26,27,32,33,35,37	21	4				
Q18	Q16	Propagation delay time (t_{PLH}), clock, left shift mode C_L = 50 pF	6	3,7,8,12,13,16,17,18,21,22, 23,24,26,27,32,33,36,37	35	38				
Q19	Q1	Propagation delay time (t_{PLH}), clock, C_L = 150 pF	6,36	3,7,8,12,13,16,17,18,21,23, 24,26,27,32,33,35,37	22	4	5.0	19.5		
Q20	Q2			3,7,8,12,13,16,17,18,21,22, 23,26,27,32,33,35,37	24	25				
Q21	Q3			7,8,12,13,16,17,18,21,22,23, 24,26,27,32,33,35,37	3	5				
Q22	Q4			3,7,8,12,13,16,17,18,21,22, 24,26,27,32,33,35,37	23	28				
Q23	Q5			3,7,12,13,16,17,18,21,22,23, 24,26,27,32,33,35,37	8	11				

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* TABLE XVII. Propagation delay time (tp_{LH}) test - Continued.

Test condition	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Connect: D to pin	Initial limits 25°±5°C ambient: Min	Initial limits 25°±5°C ambient: Max	Unit	Test figure
Q24	Q6 Propagation delay time (tp_{LH}), clock, C_L = 150 pF	6,36	3,8,12,13,16,17,18,21,22,23, 24,26,27,32,33,35,37	7	30	5.0	19.5	ns	23,24
Q25	Q7		3,7,8,12,13,16,17,18,21,22, 23,24,26,32,33,35,37	27	29				
Q26	Q8		3,7,8,12,13,16,17,18,21,22, 23,24,27,32,33,35,37	26	9				
Q27	Q9		3,7,8,13,16,17,18,21,22,23, 24,26,27,32,33,35,37	12	31				
Q28	Q10		3,7,8,12,13,16,17,18,21,22, 23,24,26,27,33,35,37	32	14				
Q29	Q11		3,7,8,12,13,16,17,18,21,22, 23,24,26,27,32,35,37	33	34				
Q30	Q12		3,7,8,12,16,17,18,21,22,23, 24,26,27,32,33,35,37	13	15				
Q31	Q13		3,7,8,12,13,16,17,21,22,23, 24,26,27,32,33,35,37	18	39				
Q32	Q14		3,7,8,12,13,16,17,18,21,22, 23,24,26,27,32,33,35	37	20				
Q33	Q15		3,7,8,12,13,16,18,21,22,23, 24,26,27,32,33,35,37	17	40				
Q34	Q16		3,7,8,12,13,17,18,21,22,23, 24,26,27,32,33,35,37	16	38				
Q35	Q1 Propagation delay time (tp_{LH}), clock, right shift mode C_L = 150 pF	36	3,6,7,8,12,13,16,17,18,22,23, 24,26,27,32,33,35,37	21	4				
Q36	Q16 Propagation delay time (tp_{LH}), clock, left shift mode C_L = 150 pF	6	3,7,8,12,13,16,17,18,21,22, 23,24,26,27,32,33,36,37	35	38				

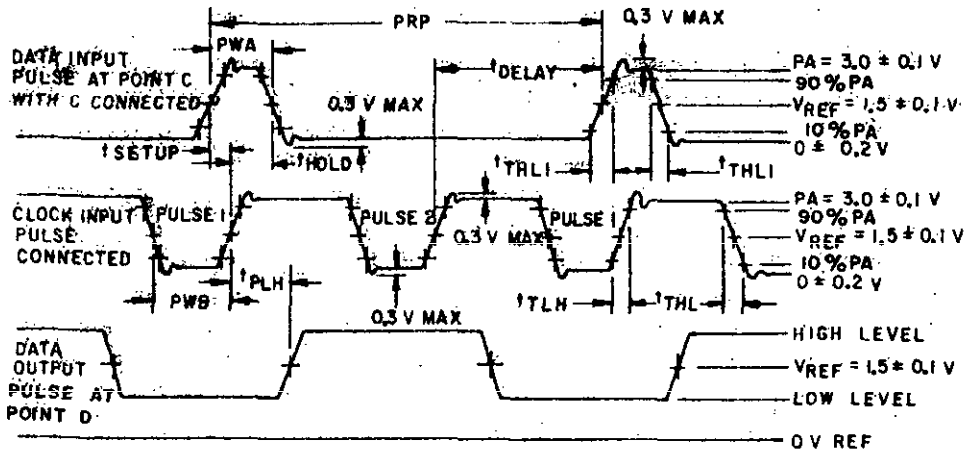
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NOTES:
1. R1 = R2 = 51.1Ω ±1%, 1/4 W.
2. Load all outputs with load as specified on figure 17.

* FIGURE 23. Propagation delay time (t_{PLH}) test circuit for table XVII.

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NOTES:
1. Data input pulse parameters: PRP = 500 ±50 ns; PWA = see note 3; $t_{TLH1} = t_{THL1} = 4$ ±2 ns.
2. Clock input pulse parameters: PWB = 100 ±10 ns; $t_{TLH} = t_{THL} = 4$ ±2 ns.
3. Item J01 is operated in double pulse mode using the negative output pulse in the dc offset mode. Using item D05 adjust the dc logic level and pulse amplitude of item J01 to obtain the required levels for the clock input pulse. Using the delay controls and pulse width controls adjust the pulse relationship of the data and clock pulses such that $t_{SETUP} = 8$ +1.0,-0 ns; $t_{HOLD} = 4.5$ +1.0 ns, -0 ns; (PWA = $t_{SETUP} + t_{HOLD}$) and $t_{DELAY} = 250$ ±50 ns. Before making any measurements, the setup and hold time requirements must be verified.
4. Verify all outputs have a high-level $\geq$ 2.5 V and low-level $\leq$ 0.5 V before making measurements.
5. Physical length of electrical conductors between resistors R1 and R2 and input pins; between items Y01-1 and Y01-2 and input pins; and between load circuits and output pins, shall be less than 1 inch (25.4 mm).
6. All voltages are referenced to ground.

* FIGURE 24. Propagation delay time (t_{PLH}) test circuit waveforms for table XVII.

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* TABLE XVIII. Propagation delay time (t_{PHL}) test.

Test condition		Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Connect: D to pin	Initial limits 25°±5°C ambient: Min	Initial limits 25°±5°C ambient: Max	Unit	Test figure
R1	Q1	Propagation delay time (t_{PHL}), clock, C_L = 50 pF	3,7,8,12,13,16,17,18,21,23, 24,26,27,32,33,35,37,6,36	Open	22	4	5.0	18.5	ns	25,26
R2	Q2		3,7,8,12,13,16,17,18,21,22, 23,26,27,32,33,35,37,6,36		24	25				
R3	Q3		7,8,12,13,16,17,18,21,22,23, 24,26,27,32,33,35,37,6,36		3	5				
R4	Q4		3,7,8,12,13,16,17,18,21,22, 24,26,27,32,33,35,37,6,36		23	28				
R5	Q5		3,7,12,13,16,17,18,21,22,23, 24,26,27,32,33,35,37,6,36		8	11				
R6	Q6		3,8,12,13,16,17,18,21,22,23, 24,26,27,32,33,35,37,6,36		7	30				
R7	Q7		3,7,8,12,13,16,17,18,21,22, 23,24,26,32,33,35,37,6,36		27	29				
R8	Q8		3,7,8,12,13,16,17,18,21,22, 23,24,27,32,33,35,37,6,36		26	9				
R9	Q9		3,7,8,13,16,17,18,21,22,23, 24,26,27,32,33,35,37,6,36		12	31				
R10	Q10		3,7,8,12,13,16,17,18,21,22, 23,24,26,27,33,35,37,6,36		32	14				
R11	Q11		3,7,8,12,13,16,17,18,21,22, 23,24,26,27,32,35,37,6,36		33	34				
R12	Q12		3,7,8,12,16,17,18,21,22,23, 24,26,27,32,33,35,37,6,36		13	15				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:00Z
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* TABLE XVIII. Propagation delay time (t_{PHL}) test - Continued.

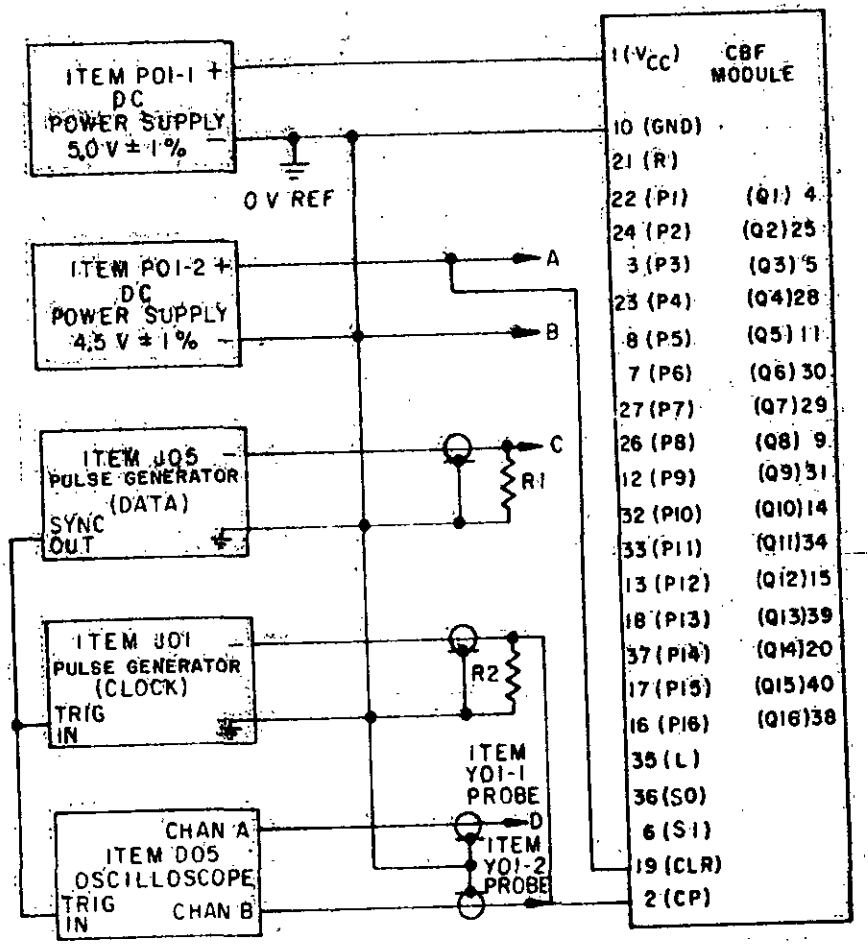
Test condition	Parameter		Connect: A to pin	Connect: B to pin	Connect: C to pin	Connect: D to pin	Initial limits 25°±5°C ambient: Min	Initial limits 25°±5°C ambient: Max	Unit	Test figure
R13	Q13	Propagation delay time (t_{PHL}), clock, C_L = 50 pF	3,7,8,12,13,16,17,21,22,23, 24,26,27,32,33,35,37,6,36	Open	18	39	5.0	18.5	ns	25,26
R14	Q14		3,7,8,12,13,16,17,18,21,22, 23,24,26,27,32,33,35,6,36		37	20				
R15	Q15		3,7,8,12,13,16,18,21,22,23, 24,26,27,32,33,35,37,6,36		17	40				
R16	Q16		3,7,8,12,13,17,18,21,22,23, 24,26,27,32,33,35,37,6,36		16	38				
R17	Q1	Propagation delay time (t_{PHL}), clock, right shift mode, C_L = 50 pF	3,7,8,12,13,16,17,18,22,23, 24,26,27,32,33,35,37,36	6	21	4				
R18	Q16	Propagation delay time (t_{PHL}), clock, left shift mode, C_L = 50 pF	3,7,8,12,13,16,17,18,21,22, 23,24,26,27,32,33,37,6	36	35	38				
R19	Q1	Propagation delay time (t_{PHL}), clock, C_L = 150 pF	3,7,8,12,13,16,17,18,21,23, 24,26,27,32,33,35,37,6,36	Open	22	4	5	23		
R20	Q2		3,7,8,12,13,16,17,18,21,22, 23,26,27,32,33,35,37,6,36		24	25				
R21	Q3		7,8,12,13,16,17,18,21,22,23, 24,26,27,32,33,35,37,6,36		3	5				
R22	Q4		3,7,8,12,13,16,17,18,21,22, 24,26,27,32,33,35,37,6,36		23	28				
R23	Q5		3,7,12,13,16,17,18,21,22,23, 24,26,27,32,33,35,37,6,36		8	11				

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* TABLE XVIII. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Connect: D to pin	Initial limits 25°±5°C ambient: Min	Initial limits 25°±5°C ambient: Max	Unit	Test figure
R24	Q6 Propagation delay time (t_{PHL}), clock, C_L = 150 pF	3,8,12,13,16,17,18,21,22,23, 24,26,27,32,33,35,37,6,36	Open	7	30	5	23	ns	25,26
R25	Q7	3,7,8,12,13,16,17,18,21,22, 23,24,26,32,33,35,37,6,36		27	29				
R26	Q8	3,7,8,12,13,16,17,18,21,22, 23,24,27,32,33,35,37,6,36		26	9				
R27	Q9	3,7,8,13,16,17,18,21,22,23, 24,26,27,32,33,35,37,6,36		12	31				
R28	Q10	3,7,8,12,13,16,17,18,21,22, 23,24,26,27,33,35,37,6,36		32	14				
R29	Q11	3,7,8,12,13,16,17,18,21,22, 23,24,26,27,32,35,37,6,36		33	34				
R30	Q12	3,7,8,12,16,17,18,21,22,23, 24,26,27,32,33,35,37,6,36		13	15				
R31	Q13	3,7,8,12,13,16,17,21,22,23, 24,26,27,32,33,35,37,6,36		18	39				
R32	Q14	3,7,8,12,13,16,17,18,21,22, 23,24,26,27,32,33,35,6,36		37	20				
R33	Q15	3,7,8,12,13,16,18,21,22,23, 24,26,27,32,33,35,37,6,36		17	40				
R34	Q16	3,7,8,12,13,17,18,21,22,23, 24,26,27,32,33,35,37,6,36		16	38				
R35	Q1 Propagation delay time (t_{PHL}), clock, right shift mode, C_L = 150 pF	3,7,8,12,13,16,17,18,22,23, 24,26,27,32,33,35,37,36	6	21	4				
R36	Q16 Propagation delay time (t_{PHL}), clock, left shift mode, C_L = 150 pF	3,7,8,12,13,16,17,18,21,22, 23,24,26,27,32,33,37,6	36	35	38				

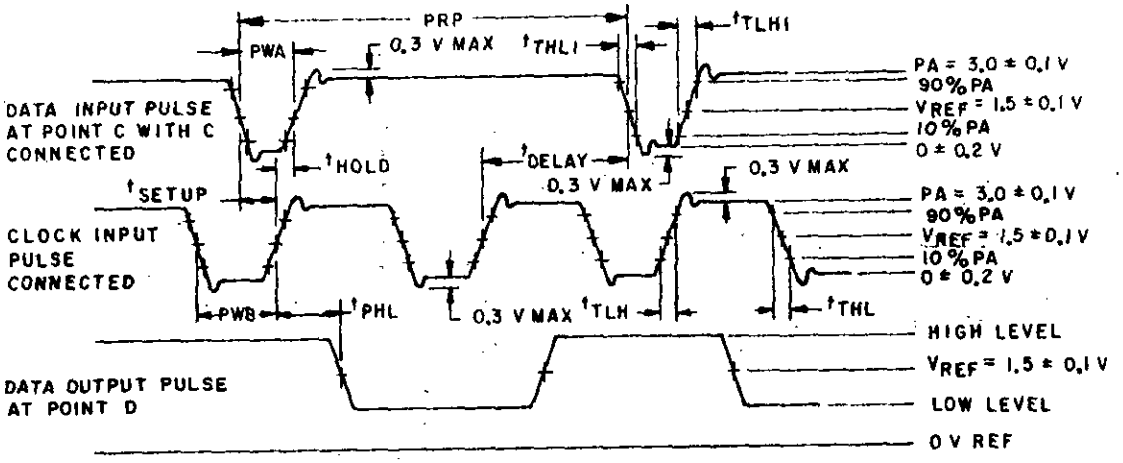
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NOTES:
1. R1 = R2 = 51.1Ω ±1%, 1/4 W.
2. Load all outputs with load as specified on figure 17.

★ FIGURE 25. Propagation delay time (t_{PHL}) test circuit for table XVIII.

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NOTES:
1. Data input pulse parameters: PRP = 500 ±50 ns; PWA = see note 3; $t_{TLH1} = t_{THL1} = 4 \pm 2$ ns.
2. Clock input pulse parameters: PWB = 100 ±10 ns; $t_{TLH} = t_{THL} = 4 \pm 2$ ns.
3. Item J01 is operated in double pulse mode. Items J01 and J05 use the negative output. Using item D05, adjust the amplitude and dc offset controls to obtain the required input logic levels of the data and clock pulses; using the delay and pulse width controls, adjust the data and clock pulses such that $t_{SETUP} = 8$ +1.0, -0 ns; $t_{HOLD} = 4.5$ +1.0, -0 ns; (PWA = $t_{SETUP} + t_{HOLD}$) and $t_{DELAY} = 250 \pm 50$ ns. Before making any measurements per table XVIII the setup and hold time requirements must be verified.
4. Verify all outputs have a high-level ≥ 2.5 V and low-level ≤ 0.5 V before making any measurements.
5. Physical length of electrical conductors between resistors R1 and R2 and input pins; between items Y01-1 and Y01-2 and input pins; and between load circuits and output pins, shall be less than 1 inch (25.4 mm).
6. All voltages are referenced to ground.

* FIGURE 26. Propagation delay time (t_{PHL}) test circuit waveforms for table XVIII.

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* TABLE XIX. Propagation delay time (t_{PHL}) test.

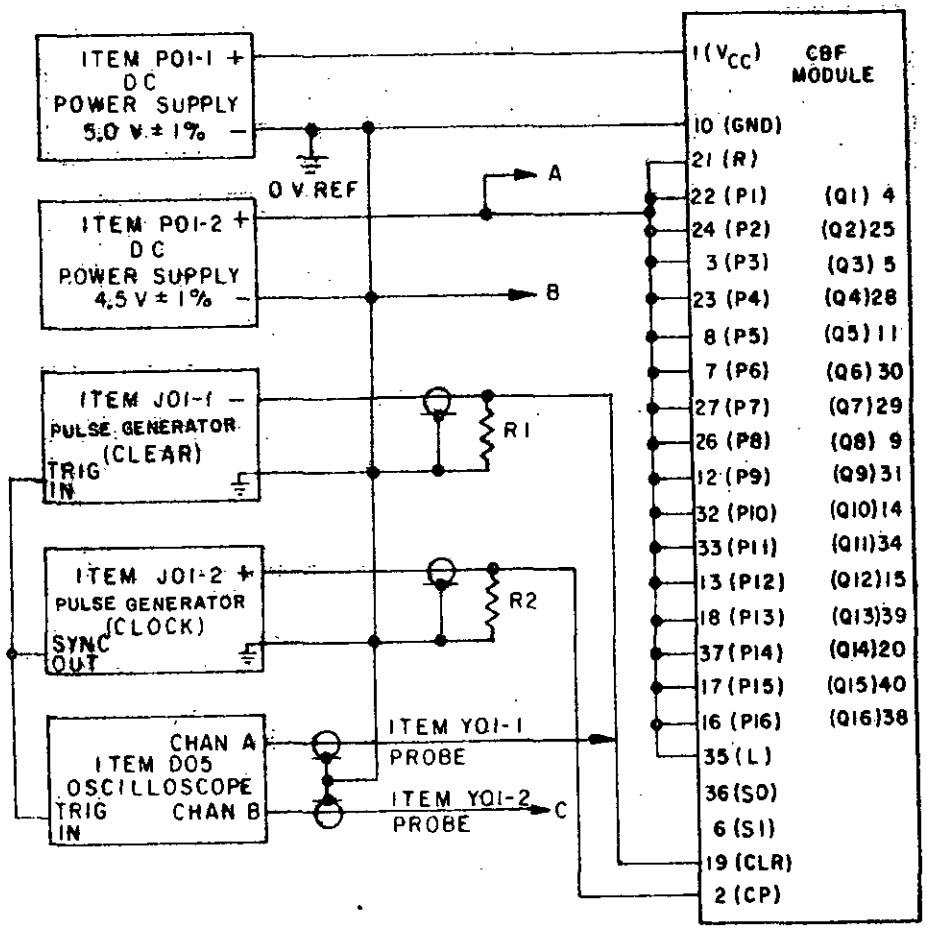
Test condition		Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
S1	Q1	Propagation delay time (t_{PHL}), clear, C_L = 50 pF	6,36	Open	4	5	21	ns	27,28
S2	Q2				25				
S3	Q3				5				
S4	Q4				28				
S5	Q5				11				
S6	Q6				30				
S7	Q7				29				
S8	Q8				9				
S9	Q9				31				
S10	Q10				14				
S11	Q11				34				
S12	Q12				15				
S13	Q13				39				
S14	Q14				20				
S15	Q15				40				
S16	Q16				38				
S17	Q1	Propagation delay time (t_{PHL}), clear, right shift mode, C_L = 50 pF	36	6	4				
S18	Q16	Propagation delay time (t_{PHL}), clear, left shift mode, C_L = 50 pF	6	36	38				

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* TABLE XIX. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
S19	Q1 Propagation delay time (t_{PHL}), clear, C_L = 150 pF	6,36	Open	4	5	25	ns	27,28
S20	Q2			25				
S21	Q3			5				
S22	Q4			28				
S23	Q5			11				
S24	Q6			30				
S25	Q7			29				
S26	Q8			9				
S27	Q9			31				
S28	Q10			14				
S29	Q11			34				
S30	Q12			15				
S31	Q13			39				
S32	Q14			20				
S33	Q15			40				
S34	Q16			38				
S35	Q16 Propagation delay time (t_{PHL}), clear, right shift mode, C_L = 150 pF	36	6	4				
S36	Q16 Propagation delay time (t_{PHL}), clear, left shift mode, C_L = 150 pF	6	36	38				

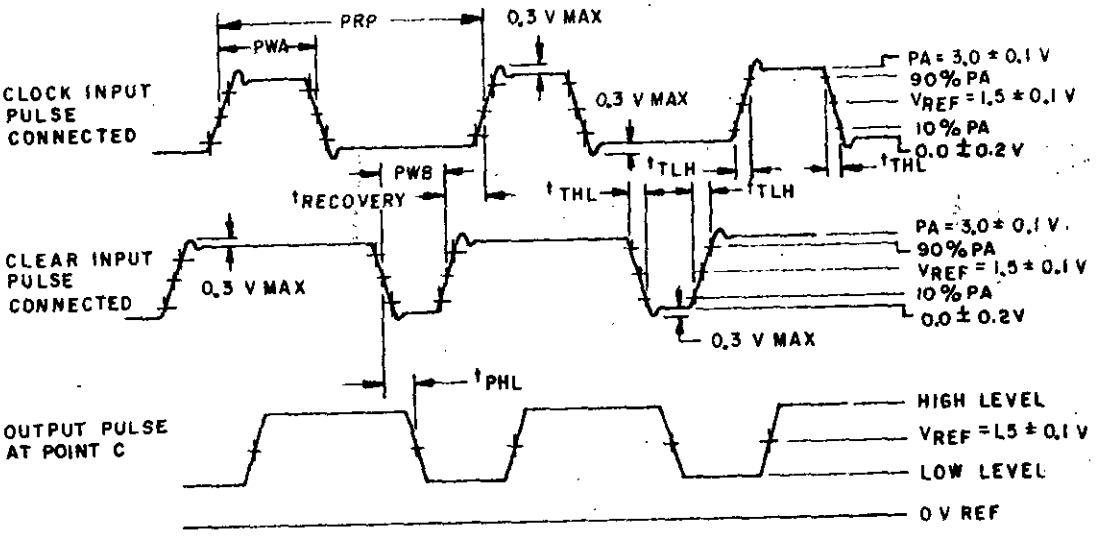
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NOTES:
1. R1 = R2 = 51.1Ω ±1%, 1/4 W.
2. Load all outputs with load s specified on figure 17.

* FIGURE 27. Propagation delay time (t_{PHL}) test circuit for table XIX.

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NOTES:
1. Clock input pulse parameters: PWA = 100 ±20 ns, PRP = 500 ±100 ns; $t_{TLH} = t_{THL} = 4 \pm 2$ ns.
2. Clear input pulse parameters: Clear pulse is taken from the negative output of item J01-1. Adjust the dc offset level of item J01-1 to obtain the required pulse voltage levels. Adjust the pulse width controls of item J01-1 to obtain PWB = 200 ±40 ns. Adjust the delay control of item J01-2 to obtain $t_{RECOVERY}$ = 10 +1.0, -0 ns (clear recovery time); $t_{TLH} = t_{THL} = 4 \pm 2$ ns.
3. Verify all outputs have a high level ≥ 2.5 V and low level ≤ 0.5 V before making measurements.
4. Physical length of electrical conductors between resistors R1 and R2 and input pins; between items Y01-1 and Y01-2 and input pins; and between load circuits and output pins, shall be less than 1 inch (25.4 mm).
5. All voltages are referenced to ground.

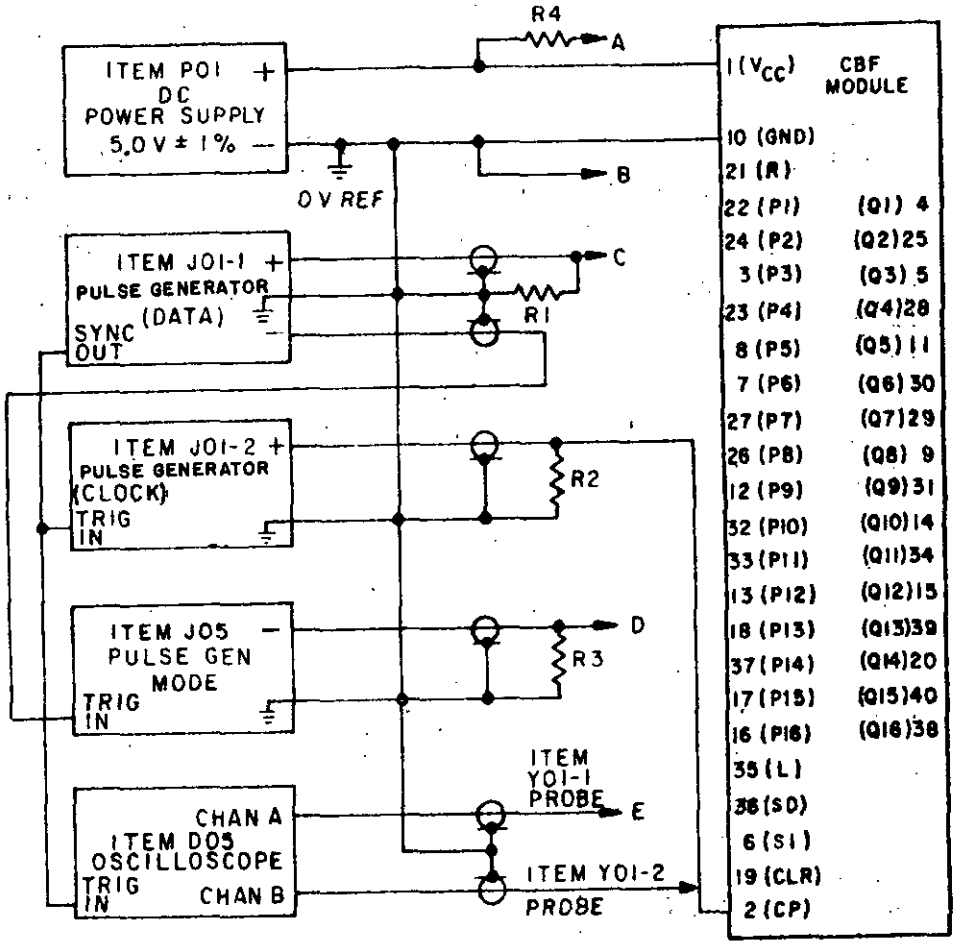
* FIGURE 28. Propagation delay time (t_{PHL}) test circuit waveforms for table XIX.

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* TABLE XX. Setup and hold time check test.

Test condition	Parameter		Connect					Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
T1	S0	Setup and hold time check (measuring Q1 output voltage, high level)	19	6,22	21	36	4	2.5	4.5	V	29,30
T2	S1	Setup and hold time check (measuring Q16 output voltage, low level)	16,19,21	36	35	6	38	0	500	mV	29,31
T3	S0,S1	Setup and hold time check (measuring Q8 output voltage, high level)	19	12,21, 27,35	26	6,36	9	2.5	4.5	V	29,30
T4	S0,S1	Setup and hold time check (measuring Q9 output voltage, low level)	19,21, 26,32, 35	Open	12	6,36	31	0	500	mV	29,31

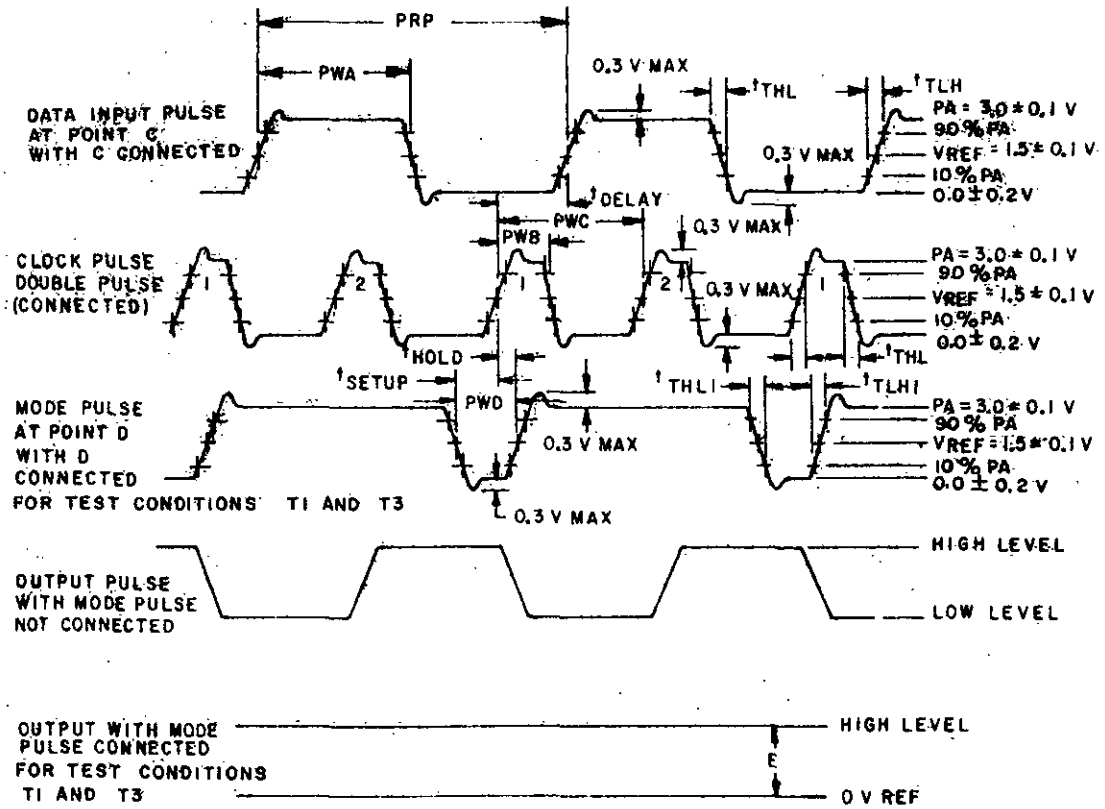
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NOTES:
1. R1 = R2 = R3 = 51.1Ω ±1%, 1/4 W.
   R4 = 1 kΩ ±5%, 1/4 W.
2. Load output under test with load as specified on figure 17.

* FIGURE 29. Setup and hold time check test circuit for table XX.

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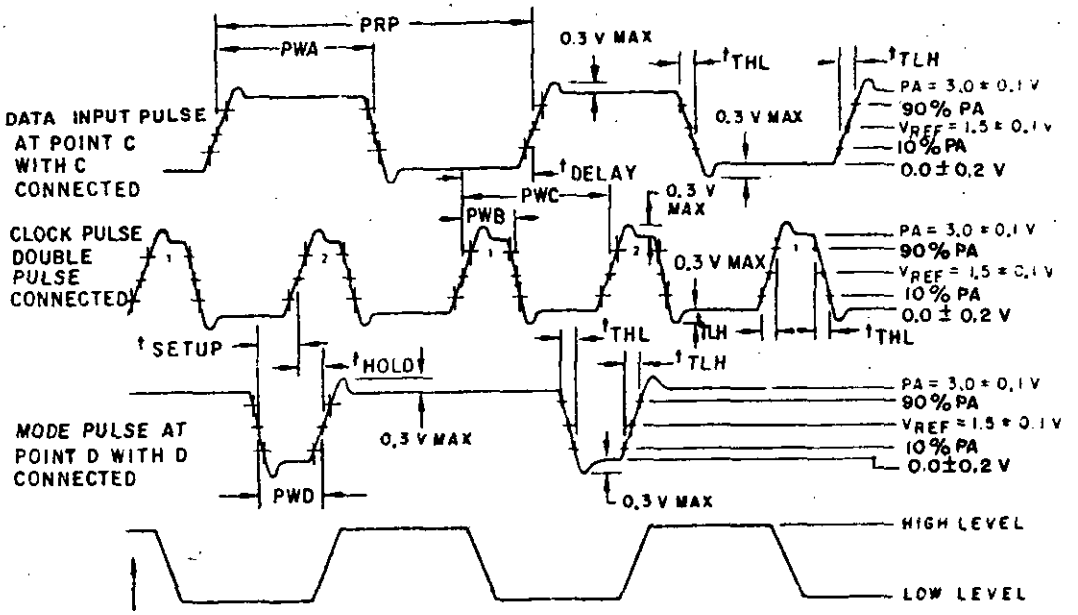


NOTES:

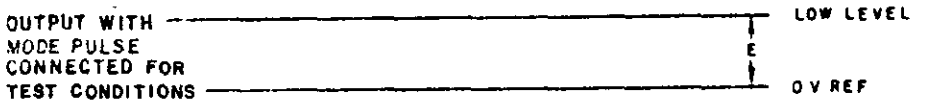
1. Data input pulse parameters: PWA = 250 ±25 ns; PRP = 500 ±50 ns; $t_{TLH} = t_{THL} = 4 \pm 2$ ns.
2. Clock pulse parameters: PWB = 100 ±10 ns; PWC = 250 ±25 ns (see note 4); $t_{TLH} = t_{THL} = 4 \pm 2$ ns.
3. Mode pulse parameters: (Pulse is taken from negative output of item J05. Adjust dc offset to obtain the required pulse amplitude and low level voltage.) $t_{TLH1} = t_{THL1} = 4 \pm 2$ ns; PWD - see note 4.
4. Item J01-2 is operated in double pulse mode. Using item D05, adjust the delay control of items J01-1 and J01-2. For $t_{DELAY} = 100 \pm 10$ ns to obtain the required data and clock pulses with item J05 not connected. Adjust the pulse width and delay controls of item J05 to obtain $t_{SETUP} = 11.0 + 1, -0$ ns and $t_{HOLD} = 4.5 +1, -0$ ns ($PWD = t_{SETUP} + t_{HOLD}$) and then connect item J05 to input. Item J05 is triggered by the negative pulse from item J01-1.
5. Physical length of electrical conductors between resistors R1 and R2 and input pins; between items Y01-1 and Y01-2 and input pins; and between load circuits and output pins, shall be less than 1 inch (25.4 mm).
6. All voltages are referenced to ground.

* FIGURE 30. Setup and hold time check test circuit waveforms for table XX.

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OUTPUT PULSE WITH MODE PULSE NOT CONNECTED FOR TEST CONDITIONS T2 AND T4



NOTES:

1. Data input pulse parameters: PWA = 250 ±25 ns; PRP = 500 ±50 ns; $t_{TLH} = t_{THL} = 4 \pm 2$ ns.
2. Clock pulse parameters: PWB = 100 ±10 ns; PWC = 250 ±25 ns (see note 4); $t_{TLH} = t_{THL} = 4 \pm 2$ ns.
3. Mode pulse parameters: (Pulse is taken from negative output of item J06. Adjust dc offset to obtain the required pulse amplitude and low level voltage.). $t_{TLH1} = t_{THL1} = 4 \pm 2$ ns; PWD = see note 4.
4. Item J01-2 is operated in double pulse mode. Using item D05, adjust the delay control of items J01-1 and J01-2. For $t_{DELAY} = 100 \pm 10$ ns to obtain the required data and clock pulses with item J05 not connected. Adjust the pulse width and delay controls of item J05 to obtain $t_{SETUP} = 10.0 + 1, -0$ ns and $t_{HOLD} = 4.5 + .5, -0$ ns ($PWD = t_{SETUP} + t_{HOLD}$) and then connect item J05 to input. Item J05 is triggered by the negative pulse from item J01-1.
5. Physical length of electrical conductors between resistors R1 and R2 and input pins; between items Y01-1 and Y01-2 and input pins; and between load circuits and output pins, shall be less than 1 inch (25.4 mm).
6. All voltages are referenced to ground.

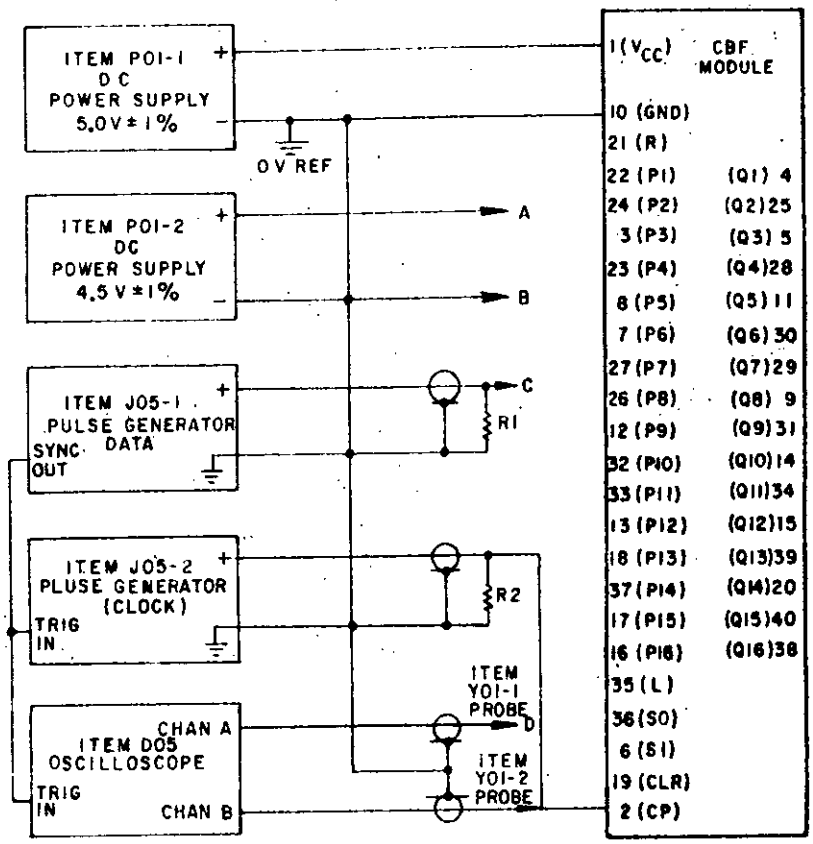
* FIGURE 31. Setup and hold time check test circuit waveforms for table XX.

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* TABLE XXI. Maximum clock frequency check test.

Test condition	Parameter		Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
U1	Q1	Maximum clock frequency check, minimum pulse width (measuring output pulse period t_{PW})	6,19,21,35, 36	Open	22	4	36	44	ns	32,33
U2	Q2				24	25				
U3	Q3				3	5				
U4	Q4				23	28				
U5	Q5				8	11				
U6	Q6				7	30				
U7	Q7				27	29				
U8	Q8				26	9				
U9	Q9				12	31				
U10	Q10				32	14				
U11	Q11				33	34				
U12	Q12				13	15				
U13	Q13				18	39				
U14	Q14				37	20				
U15	Q15				17	40				
U16	Q16				16	38				
U17	Q16	Maximum clock frequency check, minimum pulse width-right shift mode (measuring output pulse period t_{PW})	19,35,36	6	21	38				
U18	Q1	Maximum clock frequency check, minimum pulse width-left shift mode (measuring output pulse period t_{PW})	6,19,21	36	35	4				

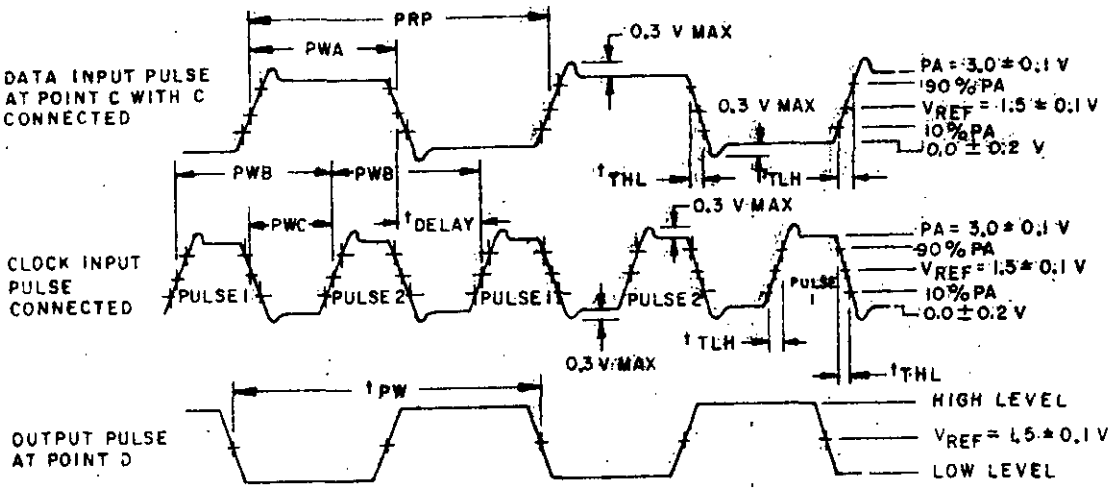
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NOTES:
1. R1= R2 = 51.1Ω ±1%, 1/4 W.
2. Load all outputs with load as specified on figure 17.

* FIGURE 32. Maximum clock frequency check test circuit for table XXI.

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NOTES:

1. Data input pulse parameters: $t_{TLH} = t_{THL} = 4 \pm 2$ ns, PWA = 20 ±1 ns; PRP = 40 ±0.5 ns.
2. Clock input pulse parameters: $t_{TLH} = t_{THL} = 4 \pm 2$ ns; PWB, PWC = see note 3.
3. Item J05-2 is operated in double pulse mode for maximum operating frequency test. Adjust pulse delay and pulsewidth to obtain PWC = 7 +1.0, -0 ns (minimum clock pulse width) and to obtain PWB = 20 +1, -0 ns (minimum clock period). Adjust delay of item J05-1 to obtain t_{DELAY} = 10 ±1 ns with output waveform as shown.
4. ★ All voltages are referenced to ground.

FIGURE 33. Maximum clock frequency check test circuit waveforms for table XXI.

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* TABLE XXII. Output crosstalk and output crosstalk duration (low-level) test.

Test condition	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Connect: D to pin	Initial limits 25°±5°C ambient: Min	Initial limits 25°±5°C ambient: Max	Unit	Test figure
V1	Q1 Output crosstalk, low level	22	3,7,8,12,13,16,17,18,23,24, 26,27,32,33,37	25	4	0.0	1.5	V	34,35
V2	Q2	24	3,7,8,12,13,16,17,18,22,23, 26,27,32,33,37	4	25				
V3	Q3	3	7,8,12,13,16,17,18,22,23,24, 26,27,32,33,37	25	5				
V4	Q4	23	3,7,8,12,13,16,17,18,22,24, 26,27,32,33,37	25	28				
V5	Q5	8	3,7,12,13,16,17,18,22,23,24, 26,27,32,33,37	30	11				
V6	Q6	7	3,8,12,13,16,17,18,22,23,24, 26,27,32,33,37	11	30				
V7	Q7	27	3,7,8,12,13,16,17,18,22,23, 24,26,32,33,37	30	29				
V8	Q8	26	3,7,8,12,13,16,17,18,22,23, 24,27,32,33,37	30	9				
V9	Q9	12	3,7,8,13,16,17,18,22,23,24, 26,27,32,33,37	14	31				
V10	Q10	32	3,7,8,12,13,16,17,18,22,23, 24,26,27,33,37	31	14				
V11	Q11	33	3,7,8,12,13,16,17,18,22,23, 24,26,27,32,37	14	34				
V12	Q12	13	3,7,8,12,16,17,18,22,23,24, 26,27,32,33,37	14	15				
V13	Q13	18	3,7,8,12,13,16,17,22,23,24, 26,27,32,33,37	20	39				
V14	Q14	37	3,7,8,12,13,16,17,18,22,23, 24,26,27,32,33	39	20				

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* TABLE XXII. Output crosstalk and output crosstalk duration (low-level) test - Continued.

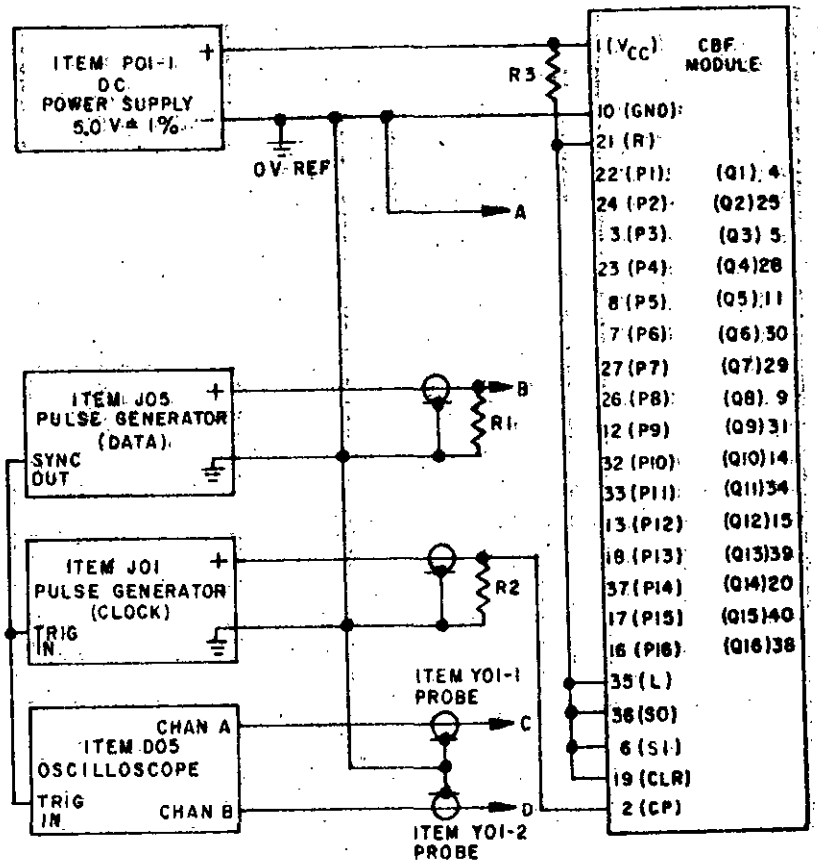
Test condition	Parameter		Connect: A to pin	Connect: B to pin	Connect: C to pin	Connect: D to pin	Initial limits 25°±5°C ambient: Min	Initial limits 25°±5°C ambient: Max	Unit	Test figure
V15	Q15	Output crosstalk, low level	17	3,7,8,12,13,16,18,22,23,24, 26,27,32,33,37	20	40	0.0	1.5	V	34,35
V16	Q16	Output crosstalk, low level	16	3,7,8,12,13,17,18,22,23,24, 26,27,32,33,37	20	38	0.0	1.5	V	34,35
W1	Q1	Output crosstalk duration test, low level	22	3,7,8,12,13,16,17,18,23,24, 26,27,32,33,37	2	4	0	500	mV	34,36
W2	Q2		24	3,7,8,12,13,16,17,18,22,23, 26,27,32,33,37		25				
W3	Q3		3	7,8,12,13,16,17,18,22,23,24, 26,27,32,33,37		5				
W4	Q4		23	3,7,8,12,13,16,17,18,22,24, 26,27,32,33,37		28				
W5	Q5		8	3,7,12,13,16,17,18,22,23,24, 26,27,32,33,37		11				
W6	Q6		7	3,8,12,13,16,17,18,22,23,24, 26,27,32,33,37		30				
W7	Q7		27	3,7,8,12,13,16,17,18,22,23, 24,26,32,33,37		29				
W8	Q8		26	3,7,8,12,13,16,17,18,22,23, 24,27,32,33,37		9				
W9	Q9		12	3,7,8,13,16,17,18,22,23,24, 26,27,32,33,37		31				
W10	Q10		32	3,7,8,12,13,16,17,18,22,23, 24,26,27,33,37		14				
W11	Q11		33	3,7,8,12,13,16,17,18,22,23, 24,26,27,32,37		34				

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* TABLE XXII. Output crosstalk and output crosstalk duration (low-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
W12	Q12 Output crosstalk duration test, low level	13	3,7,8,12,16,17,18,22,23,24, 26,27,32,33,37	2	15	0	500	mV	34,36
W13	Q13	18	3,7,8,12,13,16,17,22,23,24, 26,27,32,33,37		39				
W14	Q14	37	3,7,8,12,13,16,17,18,22,23, 24,26,27,32,33		20				
W15	Q15	17	3,7,8,12,13,16,18,22,23,24, 26,27,32,33,37		40				
W16	Q16	16	3,7,8,12,13,17,18,22,23,24, 26,27,32,33,37		38				

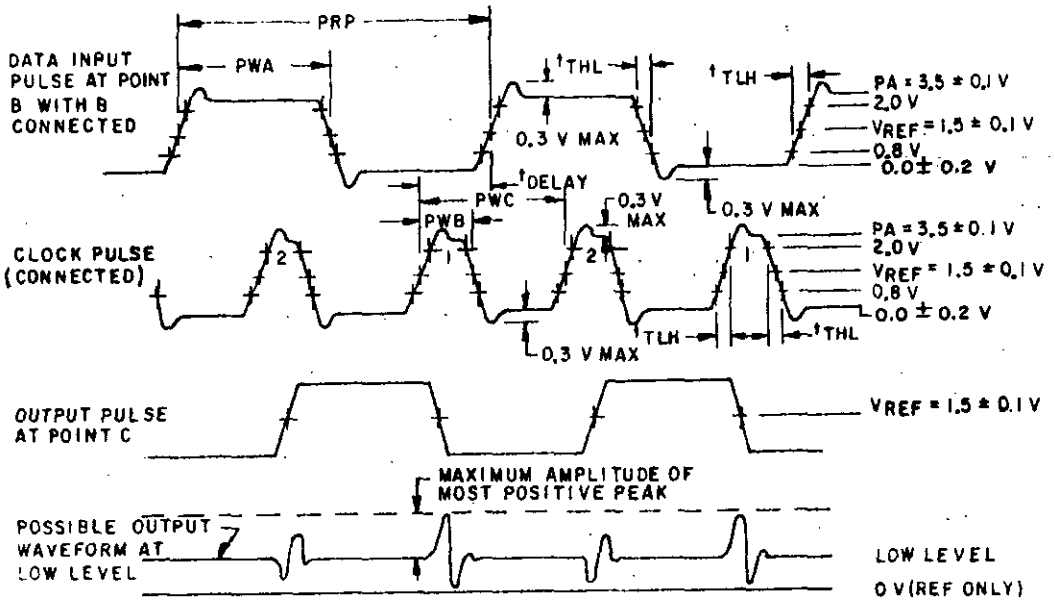
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NOTES:
1. R1 = R2 = 51.1Ω ±1%, 1/4 W.
2. R3 = 1 kΩ ±5%, 1/2 W.
3. Load all outputs with load as specified in figure 17 with the following exception: C_L = 50 ±5 pF.

* FIGURE 34. Output crosstalk and output crosstalk duration (low-level) test circuit for table XXII.

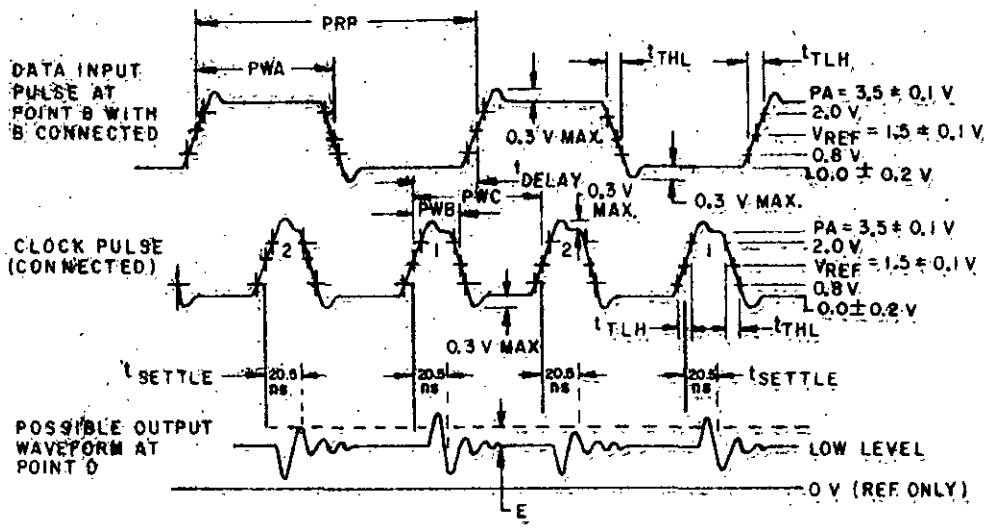
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NOTES:
1. Data input pulse parameters: PWA = 500 ±100 ns, PRP = 1,000 ±200 ns; $t_{TLH} = t_{THL}$ = 3.5 ±1.5 ns (measured at the 0.8 V and 2.0 V points of the pulse). For each test condition with the module connected to item J05, adjust item J05 for the required pulse amplitude, pulse low level voltage, and transition times. Verify the input pulse parameters at each input pin before making each measurement.
2. Clock input pulse parameters: PWB = 200 ±40 ns; PWC = 500 ±100 ns. $t_{TLH} = t_{THL}$ = 3.5 ±1.5 ns (measured at the 0.8 V and 2.0 V points of the pulse).
3. Item J01 is operated in double pulse mode. Using item D05, adjust the delay and pulse width controls of items J05, J01 to obtain the required data and clock pulse relationships with t_{DELAY} = 200 ±20 ns.
4. Physical length of electrical conductors between resistors R1 and R2 and input pins; between items Y01-1 and Y01-2 and input pins; and between load circuits and output pins, shall be less than 1 inch (25.4 mm).

* FIGURE 35. Output crosstalk (low-level) test circuit waveforms for table XXII.

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NOTES:

1. Data input pulse parameters: PWA = 500 ±100 ns, PRP = 1,000 ±200 ns. $t_{TLH} = t_{THL} = 3.5 \pm 1.5$ (measured at the 0.8 V and 2.0 V points of the pulse). For each test condition with the module connected to item J05, adjust item J05 for the required pulse amplitude, pulse low level voltage, and transition times. Verify the input pulse parameters at each input pin before making each measurement.
2. Clock input pulse parameters: PWB = 200 ±40 ns, PWC = 500 ±100 ns, $t_{TLH} = t_{THL} = 3.5 \pm 1.5$ ns (measured at the 0.8 V and 2.0 V points of the pulse).
3. Item J01 is operated in double pulse mode. Using item D05, adjust the delay and pulse width controls of items J05, J01 to obtain the required data and clock pulse relationships with $t_{DELAY} = 200 \pm 20$ ns.
4. E is the maximum amplitude of the most positive peak measured after a settle time of 20.5 ns has elapsed after any positive transition of the clock pulse and before the next positive transition of the clock pulse.
5. Physical length of electrical conductors between resistors R1 and R2 and input pins; between items Y01-1 and Y01-2 and input pins; and between load circuits and output pins, shall be less than 1 inch (25.4 mm).

* FIGURE 36. Output crosstalk duration (low-level) test circuit waveforms for table XXII.

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* TABLE XXIII. Output crosstalk and output crosstalk duration (high-level) test.

Test condition	Parameter		Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
X1	Q1	Output crosstalk, high level	22	3,7,8,12,13,16,17,18,23,24, 26,27,32,33,37	25	4	0	700	mV	37,38
X2	Q2		24	3,7,8,12,13,16,17,18,22,23, 26,27,32,33,37	4	25				
X3	Q3		3	7,8,12,13,16,17,18,22,23,24, 26,27,32,33,37	25	5				
X4	Q4		23	3,7,8,12,13,16,17,18,22,24, 26,27,32,33,37	25	28				
X5	Q5		8	3,7,12,13,16,17,18,22,23,24, 26,27,32,33,37	30	11				
X6	Q6		7	3,8,12,13,16,17,18,22,23,24, 26,27,32,33,37	11	30				
X7	Q7		27	3,7,8,12,13,16,17,18,22,23, 24,26,32,33,37	30	29				
X8	Q8		26	3,7,8,12,13,16,17,18,22,23, 24,27,32,33,37	30	9				
X9	Q9		12	3,7,8,13,16,17,18,22,23,24, 26,27,32,33,37	14	31				
X10	Q10		32	3,7,8,12,13,16,17,18,22,23, 24,26,27,33,37	31	14				
X11	Q11		33	3,7,8,12,13,16,17,18,22,23, 24,26,27,32,37	14	34				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:00Z
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* TABLE XXIII. Output crosstalk and output crosstalk duration (high-level) test - Continued.

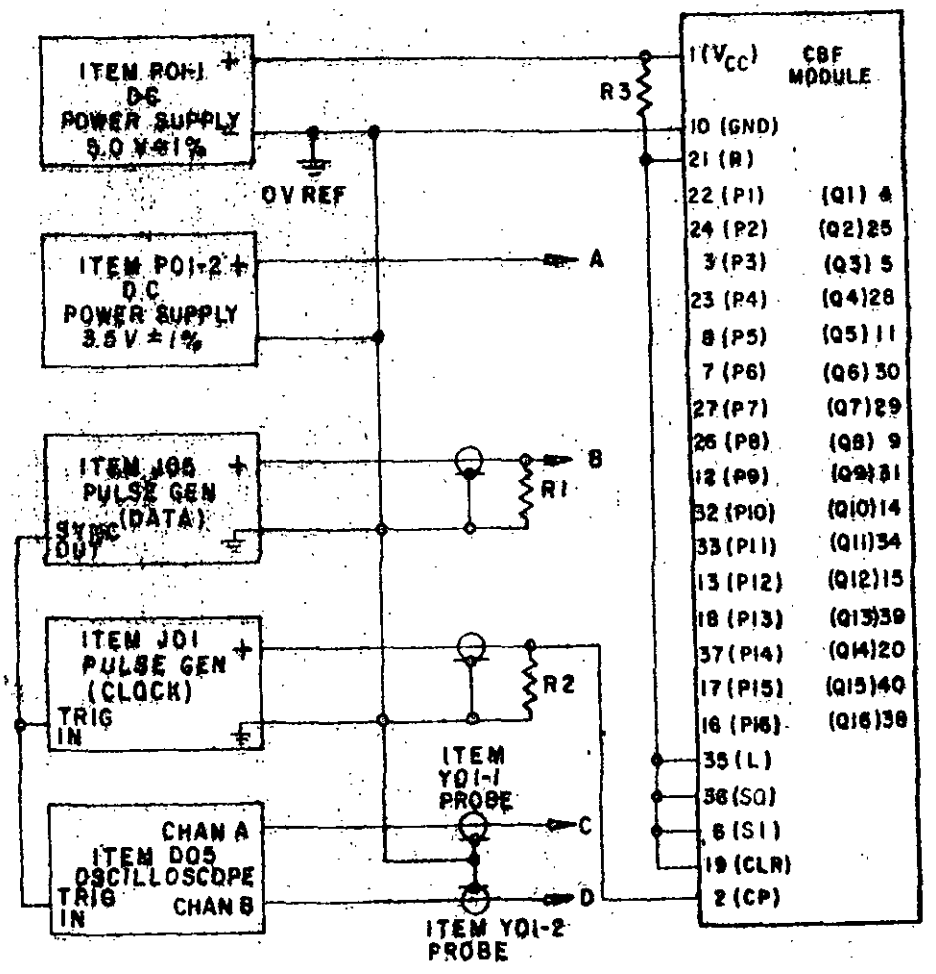
Test condition	Parameter		Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
X12	Q12	Output crosstalk, high level	13	3,7,8,12,16,17,18,22,23,24, 26,27,32,33,37	14	15	0	700	mV	37,38
X13	Q13		18	3,7,8,12,13,16,17,22,23,24, 26,27,32,33,37	20	39				
X14	Q14		37	3,7,8,12,13,16,17,18,22,23, 24,26,27,32,33	39	20				
X15	Q15		17	3,7,8,12,13,16,18,22,23,24, 26,27,32,33,37	20	40				
X16	Q16		16	3,7,8,12,13,17,18,22,23,24, 26,27,32,33,37	20	38				
Y1	Q1	Output crosstalk duration test, high level	22	3,7,8,12,13,16,17,18,23,24, 26,27,32,33,37	2	4		500		37,39
Y2	Q2		24	3,7,8,12,13,16,17,18,22,23, 26,27,32,33,37	2	25				
Y3	Q3		3	7,8,12,13,16,17,18,22,23,24, 26,27,32,33,37	2	5				
Y4	Q4		23	3,7,8,12,13,16,17,18,22,24, 26,27,32,33,37		28				
Y5	Q5		8	3,7,12,13,16,17,18,22,23,24, 26,27,32,33,37		11				
Y6	Q6		7	3,8,12,13,16,17,18,22,23,24, 26,27,32,33,37		30				
Y7	Q7		27	3,7,8,12,13,16,17,18,22,23, 24,26,32,33,37		29				

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* TABLE XXIII. Output crosstalk and output crosstalk duration (high-level) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
Y8	Q8	Output crosstalk duration test, high level	26	3,7,8,12,13,16,17,18,22,23,24, 27,32,33,37	2	9	0	500	mV	37,39
Y9	Q9		12	3,7,8,13,16,17,18,22,23,24, 26,27,32,33,37		31				
Y10	Q10		32	3,7,8,12,13,16,17,18,22,23,24, 26,27,33,37		14				
Y11	Q11		33	3,7,8,12,13,16,17,18,22,23,24, 26,27,32,37		34				
Y12	Q12		13	3,7,8,12,16,17,18,22,23,24, 26,27,32,33,37		15				
Y13	Q13		18	3,7,8,12,13,16,17,22,23,24, 26,27,32,33,37		39				
Y14	Q14		37	3,7,8,12,13,16,17,18,22,23,24, 26,27,32,33		20				
Y15	Q15		17	3,7,8,12,13,16,18,22,23,24, 26,27,32,33,37		40				
Y16	Q16		16	3,7,8,12,13,17,18,22,23,24, 26,27,32,33,37		38				

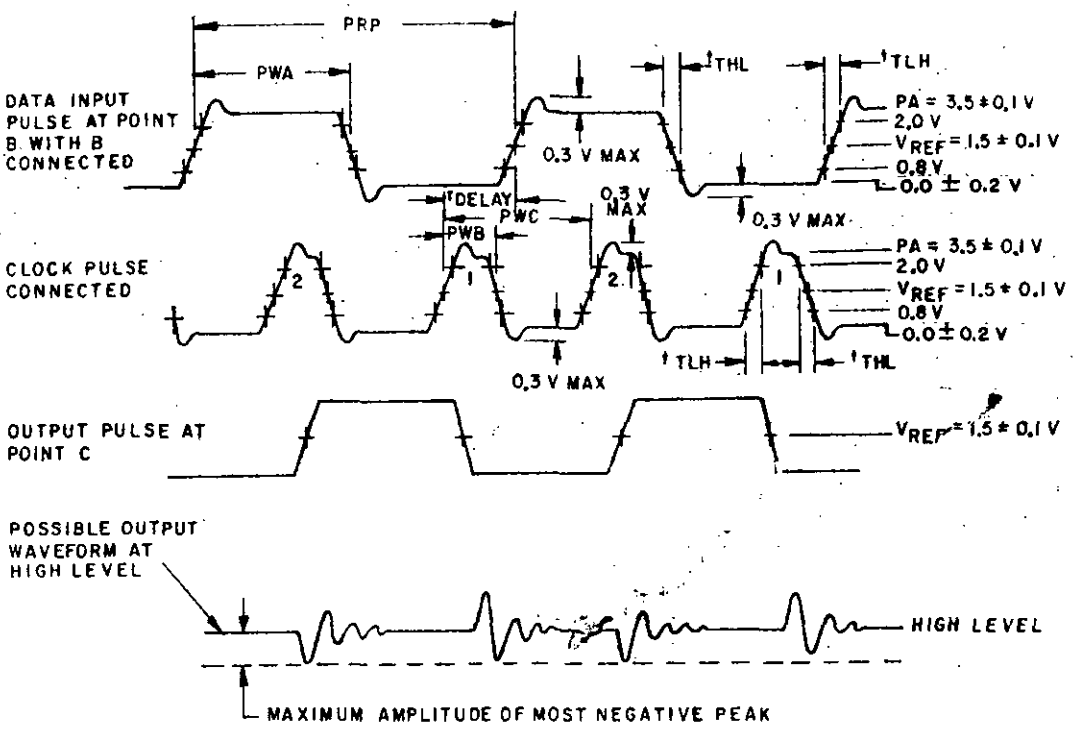
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:00Z
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NOTES:
1. R1 = R2 = 51.1Ω ±1%, 1/4 W.
2. R3 = 1 kΩ ±5%, 1/2 W.
3. Load all outputs as specified on figure 17 with the following exception: C_L = 50 ±5 pF.

* FIGURE 37. Output crosstalk and output crosstalk duration (high-level) test circuit for table XXIII.

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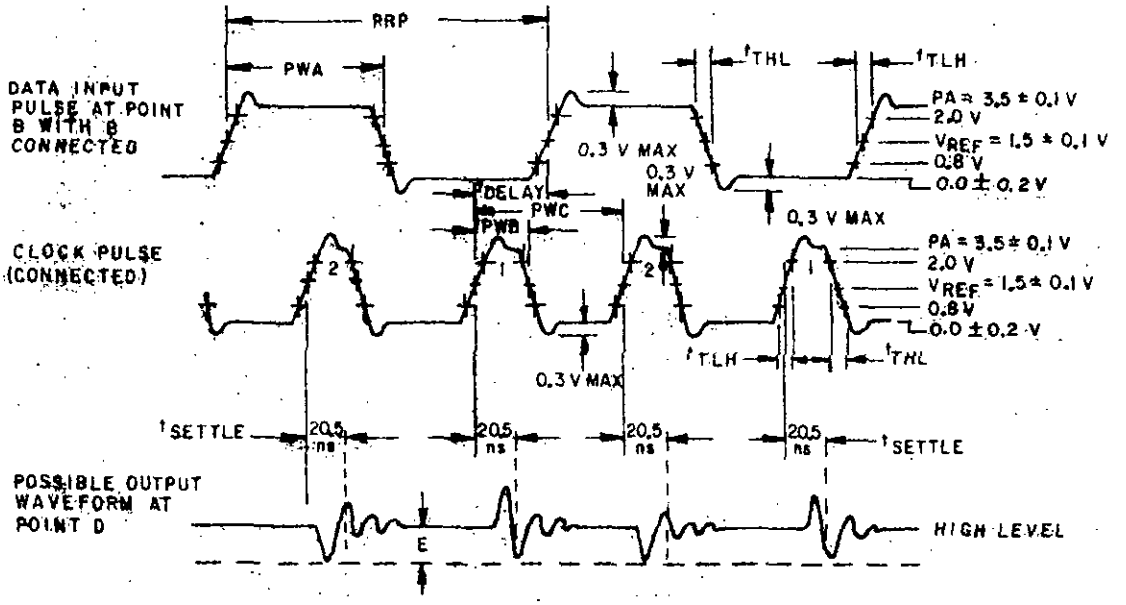


NOTES:

1. Data input pulse parameters: PWA = 500 ±100 ns; PRP = 1,000 ±200 ns. $t_{TLH} = t_{THL} = 3.5 \pm 1.5$ ns (measured at the 0.8 V and 2.0 V points of the pulse). For each test condition with the module connected to item J05, adjust item J05 for the required pulse amplitude, pulse low level voltage, and transition times. Verify the input pulse parameters at each input pin before making each measurement.
2. Clock input pulse parameters: PWB = 200 ±40 ns; PWC = 500 ±100 ns. $t_{TLH} = t_{THL} = 3.5 \pm 1.5$ ns (measured at the 0.8 V and 2.0 V points of the pulse).
3. Item J01 is operated in double pulse mode. Using item D05, adjust the delay and pulse width controls of items J05, J01 to obtain the required data and clock pulse relationships with $t_{DELAY} = 200 \pm 20$ ns.
4. Physical length of electrical conductors between resistors R1 and R2 and input pins; between items Y01-1 and Y01-2 and input pins; and between load circuits and output pins, shall be less than 1 inch (25.4 mm).

★ FIGURE 38. Output crosstalk (high-level) test circuit waveforms for table XXIII.

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NOTES:

1. Data input pulse parameters: PWA = 500 ±100 ns, PRP = 1,000 ±200 ns. $t_{TLH} = t_{THL}$ = 3.5 ±1.5 ns (measured at the 0.8 V and 2.0 V points of the pulse). For each test condition with the module connected to item J05, adjust item J05 for the required pulse amplitude, pulse low level voltage, and transition times. Verify the input pulse parameters at each input pin before making each measurement.
2. Clock input pulse parameters: PWB = 200 ±40 ns; PWC = 500 ±100 ns. $t_{TLH} = t_{THL}$ = 3.5 ±1.5 ns (measured at the 0.8 V and 2.0 V points of the pulse).
3. Item J01 is operated in double pulse mode. Using item D05, adjust the delay and pulse width controls of items J05, J01 to obtain the required data and clock pulse relationships with t_{DELAY} = 200 ±20 ns.
4. E is the maximum amplitude of the most negative peak measured after a settle time of 20.5 ns has elapsed after any positive transition of the clock pulse and before the next positive transition of the clock pulse.
5. Physical length of electrical conductors between resistors R1 and R2 and input pins; between items Y01-1 and Y01-2 and input pins; and between load circuits and output pins, shall be less than 1 inch (25.4 mm).

* FIGURE 39. Output crosstalk duration (high-level) test circuit waveforms for table XXIII.

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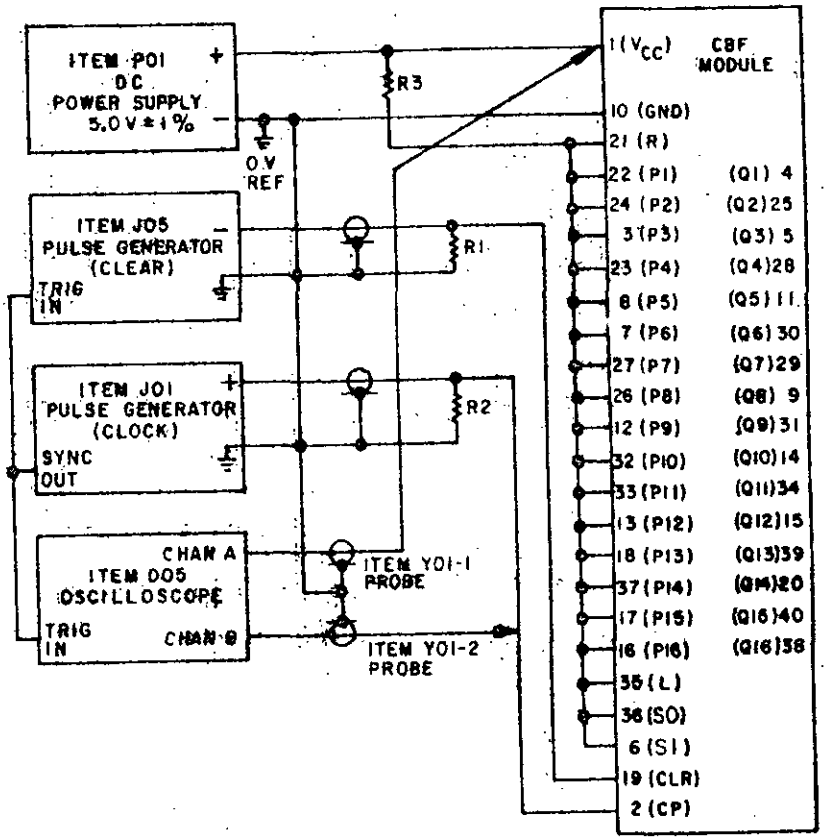
TABLE XXIV. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
Z	Power supply transient amplitude	0	600	mV	40,41

TABLE XXV. Power supply filter capacitance test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
AA	Power supply filter capacitance	Open	1	3.7	7.2	µF	42

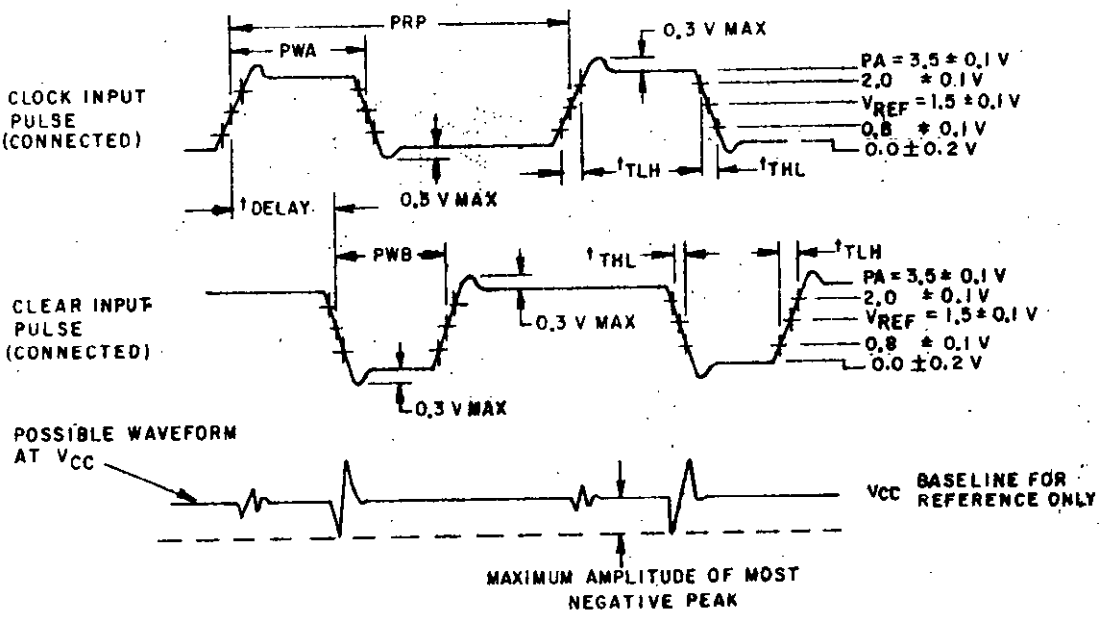
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NOTES:
1. R1 = R2 = 51.1Ω ±1%, 1/4 W.
2. R3 = 1 kΩ ±5%, 1/2 W.

FIGURE 40. Power supply transient amplitude test circuit for table XXIV.

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NOTES:
1. Clock input pulse parameters:

   $t_{TLH} = t_{THL} = 3.5 \pm 1.5$ ns (verified at each input pin)
   PWA = 500 ±100 ns.
   PRP = 1,000 ±200 ns.

2. Clear input pulse parameters:

   $t_{TLH} = t_{THL} = 3.5 \pm 1.5$ ns (verified at each input pin); PWB = 400 ±100 ns
   Negative output of item J05 is used for clear pulse.
   Adjust the DC offset and pulse amplitude to obtain the required waveform. Adjust the delay control of item J01 to obtain $t_{DELAY} = 250 \pm 50$ ns.

FIGURE 41. Power supply transient amplitude test circuit waveforms for table XXIV.

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TABLE XXVI. *Input capacitance tests.*

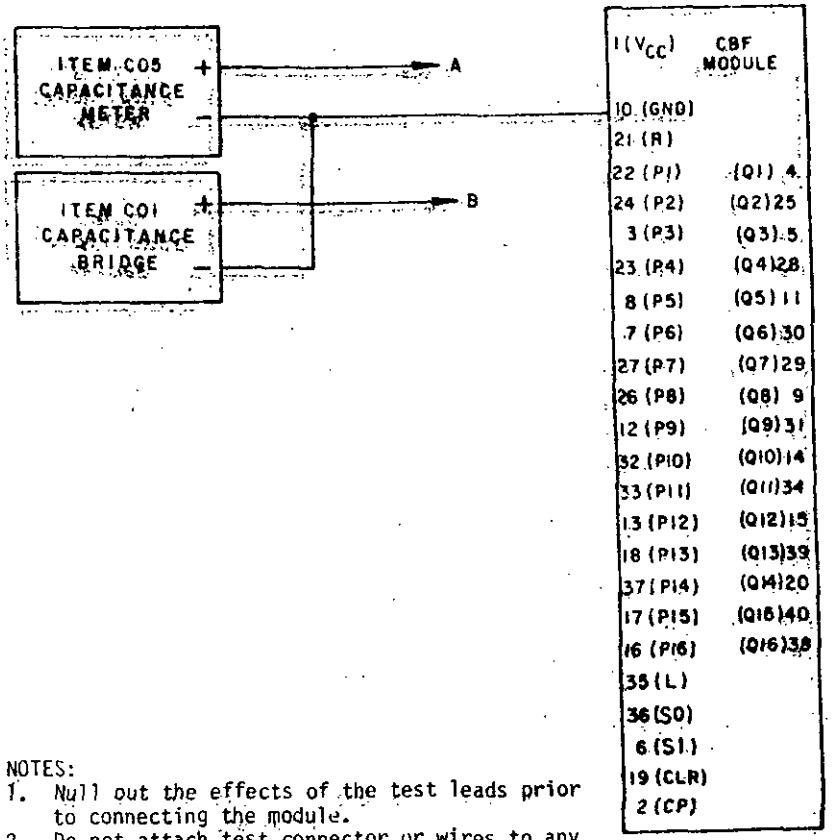
Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
BB1	R Input capacitance	21	Open	5	15	pF	42
BB2	P1	22					
BB3	P2	24					
BB4	P3	3					
BB5	P4	23					
BB6	P5	8					
BB7	P6	7					
BB8	P7	27					
BB9	P8	26					
BB10	P9	12					
BB11	P10	32					
BB12	P11	33					
BB13	P12	13					
BB14	P13	18					
BB15	P14	37					
BB16	P15	17					

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TABLE XXVI. Input capacitance tests - Continued.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
BB17	P16 Input capacitance	16	Open	5	15	pF	42
BB18	L	35			15		
BB19	SO	36			60		
BB20	S1	6			60		
BB21	CLR	19			72		
BB22	CP	2			60		

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NOTES:
1. Null out the effects of the test leads prior to connecting the module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.
* 3. Item C01 setup:
Frequency = 120 Hz; output
Vpk = 0.3 maximum.

FIGURE 42. Input and filter capacitance test circuit for tables XXV and XXVI.

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TABLE XXVII. Noise sensitivity (PW) test.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
CC1	Noise sensitivity (PW), CLR	19	4	2	19	36,6	450	550	ns	43,44
CC2			25							
CC3			5							
CC4			28							
CC5			11							
CC6			30							
CC7			29							
CC8			9							
CC9			31							
CC10			14							
CC11			34							
CC12			15							
CC13			39							
CC14			20							
CC15			40							
CC16			38							

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TABLE XXVII. Noise sensitivity (PW) test - Continued.

Test condition	Parameter	Connect					Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
CC17	Noise sensitivity (PW), SO	36	4	2	36	6,19	450	550	ns	43,44
CC18			25							
CC19			5							
CC20			28							
CC21			11							
CC22			30							
CC23			29							
CC24			9							
CC25			31							
CC26			14							
CC27			34							
CC28			15							
CC29			39							
CC30			20							
CC31			40							
CC32			38							

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TABLE XXVII. Noise sensitivity (PW) test - Continued.

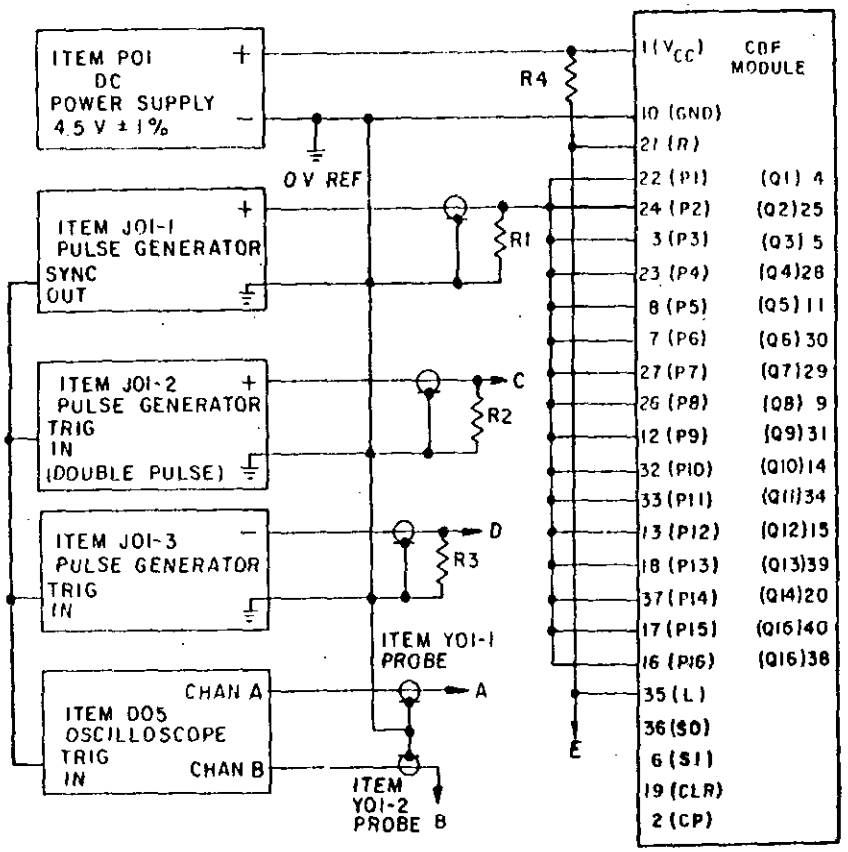
Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
CC33	Noise sensitivity (PW), S1	6	4	2	6	19,36	450	550	ns	43,44
CC34			25							
CC35			5							
CC36			28							
CC37			11							
CC38			30							
CC39			29							
CC40			9							
CC41			31							
CC42			14							
CC43			34							
CC44			15							
CC45			39							
CC46			20							
CC47			40							
CC48			38							

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TABLE XXVII. Noise sensitivity (PW) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
CC49	Noise sensitivity (PW), CP	2	4	Open	2	36,6,19	0.0	1.5	V	43,45
CC50			25							
CC51			5							
CC52			28							
CC53			11							
CC54			30							
CC55			29							
CC56			9							
CC57			31							
CC58			14							
CC59			34							
CC60			15							
CC61			39							
CC62			20							
CC63			40							
CC64			38							

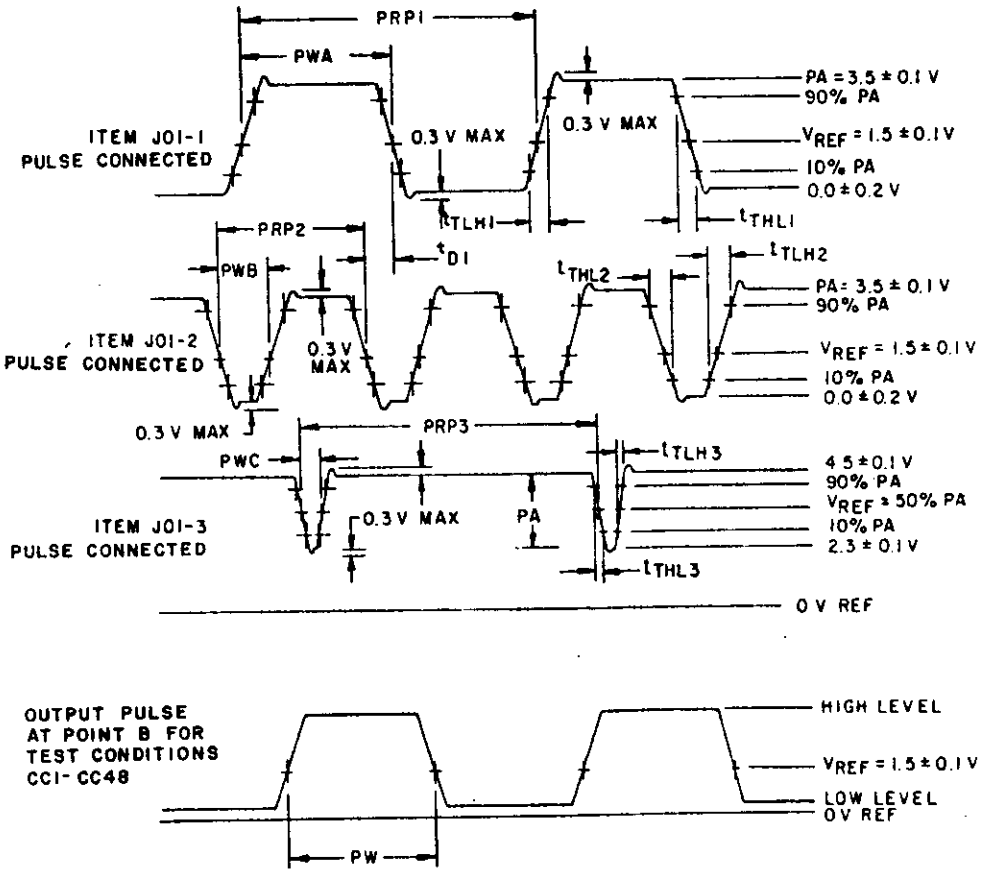
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NOTES:
1. R1 = R2 = R3 = 51.1Ω ±1%, 1/4 W.
2. R4 = 1 kΩ ±1%, 1/2 W.
* 3. Load all outputs as specified on figure 17 with the exception that C_L = 50 ±5 pF.
4. Clear the register before making the measurements for test conditions CC49-CC64.

FIGURE 43. Noise sensitivity (PW) test circuit for table XXVII.

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* FIGURE 44. Noise sensitivity (PW) test circuit waveforms for table XXVII.

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NOTES:

1. J01-1 pulse parameters:

   PRP1 = 1000 ± 100 ns.

   PWA = 500 ± 50 ns.

   $t_{TLH1} = t_{THL1} = 10 \pm 2$ ns.

2. J01-2 pulse parameters:

   PRP2 = 500 ± 50 ns.

   PWB = 200 ± 20 ns.

   $t_{TLH2} = t_{THL2} = 10 \pm 2$ ns.

   $t_{D1} = 100 \pm 10$ ns.

3. J01-3 pulse parameters:

   PRP3 = 1000 ± 100 ns.

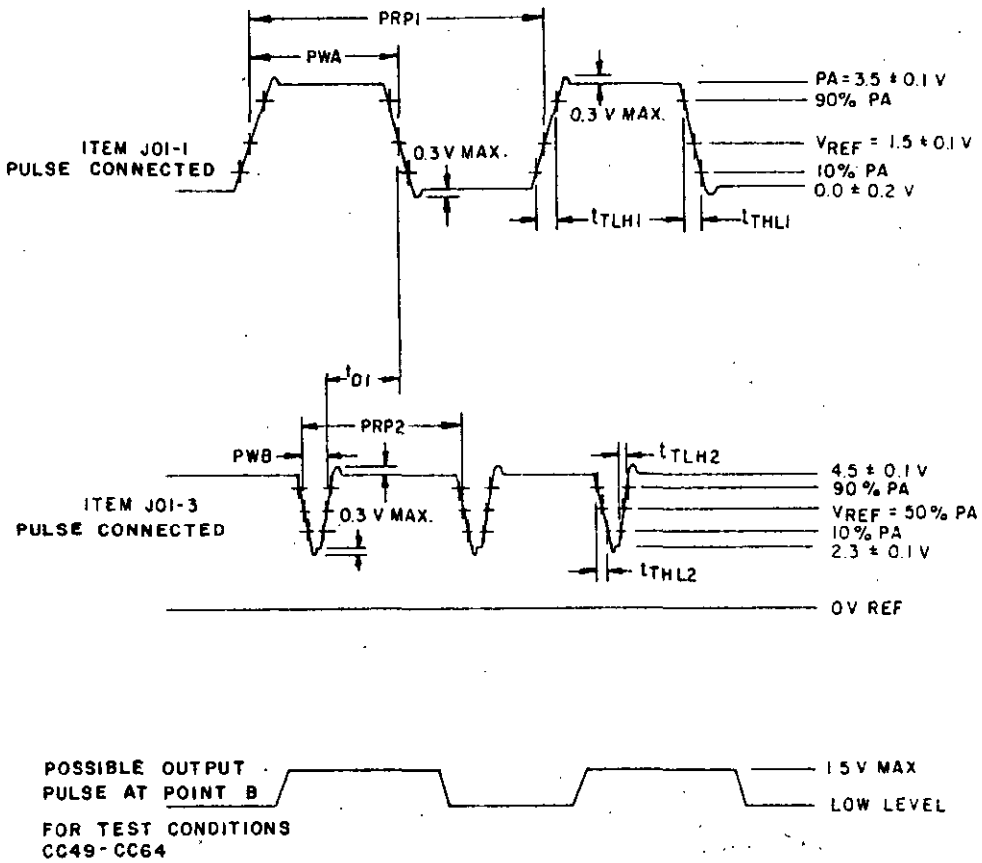
   PWC = 20 ± 5 ns.

   $t_{TLH3} = t_{THL3} = 6 \pm 2$ ns.

4. Sweep item J01-3's pulse through one cycle. The output pulse at point B shall meet the requirements of table XXVII.

* FIGURE 44. Noise sensitivity (PW) test circuit waveforms for table XXVII - *Continued*.

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* FIGURE 45. Nosie sensitivity (PW) test circuit waveforms for table XXVII.

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NOTES:

1. J01-1 pulse parameters:

   PRP1 = 1000 ± 100 ns.

   PWA = 500 ± 50 ns.

   $t_{TLH1} = t_{THL1} = 10 \pm 2$ ns.

   $t_{D1} = 100 \pm 10$ ns.

2. J01-3 pulse parameters:

   PRP2 = 500 ± 100 ns.

   PWB = 20 ± 5 ns.

   $t_{TLH2} = t_{THL2} = 6 \pm 2$ ns.

3. The output at point B shall satisfy the limits in table XXVII, test conditions CC49-CC64.

* FIGURE 45. Noise sensitivity (PW) test circuit waveforms for table XXVII - Continued.

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